

CSE 430 Architectural Design

Career Bound

Software Design Document

Name (s): Jeremy Troy Suchanski, Robert Anderson, Danny Capps,
Savannah Harvey, Spencer Beals, Kimberly Uresti, Adam Sheridan, Marco
Quintero, Corbin Probasco, Nathan Lipovich, Abraham Moreno-Sanchez,
Kaleb Swertfeger, Mike Olson, Kyle Jones, Yat Lam, Ernesto Quispe Cajan,
Jacob Mychal Kertesz, Pavel Shvets, Peyton Markus, Carson Roberts, Joe
Allen, Sam Evans, Zach Crittenden, Caleb Schmid, Jackson Aden, Scott
Shumway, Zachary Humphreys, Braulio Garcia Gonzalez, Ethan Houser,
Wessly Green Contreras, William Torman, Ethan Seegmiller, Sergey
Tsybenko, Sergio Bravo Bravo Moran, Dallin Stephens, Sunseehtay
Tirazona, Teph Milford

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Revision History

Version	Date	Author	Change Description
1.0	5/14/26	Jeremy Troy Suchanski	Filled in work done for Milestone 1 and started filling in sections relevant to Milestone 2 for Initial SWDD draft
2.0	5/22/26	Nathan Lipovich	Filled in the diagrams for 6.2 and 7 as well as the 6.1 introduction
3.0	5/24	Corbin Probasco	Fill in the missing information for Component Diagram Design Information Tables & other fixes.
4.0	5/28/26	Robert Anderson	Updated Component Diagrams and associated tables with missing information
5.0	6/6/26	Nathan Lipovich	Filled in Entity Identification Tables

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1.0 INTRODUCTION

1.1 Purpose

This Software Design Document (SWDD) describes the architecture, subsystem breakdown, data design, component structure, and human interface design of the Career Bound system. The document provides the design foundation necessary for implementation and integration of the Career Bound software platform.

The intended audience of this document includes:

- Developers
- Designers
- QA reviewers
- Project managers
- Instructors
- Stakeholders

This document serves as the primary design reference for the Career Bound project and establishes standardized architecture, interface, workflow, and UI design patterns used throughout the system.

1.2 Scope

Career Bound is a web and mobile application for career networking and job searching designed to connect students and employers through profiles, resumes, job postings, notifications, and search capabilities.

The system supports:

- Student profile management
- Company profile management
- Resume creation and submission
- Job posting management
- Candidate and job search
- Notifications and news feed updates
- Responsive desktop and mobile interfaces

The goals of the project are to:

- Simplify finding jobs and candidates
- Improve communication between students and employers
- Provide a centralized career development platform
- Support scalable and modular software architecture

- Demonstrate standardized software engineering practices

The project delivers the following key benefits:

- Improved organization of career-related information
 - All student profiles, resumes, job postings, company information, news, and notifications are centralized in one easy-to-access platform instead of being scattered across emails, spreadsheets, and multiple websites.
- Streamlined resume and application workflows
 - Students can create, update, and submit resumes quickly and efficiently, while companies can easily review candidates and manage job postings, which reduces the manual work and speeds up the entire hiring process.
- Improved accessibility on desktop and mobile devices
 - Students and companies can access all features (profiles, search, notifications, resumes, etc.) conveniently from any device. The platform provides consistent, high-quality experiences whether on a desktop computer, laptop, tablet, or mobile phone.
- Modular architecture supporting maintainability and scalability
 - The system is built using well-defined and independent components (as shown in the component diagram). This modularity makes the application easier to maintain, update, and expand in the future without disrupting the entire system.

1.3 Overview

This document is organized into the following major sections:

- Introduction and project overview
- System architecture and subsystem decomposition
- Data design and storage structure
- Component design
- Human interface design and screen views
- Requirements traceability
- Supporting appendices and standards

The SWDD combines architectural diagrams, interface views, workflow standards, and UI design patterns to provide a complete description of the Career Bound software system.

1.4 Reference Material

The following references were used in the design of Career Bound:

- Software Requirements Specification (SRS)
- IEEE 1016 Software Design Description guidance
- Course textbook Chapter 4.1 – Design Description
- GitHub CareerBound-SDD Repository Standards and Workflow Documentation

- Career Bound Component Diagram Standards
- Career Bound UI and Wireframe Standards

1.5 Definitions and Acronyms

Term / Acronym	Definition
API	Application Programming Interface
UI	User Interface
UX	User Experience
QA	Quality Assurance
IC	Individual Contributor
SWDD	Software Design Document
SRS	Software Requirements Specification
REST	Representational State Transfer
Wireframe	Simplified UI layout representation
Component Diagram	Diagram representing major system components and interfaces
Responsive Design	UI design adapting to multiple screen sizes

2.0 SYSTEM OVERVIEW

Career Bound is designed as a modular career networking platform that supports interaction between students and employers. The system provides profile management, resume handling, job posting functionality, notifications, and advanced search features.

The system is divided into multiple major services and interface layers in order to improve maintainability, scalability, and separation of responsibilities.

The primary system users include:

- Students
- Employers / Companies
- Administrators

The system supports both desktop and mobile user interfaces through responsive UI design patterns.

Major functional areas include:

- Authentication and user management
- Profile management
- Resume services
- Job posting services
- Search services
- Communication services
- Data storage services

3.0 SYSTEM ARCHITECTURE

3.1 Architectural Design

The Career Bound system is designed using modular, component-based architecture to clearly separate concerns and thus enable scalability and maintainability. The overall system is decomposed into well-defined, replaceable components that communicate through explicit interfaces and ports.

Figure 1.1 below shows the high-level component diagram for the Career Bound system. This diagram shows how the system responsibilities have been partitioned into major subsystems and how those subsystems collaborate to deliver the complete functionality required by the SRS.

Figure 1.1 — Career Bound System Master Component Diagram (UML 2.0)

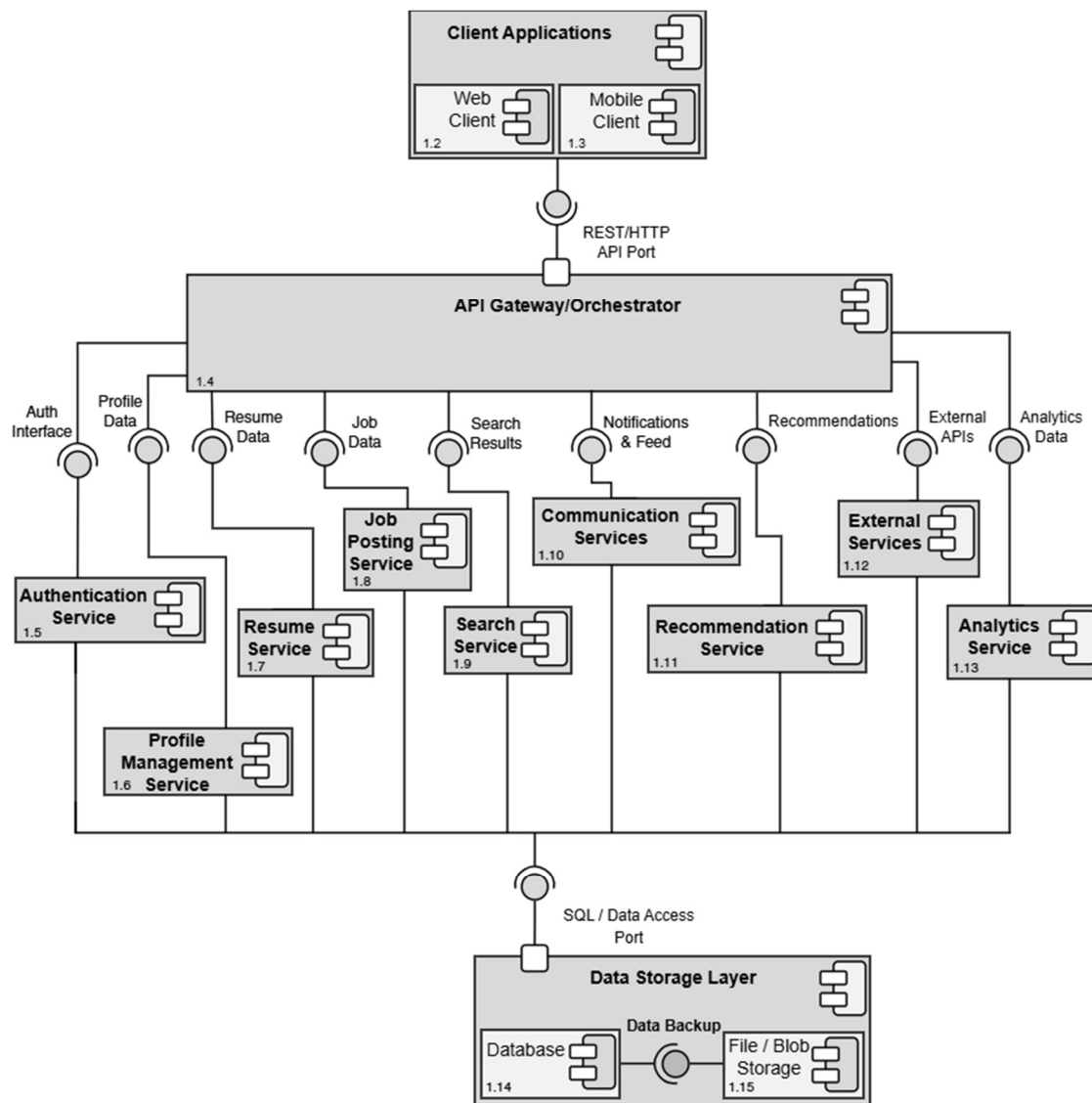


FIGURE 1.1

Design Information – Career Bound System Master Component Diagram

Field	Description / Purpose
View Name & Number	1.1 – Career Bound System Master Component Diagram (UML 2.0)
Purpose	Describe the overall system view.
Description	Basic client, application, database architecture with two types of clients: mobile and web.
Requirements	All SRS sections — 2.1 Company Profile, 2.2 Notifications, 2.3 Search, 2.4 News Feed, 2.5 Job Postings, 3.1 Student Profile, 3.2 Notifications, 3.3 Search, 3.4 News Feed, 3.5 Resume, 4.0 Hardware.
Elements	Client Applications: Web Client and Mobile Client
	API Gateway/Orchestrator
	Auth Service
	Profile Management Service
	Resume Service
	Job Posting Service
	Search Service
	Communications Service
	Recommendation Service
	External Services
	Analytics Service
Navigation / References	Figures 1.2–1.15 all component detail views
Viewpoint	Component Diagram

The interfaces and ports shown in Figure 1.1 are formally defined in the Interface Glossary below. All components communicate exclusively through these explicit interfaces, ensuring low coupling and high replaceability.

Interface Glossary – Career Bound System Master Component Diagram

Interface Name	Description	Provided By	Required By
REST/HTTP API Port	Formal, well-defined entry point for all client (Web + Mobile) communication	API Gateway / Orchestrator	Client Applications (Web Client + Mobile Client)
Auth Interface	Login, registration, role management, and session handling	Authentication Service	API Gateway
Profile Data	Read/write access to company and student profiles (locations, following, skills, GPA, etc.)	Profile Management Service	API Gateway, Search Service, Recommendation Service
Resume Data	Resume creation, submission, storage, retrieval, and access control	Resume Service	API Gateway, Search Service, Company profiles

Interface Name	Description	Provided By	Required By
Job Data	Job posting creation, editing, deletion, required skills, and job details	Job Posting Service	API Gateway, Search Service, Recommendation Service
Search Results	Candidate and job search results, filters, autocomplete, and wildcards	Search Service	API Gateway
Notifications & Feed	In-app notifications, push notifications, news feed posts, and updates	Communication Services	API Gateway
Recommendations	Job and candidate recommendations based on profiles, resumes, and job data	Recommendation Service	API Gateway
External APIs	Integration with third-party services (USPS address validation, autocomplete APIs, push notification providers, etc.)	External Services	API Gateway
Analytics Data	System usage metrics, reporting, and analytics data	Analytics Service	API Gateway
SQL / Data Access Port	Formal port — the single standardized interface through which all services access persistent data (the only way any service talks to the database or file storage)	Data Storage Layer	All services

The architecture is organized into four primary layers:

- **Client Applications (parent)** – Encloses the Web Client and Mobile Client. These components provide the unified user interface for both students and companies, supporting all features described in SRS sections 2.x and 3.x.
- **API Gateway / Orchestrator** – Serves as the single, centralized entry point for all client requests. It routes traffic to the appropriate backend services and enforces cross-cutting concerns such as authentication and request orchestration.
- **Core Services Layer** – Contains the business logic components:
 - Authentication Service
 - Profile Management Service
 - Resume Service
 - Job Posting Service
 - Search Service
 - Communication Services (Notifications + News Feed)
 - Recommendation Service
 - External Services
 - Analytics Service

Each service is responsible for a cohesive set of related functions and exposes well-defined interfaces to the rest of the system.

- **Data Storage Layer (parent)** – Acts as the single source of truth for all persistent data. It

encloses the Database and File / Blob Storage components and exposes the formal SQL / Data Access Port used by every service in the system.

All communication between the Client Applications and the backend services flows through the **REST/HTTP API Port** provided by the API Gateway. Every service communicates with the Data Storage Layer exclusively through the **SQL / Data Access Port**. This design ensures reduced dependencies between components, clear separation of responsibilities, and makes individual components replaceable and maintainable. Detailed views of each major component are provided in Figures 1.2 through 1.15. This modular architecture directly supports the system's requirements for accessibility, scalability, and maintainability while providing a clear roadmap for implementation and future extension.

3.2 Decomposition Description

The master component diagram (Figure 1.1) provides a high-level view of the overall system architecture. This section presents the detailed decomposition of each major component identified in the master diagram. Each sub-diagram shows the internal structure of the component, its sub-components, and the interfaces through which it interacts with the rest of the system. All sub-diagrams maintain the same interface names and ports as defined in the master diagram and the Interface Glossary to ensure consistency.

The following figures provide the detailed decomposition of each major component shown in the master diagram (Figure 1.1).

Additional Decomposition Artifacts

In addition to the component decomposition diagrams and information tables, this section also includes:

- Entity Identification Tables,
- Structure Charts,
- and selected Data Flow Diagrams (DFDs)

for major subcomponents within the system architecture.

These artifacts further decompose the:

- internal functionality,
- entities,
- data movement,
- and functional organization

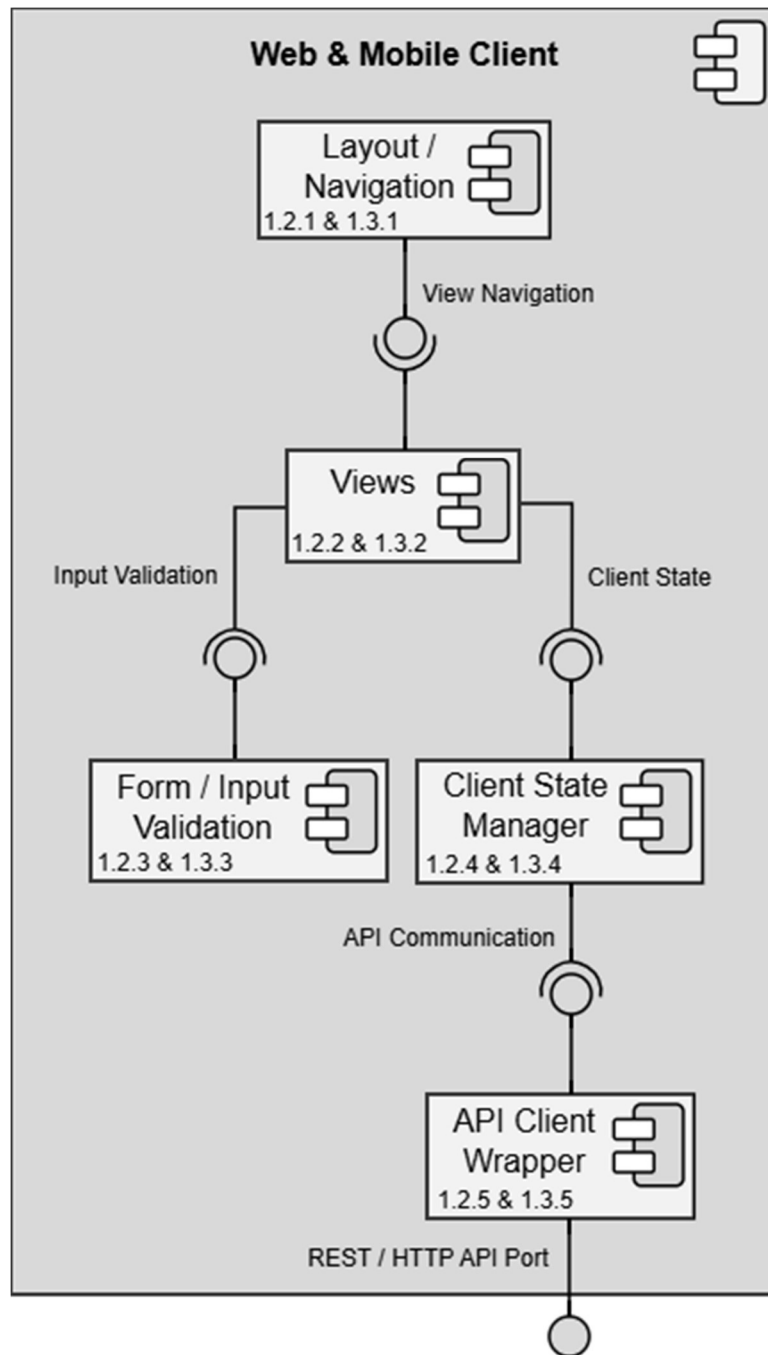
of the system components identified in the master component diagram.

The Profile, Resume, Job Posting, Search, Communication Services, Recommendation, External Services, and Analytics Orchestrators follow a common orchestration pattern. Each receives a validated request from the API Gateway, coordinates with its assigned downstream service, handles service responses and errors, and returns a normalized response. Because their control responsibilities are similar, their shared entities are documented once within a shared Service Orchestrator Pattern Entity Identification Table, while domain-specific entities are documented separately under each related orchestrator component.

Figure 1.2 & 1.3 — Web Client and Mobile Client Shared Architecture

The Web Client and Mobile Client share the same internal architecture and corresponding subcomponents. Therefore, the shared subcomponents are labeled with both their Web Client (1.2.x) and Mobile Client (1.3.x) numbering references.

Differences between the Web Client and Mobile Client are primarily related to layout and presentation. The Web Client uses a desktop layout with sidebar navigation, while the Mobile Client uses a mobile layout with hamburger navigation and responsive screens.



Design Information – Web & Mobile Client

Field	Description / Purpose
View Name & Number	1.2 – Web Client & 1.3 – Mobile Client
Purpose	Describe the overall system view.
Description	Basic client, application, database architecture with two types of clients: mobile and web.
Requirements	Account
Elements	1.2.1 & 1.3.1 Layout / Navigation
	1.2.2 & 1.3.2 Views
	1.2.3 & 1.3.3 Form / Input Validation
	1.2.4 & 1.3.4 Client State Manager
	1.2.5 & 1.3.5 API Client Wrapper
Navigation / References	1.4 API Gateway / Orchestrator; all screen views (1.1–4.2)
Viewpoint	Component Diagram

Entity Identification Tables – Web and Mobile Client**EI Table 1.2.1 & 1.3.1 – Layout / Navigation**

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Layout/Navigation	Menu	Class	Represents a menu	1.2.2 & 1.3.2	Views
Layout/Navigation	currentView	Object	Represents the current view shown on the page	1.2.2 & 1.3.2	Views
Layout/Navigation	updateView()	Method	Navigates to the chosen view when the button is selected	1.2.2 & 1.3.2	Views
Layout/Navigation	navButton	Object	Represents an interactable button on the screen that navigates to a paired view, calls updateView()	1.2.2 & 1.3.2	Views
Layout/Navigation	Favicon	Data Store	Stores a logo or otherwise small PNG or JPEG image to be used in locations such as the browser tab		
Layout/Navigation	menuTab	Object	Represents a dropdown tab featuring a collection of buttons		
Layout/Navigation	style	Data Store	Stores the color code, fonts, and other aesthetic details of the navigation		

			menu		
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EI Table 1.2.2 & 1.3.2 – Views

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Views	View	Class	Represents a view		
Views	component	Object	Represents a reusable UI element		
Views	render()	Method	Displays the view		
Views	onInit()	Method	Executes code at the initial loading of the page		
Views	language	Data Store	Stores the native language of the view, such as English		
Views	LocalizationService	Data Store	Provides manual language translations of elements on the page		
Views	refresh()	Method	Reload the screen		
Views	Style	Data Store	Stores the color code, fonts, and other aesthetic details of the view		
Views	State	Method	Receives layout information from Client State Manager for resizing the page, desktop vs mobile layout, etc.	1.2.4 & 1.3.4	Client State Manager

EI Table 1.2.3 & 1.3.3 – Form / Input Validation

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Form / Input Validation	Form	Class	Represents a form	1.2.2 & 1.3.2	Views
Form / Input Validation	Fields	Object	Represents the collection of fields in a form		
Form / Input Validation	Field	Class	Represents a reusable field element	1.2.2 & 1.3.2	Views

Form / Input Validation	input	Variable	Represents a field's input		
Form / Input Validation	fieldName	Variable	Represents the name of a field		
Form / Input Validation	requiredDataType	Variable	Determines the data type required by a specific field, such as String or Int		
Form / Input Validation	submit()	Method	Submits the form and the inputs in each field	1.2.2 & 1.3.2	Views
Form / Input Validation	populate()	Method	Populates the form with user information saved in the browser	1.2.2 & 1.3.2	Views
Form / Input Validation	Style	Data Store	Stores the color code, fonts, and other aesthetic details of the form	1.2.2 & 1.3.2	Views
Form / Input Validation	checkForm()	Method	Checks the validity of the form when a submission is attempted (ex. are all required fields filled out)		

EI Table 1.2.4 & 1.3.4 – Client State Manager

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Client State Manager	State	Class	Represents the client state	1.2.2 & 1.3.2	Views
Client State Manager	width	Variable	Represents the screen width in pixels	1.2.2 & 1.3.2	Views
Client State Manager	height	Variable	Represents the screen height in pixels	1.2.2 & 1.3.2	Views
Client State Manager	onUpdate()	Method	Checks for screen updates and acts accordingly (ex. updating width and height)	1.2.2 & 1.3.2	Views
Client State Manager	DynamicUI	Method	Provides layout details when certain requirements are met (ex. updates between mobile and desktop based on screen width)	1.2.2 & 1.3.2	Views

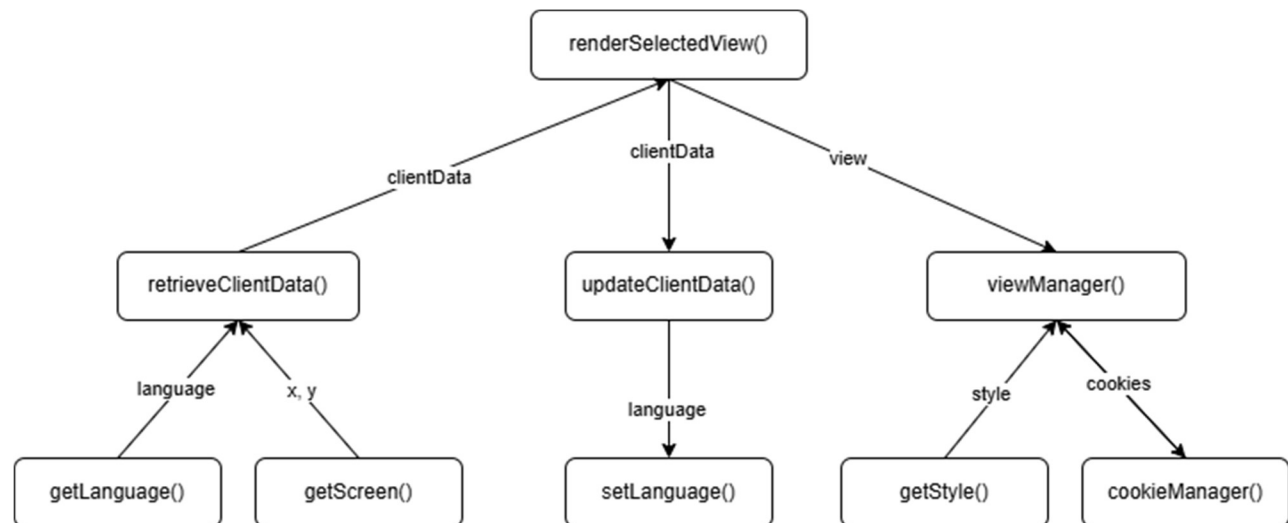
Client State Manager	updateAccessibility()	Method	Updates the client based on user accessibility settings	1.2.2 & 1.3.2	Views
Client State Manager	BackgroundHandler	Method	Runs background tasks such as caching, API calls, etc	1.2.2 & 1.3.2, 1.2.5 & 1.3.5	Views, API Client Wrapper

EI Table 1.2.5 & 1.3.5 – API Client Wrapper

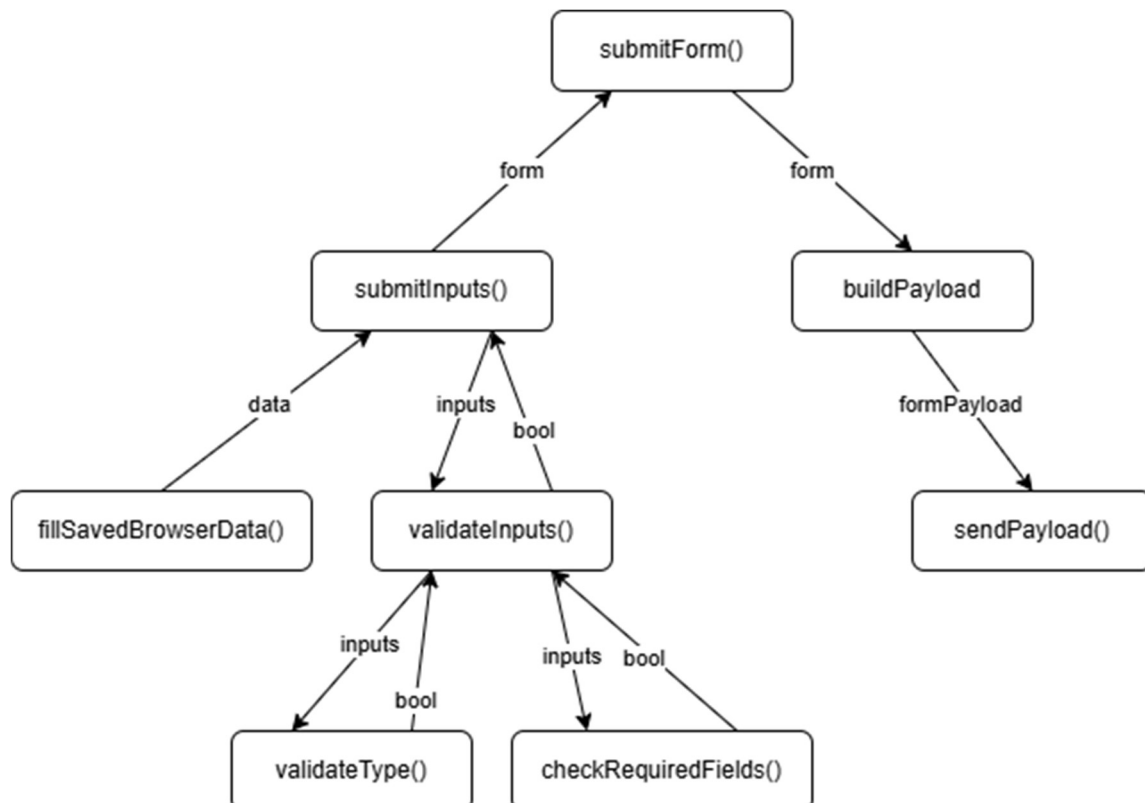
Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
API Client Wrapper	sendRequest()	Method	Sends a request to the API		
API Client Wrapper	Endpoint	Class	Represents an API endpoint		
API Client Wrapper	response	Object	Represents a response from the API		
API Client Wrapper	ResponseParser()	Method	Parses API calls		
API Client Wrapper	HttpClient	Class	Represents a raw API engine	1.2.4 & 1.3.4	Client State Manager
API Client Wrapper	NetworkConfig	Object	Represents the network configuration details		
API Client Wrapper	JsonSerializer	Object	Represents a JSON encoder and decoder	1.2.4 & 1.3.4	Client State Manager
API Client Wrapper	CookieStore	Data Store	Stores cookies in the browser		
API Client Wrapper	timeout()	Method	Executes a timeout	1.2.4 & 1.3.4	Client State Manager
API Client Wrapper	TimeoutPolicy	Object	Determines the criteria for a timeout		

Structure Charts – Web & Mobile Client

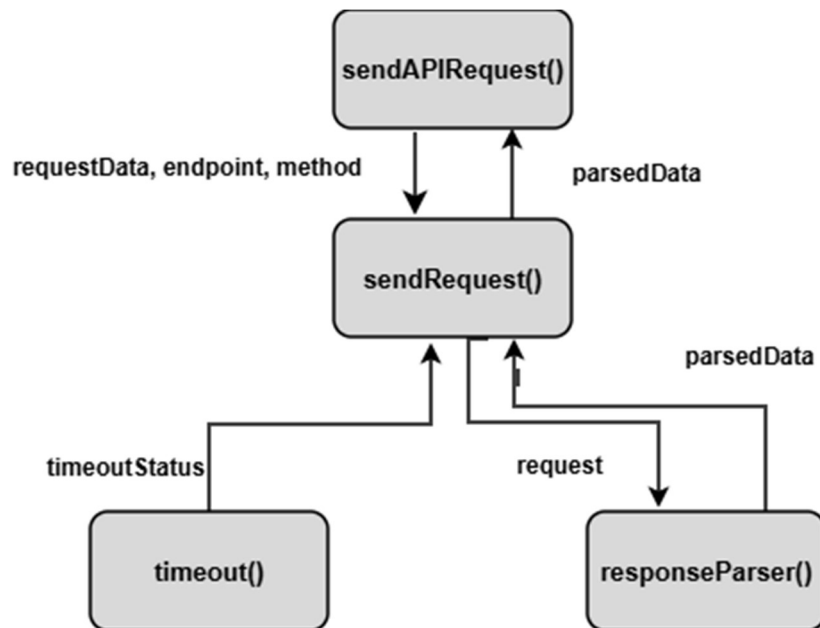
SC Figure 1.2.0.1 & 1.3.0.1 – renderSelectedView()



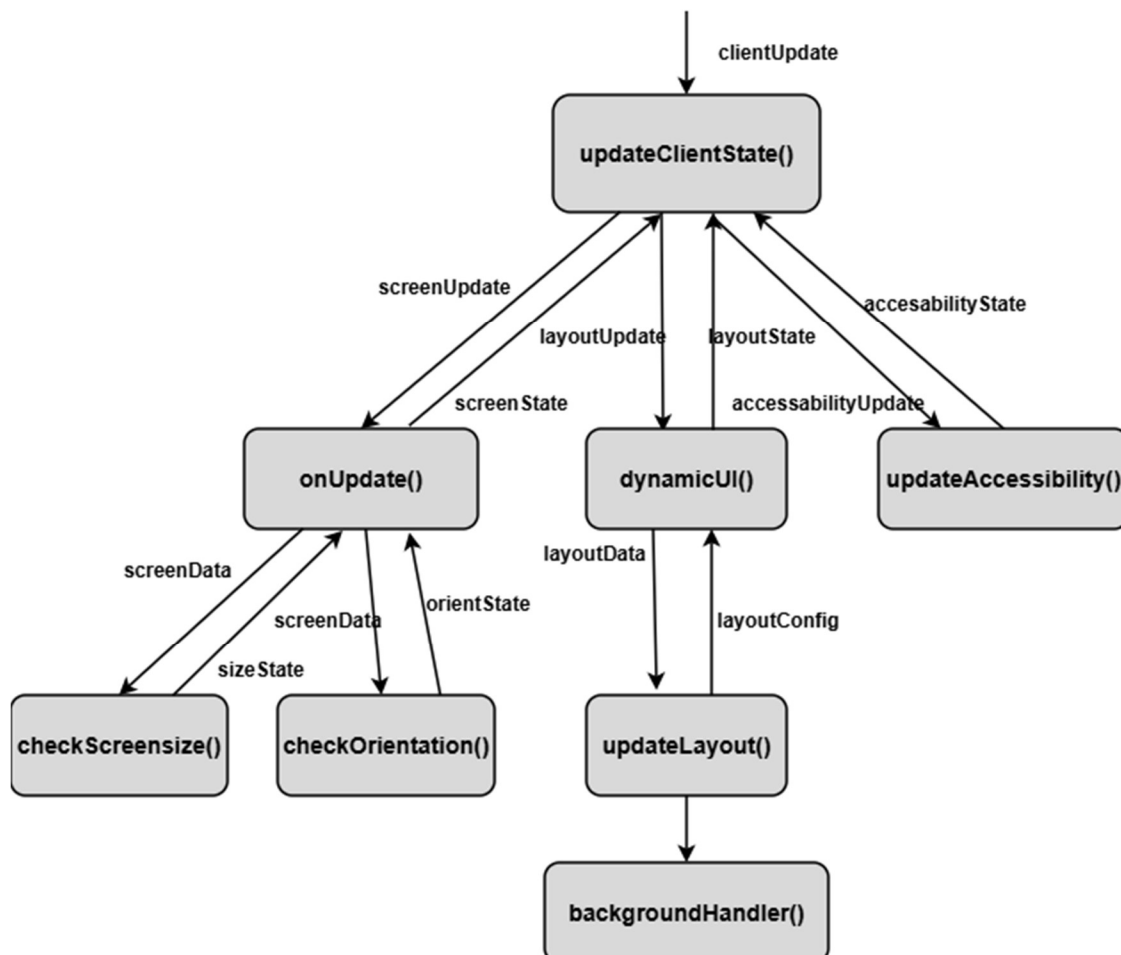
SC Figure 1.2.0.2 & 1.3.0.2 – submitForm()



SC Figure 1.2.0.3 & 1.3.0.3 – sendAPIRequest()

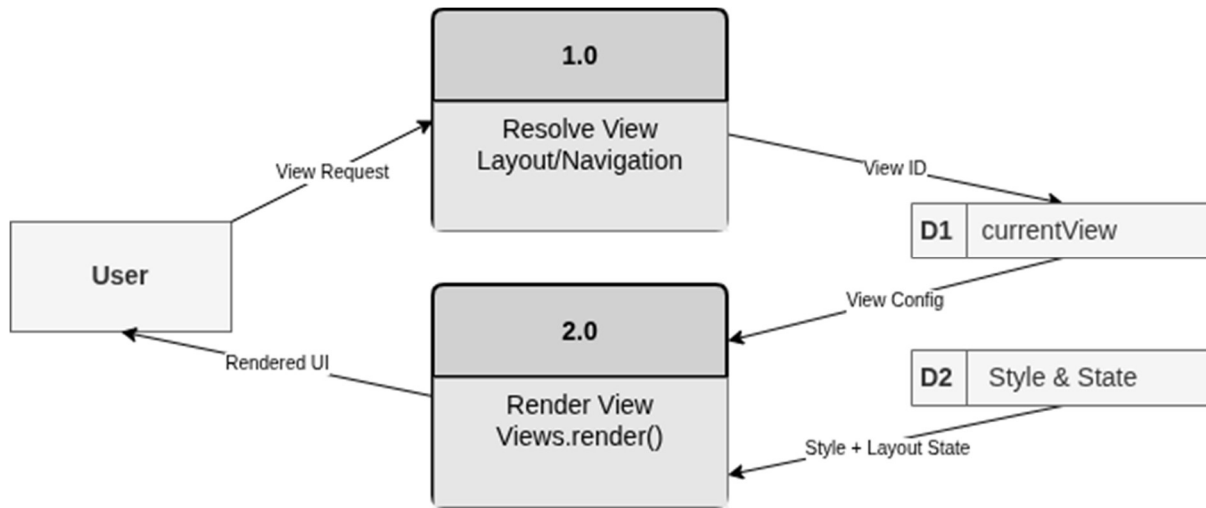


SC Figure 1.2.0.4 & 1.3.0.4 – updateClientState()

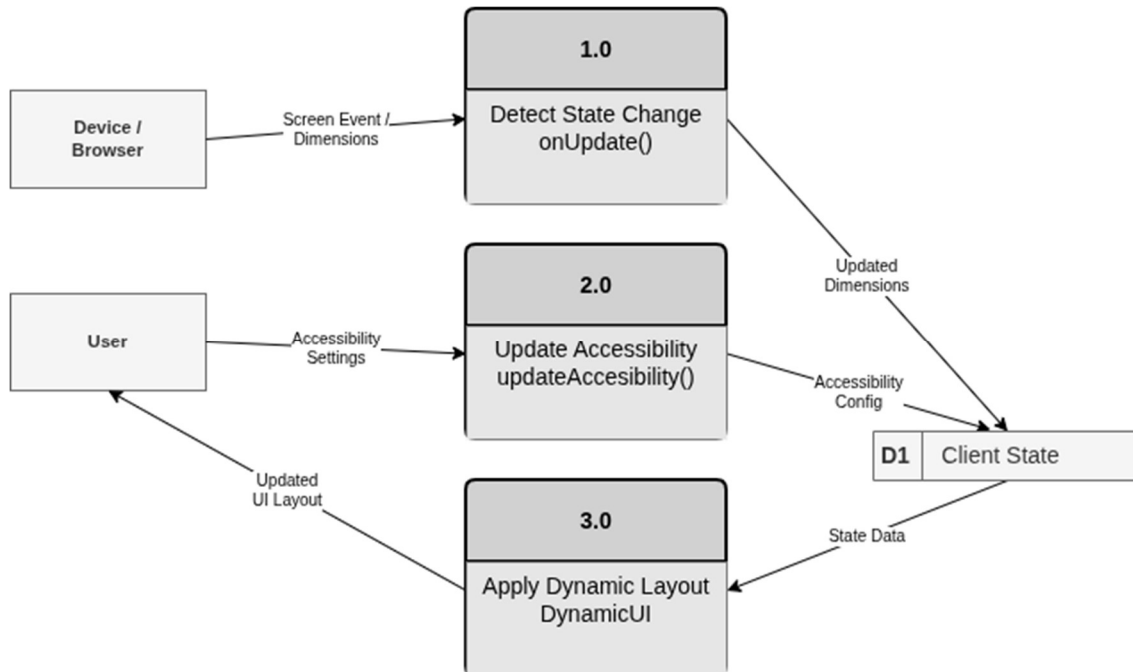


Data Flow Diagrams - Web & Mobile Client

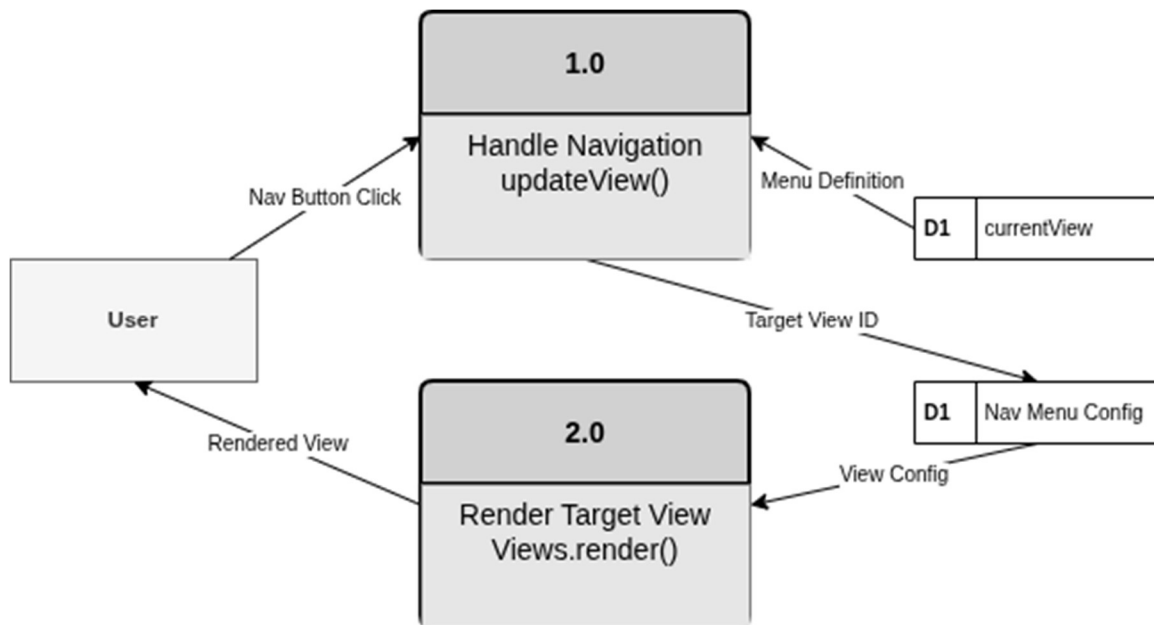
DFD Figure 1.2.0.0.1 & 1.3.0.0.1 – UI View Rendering Data Flow



DFD Figure 1.2.0.0.2 & 1.3.0.0.2 – Client State Update Data Flow



DFD Figure 1.2.0.0.3 & 1.3.0.0.3 – Navigation Selection Data Flow



DFD Figure 1.2.0.0.4 & 1.3.0.0.4 – Client API Response Display Data Flow

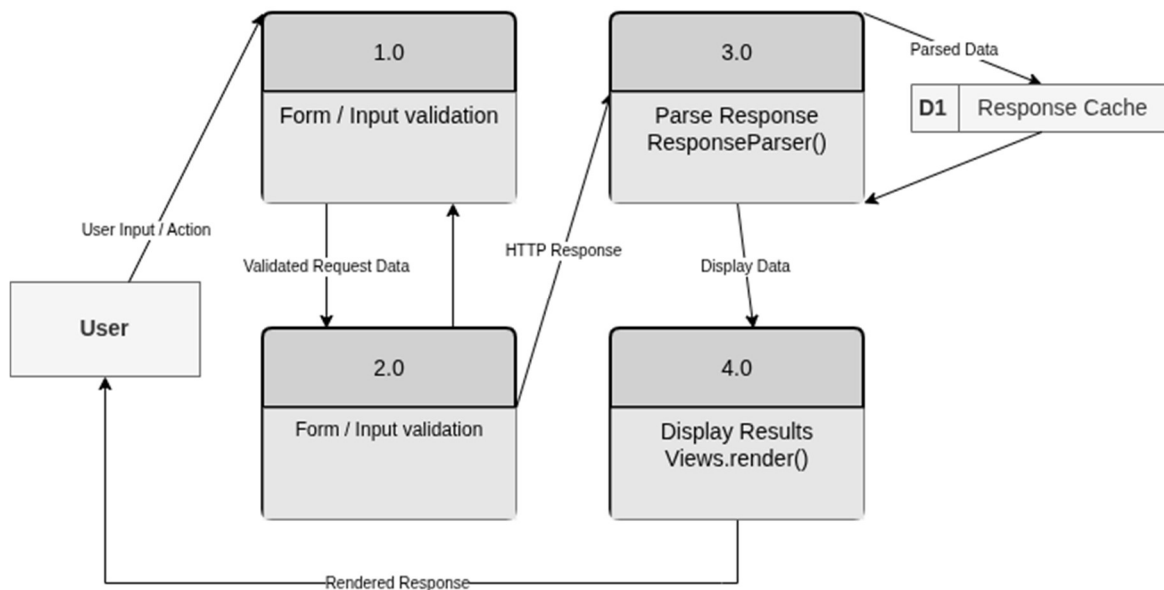
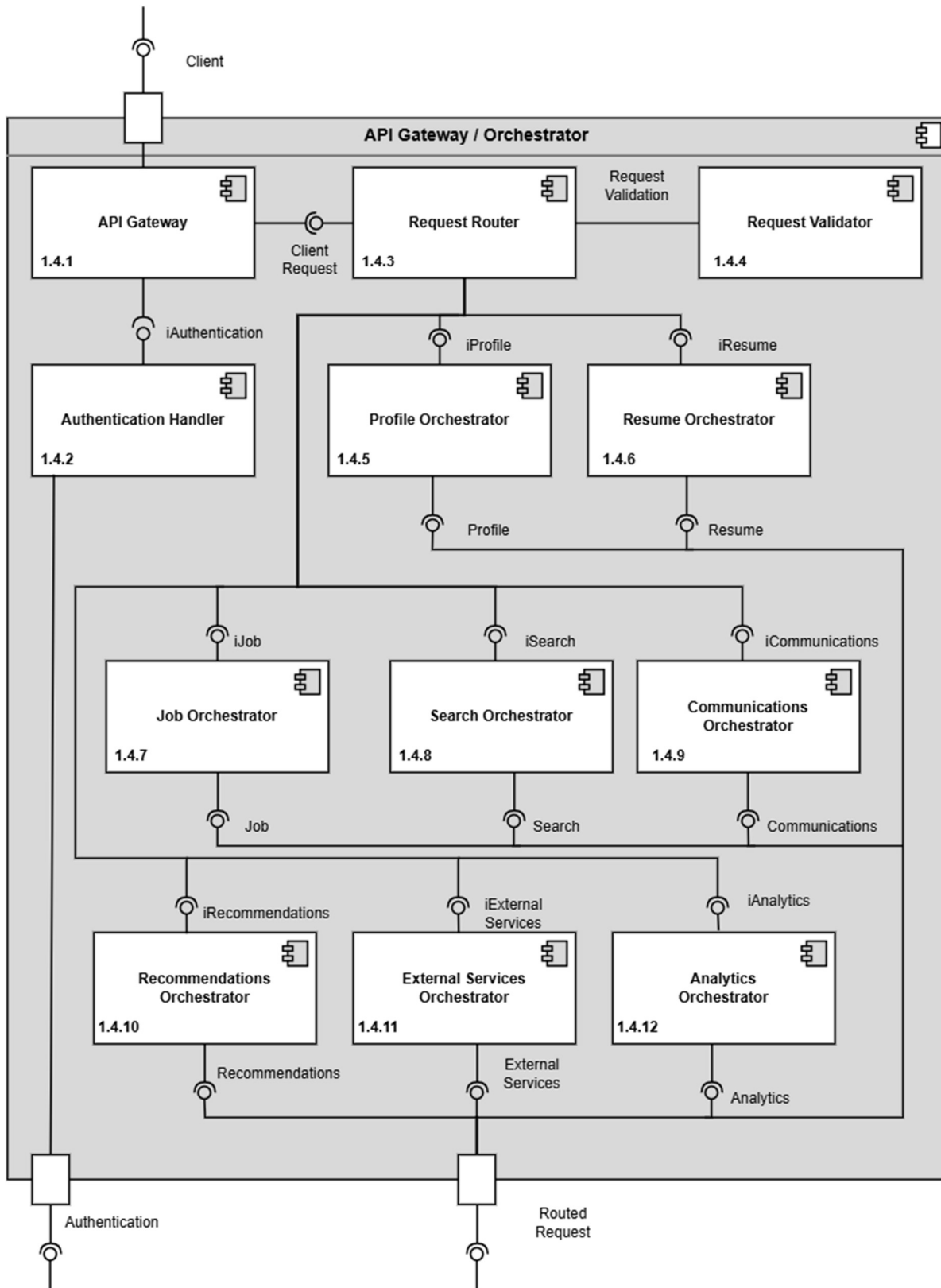


Figure 1.4 — API Gateway / Orchestrator Detail

Design Information – API Gateway / Orchestrator

Field	Description / Purpose
View Name & Number	1.4 – API Gateway / Orchestrator
Purpose	Describe the API/ Orchestrator View
Description	API Gateway with authentication Handling and request routing to various component orchestration. The service-specific orchestrators communicate with their assigned downstream services through the Downstream Service Interfaces port. The individual downstream service relationships are shown in the master component diagram and Interface Glossary.
Requirements	Cross-cutting for all SRS sections — routes all client requests to backend services
Elements	1.4.1 API Gateway
	1.4.2 Authentication Handler
	1.4.3 Request Router
	1.4.4 Request Validator
	1.4.5 Profile Orchestrator
	1.4.6 Resume Orchestrator
	1.4.7 Job Orchestrator
	1.4.8 Search Orchestrator
	1.4.9 Communications Orchestrator
	1.4.10 Recommendation Orchestrator
	1.4.11 External Orchestrator
	1.4.12 Analytics Orchestrator
Navigation / References	1.2 Web Client, 1.3 Mobile Client, 1.5–1.13 all backend service detail views
Viewpoint	Component Diagram

Entity Identification Tables – API Gateway / Orchestrator**EI Table 1.4.1 — API Gateway**

Subcomponent	Entity Name	Entity Type	Description / Purpose	Collaborators / Related Entities
API Gateway	GatewayRequest	Class	Represents the request received from the Request Router	RouterRequest, RequestContext
API Gateway	gatewayRequestObj	Object	An instance of GatewayRequest created per	GatewayRequest

Subcomponent	Entity Name	Entity Type	Description / Purpose	Collaborators / Related Entities
			incoming request	
API Gateway	parseIncomingRequest()	Method	Extracts headers, body, and parameters from RouterRequest	RouterRequest, GatewayRequest
API Gateway	GatewayResponse	Class	Represents the response sent back to the client	OrchestratorResponse, ErrorObject
API Gateway	AuthenticationResult	Class	Represents the results of the authentication request that are sent to the client.	AuthenticationHandler, GatewayResponse
API Gateway	RequestContext	Class	Represents and stores the metadata contained within an incoming request.	GatewayRequest, AuthenticationResult
API Gateway	gatewayConfig	Variable	Holds timeout, retry, and routing settings.	GatewayRequest, RoutingDecision
API Gateway	/api/ (Gateway Entry Point)	API Endpoint	Public entry point for all client requests.	Request Router

EI Table 1.4.2 — Authentication Handler

Subcomponent	Entity Name	Entity Type	Description / Purpose	Collaborators / Related Entities
Authentication Handler	AuthenticationRequest	Class	Represents the normalized Login/signup/token	UserCredential

Subcomponent	Entity Name	Entity Type	Description / Purpose	Collaborators / Related Entities
			refresh request	
Authentication Handler	authenticationRequest Obj	Object	Instance of AuthenticationRequest created per incoming request	AuthenticationRequest
Authentication Handler	UserCredential	Class	Represents stored user information: hashed password, salt, MFA settings, etc.	AuthenticationRequest
Authentication Handler	AccessToken	Class	Represents a short-lived token for authenticated sessions	AuthenticationResult
Authentication Handler	RefreshToken	Class	Represents a long-lived token for session renewal.	AccessToken
Authentication Handler	AuthenticationResult	Class	Represents Authentication outcome sent to the API Gateway	AccessToken, PermissionSet
Authentication Handler	validateCredentials()	Method	Verifies username and password against stored entries	UserCredentials
Authentication Handler	authDB	Data Store	Stores user credentials, tokens, roles, etc.	UserCredential
Authentication Handler	/auth/login	API Endpoint	Handles login requests	API Gateway

EI Table 1.4.3 — Request Router

Subcomponent	Entity Name	Entity Type	Description / Purpose	Collaborators / Related Entities
Request Router	RouteDefinition	Class	Defines a mapping between a URL path and a target orchestrator	RoutingRule, BackendOrchestrator
Request Router	RoutingDecision	Class	Contains result data obtained from routing requests to target orchestrator.	RoutingRule, BackendOrchestrator
Request Router	RouterRequest	Class	Normalized representation of an incoming request made before routing.	GatewayRequest, RoutingRule
Request Router	routerRequestObj	Object	Instance of RouterRequest created for each incoming request	RouterRequest
Request Router	RoutingRule	Class	Defines rules/conditions used to determine which orchestrator should handle a request.	RouteDefinition, RoutingDecision, RouterRequest
Request Router	requestRoute()	Method	Evaluates RoutingRules and produces a RoutingDecision	RoutingRule, RoutingDecision
Request Router	applyRoutingRules()	Method	Iterates through RoutingRules objects to find rules that match the needs of the request.	RoutingRule, RouterRequest
Request Router	BackendOrchestrator	Class	Contains metadata for orchestrators	RouteDefinition
Request Router	routerConfig	Variable	Stores routing table and orchestrator registry	RouteDefinition
Request Router	routingTable	Data Store	Persistent storage of RouteDefinitions and RoutingRules	RouteDefinition, RoutingRule

Subcomponent	Entity Name	Entity Type	Description / Purpose	Collaborators / Related Entities
Request Router	/route	API Endpoint	Internal endpoint used by the Gateway to forward requests.	API Gateway

EI Table 1.4.4 — Request Validator

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Search Query Manager	searchJobs()	Function	Processes job search requests submitted by the user	2.3.1–2.3.8	SearchFilter, JobPosting, SearchResults
Search Query Manager	SearchFilter	Class	Represents the filters selected by the user for job searching	2.3.2, 2.3.4	searchJobs(), SearchResults
Request Validator	validateRequest()	Function	Checks requests before sending them to services.	1.2.1	Authentication Handler, validateRequiredFields()
Request Validator	validateRequiredFields()	Function	Checks that all needed fields are included.	3.5.5.1; 3.5.5.2; 3.5.5.3	ValidationRuleSet, orchestrateResumeRequest()
Request Validator	validateSearchRequest()	Function	Checks search requests	2.3.1–2.3.8;	orchestrateSearchRequest(), Filter Processor

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
			before processing.	3.3.1–3.3.4	
Request Validator	validateNotificationSettings()	Function	Checks notification settings requests.	3.2.6; 3.2.7	orchestrateCommunicationRequest(), Notification Controller
Request Validator	validateNewsFeedPost()	Function	Checks news feed posts before submission.	2.4.1; 2.4.3; 2.4.6; 3.4.2	orchestrateCommunicationRequest(), News Feed Manager
Request Validator	sanitizeInput()	Function	Removes unsafe input from requests.	2.1.3.10; 2.4.1; 2.4.3; 2.5.1; 2.5.7; 3.1.2; 3.4.2; 3.5.4.1	Service Orchestrators, validateRequest()
Request Validator	ValidationRuleSet	Data Entity	Stores validation rules for requests.	2.1.3.7–2.1.3.9; 3.5.5.2; 3.5.5.3	validateRequiredFields(), validateRequest()

EI Table 1.4.5–1.4.12— Service Orchestrator Pattern

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Profile Orchestrator	orchestrateProfileRequest()	Function	Handles profile requests.	2.1.1 – 2.1.3	validateRequest(), Profile

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
				.10; 3.1.1 – 3.1.3 .4	Management Service
Resume Orchestrator	orchestrateResumeRequest()	Function	Handles resume requests.	3.5.1 – 3.5.5 .3; 2.1.2 ; 2.2.5 ; 2.3.3 ; 2.3.5 ;	validateRequest(), Resume Service
Job Orchestrator	orchestrateJobRequest()	Function	Handles job posting requests.	2.5.1 – 2.5.9 ; 3.3.1 ; 3.4.1 ; 2.2.5	validateRequest(), Job Posting Service
Search Orchestrator	orchestrateSearchRequest()	Function	Handles search requests.	2.3.1 – 2.3.8 ; 3.3.1 – 3.3.4	validateSearchRequest(), Search Service
Communications Orchestrator	orchestrateCommunicationRequest()	Function	Handles notification and	2.2.1 – 2.2.5 ;	validateNotificationSettings(), Communication Services

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
			news feed requests.	2.4.1 – 2.4.6 ; 3.2.1 – 3.2.7 ; 3.4.1 – 3.4.2	
Recommendation Orchestrator	orchestrateRecommendationRequest()	Function	Handles recommendation requests.	2.3.1 – 2.3.8 ; 2.5.1 – 2.5.9 ; 3.1.1 – 3.1.3 .4; 3.5.1 – 3.5.5 .3	orchestrateSearchRequest(), Recommendation Service
External Orchestrator	orchestrateExternalRequest()	Function	Handles requests to outside services.	2.1.3 .9; 2.3.6 ; 2.2.2 ; 3.2.1 ; 3.2.4	External Services, validateRequest()

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Service Orchestrator Pattern	routeToService()	Function	Sends requests to the correct service.	1.2.1	validateRequest(), Service Orchestrators

EI Table 1.4.5 – Profile Orchestrator

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Profile Orchestrator	ProfileOrchestrator	Class	Coordinates profile-related workflows between the profile page and supporting managers.	3.1.2, 3.1.3.3, 3.1.3.4, 2.1.2, 2.1.3.6	Profile Manager, Resume Manager, Privacy Settings Manager, Authentication Service
Profile Orchestrator	ProfileRequest	Object	Represents a request to view, update, or access profile-related information.	3.1.2, 3.1.3.3, 3.1.3.4	StudentUser, CompanyRepresentative, Profile Manager
Profile Orchestrator	ProfileResponse	Object	Represents the profile information or access result returned after coordination	3.1.3.1, 3.1.3.4, 2.1.3.1, 2.1.3.6	Profile Manager, Resume Manager

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Profile Orchestrator	AccessContext	Object	Represents the requesting user, their role, and the action being requested	3.1.3.3, 3.1.3.4, 2.1.3.6	Authentication Service, Privacy Settings Manager
Profile Orchestrator	ProfileNavigationReference	Data Store	Represents profile page links or reference to related features managed by other subcomponents	3.1.3.3, 3.1.3.4, 2.1.3.6	Resume Manager, Profile Manager
Profile Orchestrator	ProfileWorkflow	Class	Represents a coordinated profile process such as viewing a student profile or accessing a resume link.	3.1.2, 3.1.3.3, 3.1.3.4, 2.1.2	Profile Manager, Resume Manager, Job Posting Service

EI Table 1.4.6 – Resume Orchestrator

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
API Gateway	resumeServiceRouter()	Module	Routes to requested Resume Service for authenticated users	3.5.1, 2.5.6	Resume Services, Authentication Services, Web Client, Database

EI Table 1.4.7 – Job Orchestrator

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Job Orchestrator	Job Orchestrator	Class	Coordinates all job-related requests and responses.	2.5.1-2.5.4	RequestRouter, Database
Job Orchestrator	RequestJobs	Method	Sends a request to retrieve job listings	2.5.1-2.5.4	Database
Job Orchestrator	GetJobDetails	Method	Requests detailed information for a selected job.	2.5.1-2.5.4	Database
Job Orchestrator	FilterJobs	Method	Applies user-selected filters to job search	2.5.1-2.5.4	
Job Orchestrator	JobResults	Object	Stores job listing results returned from services.	2.5.1-2.5.4	Database
Job Orchestrator	JobRequest	Object	Contains search criteria and request parameters.	2.5.1-2.5.4	RequestRouter

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Job Orchestrator	SearchCriteria	Object	Stores keywords, location, and filter values.	2.5.1-2.5.4	JobRequest
Job Orchestrator	JobListing	Object	Represents an individual job posting.	2.5.1-2.5.4	JobResults
Job Orchestrator	RequestRouter	Class	Routes incoming requests to the Job Orchestrator.	2.5.1-2.5.4	JobOrchestrator

EI Table 1.4.8 – Search Orchestrator

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Search Orchestrator	SearchRequest	Class	Contains the search query along with filters used when it enters the orchestrator.	2.3.1 – 2.3.8, 3.3.1 – 3.3.4	SearchHandler
Search Orchestrator	SearchHandler	Class	Handles requests and returns responses.	2.3.1 – 2.3.8, 3.3.1 – 3.3.4	SearchRequest, SearchResponse, ServiceHandler
Search Orchestrator	SearchResponse	Class	Contains the final, aggregated search result	2.3.1 – 2.3.8, 3.3.1	SearchResult

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
			after the services have finished.	– 3.3.4	
Search Orchestrator	SearchResult	Class	Contains a singular search result within the final result.	2.3.1 – 2.3.8, 3.3.1 – 3.3.4	SearchResponse
Search Orchestrator	ServiceTarget	Class	A representation of a downstream service the orchestrator can send tasks to and receive results from.	2.3.1 – 2.3.8, 3.3.1 – 3.3.4	ServiceHandler, determineTargets()
Search Orchestrator	ServiceRequest	Class	Represents the transformed request to be sent to a service.	2.3.1 – 2.3.8, 3.3.1 – 3.3.4	ServiceTarget, ServiceHandler
Search Orchestrator	ServiceResult	Class	Contains the untransformed result directly from the service.	2.3.1 – 2.3.8, 3.3.1 – 3.3.4	ServiceTarget, ServiceHandler
Search Orchestrator	ServiceHandler	Class	Handles service timeouts/retries, calls, and responses.	2.3.1 – 2.3.8, 3.3.1 – 3.3.4	ServiceTarget, ServiceRequest, ServiceResult, SearchHandler
Search Orchestrator	routingTable	Data Table	Identifies separate services for job queries and	2.3.1 – 2.3.8, 3.3.1	determineTargets(), ServiceHandler

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
			candidate queries.	– 3.3.4	
Search Orchestrator	correlationId	Variable	Tracks a request across systems.	2.3.1 – 2.3.8, 3.3.1 – 3.3.4	SearchRequest, SearchResponse, ServiceRequest, ServiceResponse
Search Orchestrator	handleRequest()	Method	Transforms request, sends request, and returns result.	2.3.1 – 2.3.8, 3.3.1 – 3.3.4	SearchHandler, SearchRequest
Search Orchestrator	determineTargets()	Method	Determines which service to send the request.	2.3.1 – 2.3.8, 3.3.1 – 3.3.4	ServiceHandler, routingTable
Search Orchestrator	dispatchRequests()	Method	Sends request to a service.	2.3.1 – 2.3.8, 3.3.1 – 3.3.4	ServiceHandler, ServiceRequest, ServiceTarget
Search Orchestrator	aggregateResults()	Method	Aggregates multiple results into one result.	2.3.1 – 2.3.8, 3.3.1 – 3.3.4	SearchHandler, SearchResult, SearchResponse
Search Orchestrator	/search	API Endpoint	The entry point into the orchestrator.	2.3.1 – 2.3.8, 3.3.1 – 3.3.4	SearchHandler, SearchRequest

EI Table 1.4.9 — Communications Orchestrator

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Communications Orchestrator	routeCommunicationRequest()	Function	Sends communication requests to services.	2.2.1–2.2.5; 2.4.1–2.4.6; 3.2.1–3.2.7; 3.4.1–3.4.2	validateRequest(), Communication Services
Communications Orchestrator	getNotifications()	Function	Gets notifications for a user.	2.2.3; 2.2.4; 2.2.5; 3.2.1–3.2.5	Notification Controller, Notification Repository
Communications Orchestrator	createNotification()	Function	Creates a notification.	2.2.1–2.2.5; 3.2.1–3.2.5	Event Listener, Notification Orchestrator

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Communications Orchestrator	muteNotifications()	Function	Handles notification mute requests.	3.2.6	Notification Controller, Notification Repository
Communications Orchestrator	updateNotificationFilters()	Function	Updates notification filters.	3.2.7	Notification Controller, Recipient Resolver
Communications Orchestrator	resolveNotificationRecipients()	Function	Finds users who should receive notifications.	2.2.2; 2.2.4; 2.2.5; 3.2.2; 3.2.3; 3.2.5	Recipient Resolver, orchestrateJobRequest(), orchestrateResumeRequest()
Communications Orchestrator	formatNotificationMessage()	Function	Formats notification messages.	2.2.1	Notification Formatter, createNotification()
Communications Orchestrator	sendPushNotification()	Function	Sends push notifications.	2.2.2; 3.2.1; 3.2.2; 3.2.3; 3.2.4	Push Delivery Component, orchestrateExternalRequest()

EI Table 1.4.10 – Recommendation Orchestrator

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
API Gateway	recommendationServiceRouter()	Module	Routes to requested Recommendation Service for authenticated users	2.2.5, 2.3.5, 3.3.3	Recommendation Services, Authentication Services, Web Client, Database, Analytics Services, Notifications Services

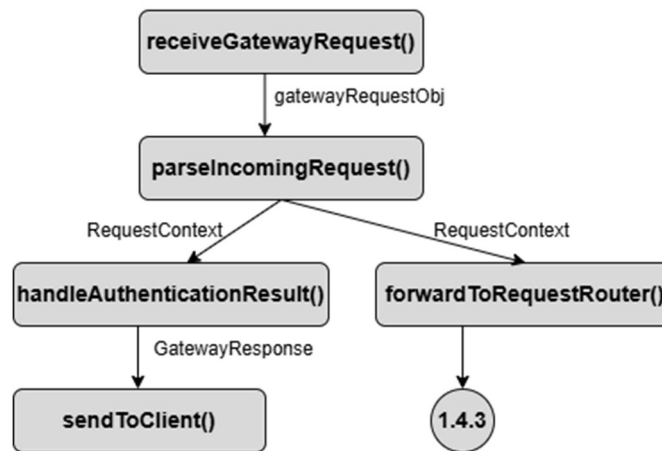
EI Table 1.4.11 – External Services Orchestrator

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
External Services Orchestrator	ExternalServicesOrchestrator	Class	Receives routed requests from the API Gateway and dispatches them to the correct external service adapter.	2.1.3 .9, 2.3.6 , 3.2.4	APIGateway, USPSAddressValidationAdapter, AutocompleteAPIAdapter, PushNotificationGateway
External Services Orchestrator	ExternalServicesRequest	Class	Encapsulates an inbound request destined for an external service, holding the service target and payload.	2.1.3 .9, 2.3.6 , 3.2.4	ExternalServicesOrchestrator, routeRequest(), validateRequest()

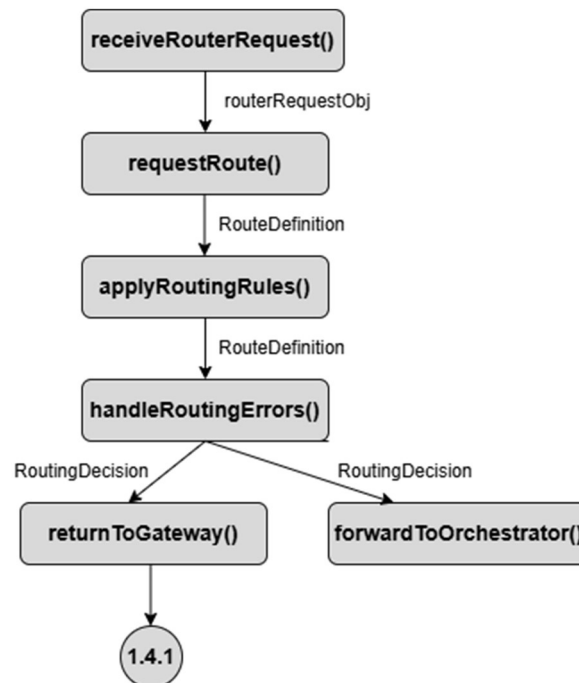
Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
External Services Orchestrator	ExternalServiceResponse	Class	Encapsulates the result returned from an external adapter and carries it back upstream to the API Gateway	2.1.3.9, 2.3.6, 3.2.4	handleResponse(), APIGateway
External Services Orchestrator	routeRequest()	Method	Inspects and inbound request and dispatches it to the correct external adapter based on the service target.	2.1.3.9, 2.3.6, 3.2.4	ExternalServiceRequest, USPSAddressValidationAdapter, AutocompleteAPIAdapter, PushNotificationGateway
External Services Orchestrator	validateRequest()	Method	Validates that an inbound request contains required fields and a recognizer service target before routing proceeds.	2.1.3.9, 2.3.6, 3.2.4	ExternalServiceRequest, ExternalServiceResponse
External Services Orchestrator	handleResponse()	Method	Wraps an adapter's result into an ExternalServiceResponse and returns it upstream to the API Gateway.	2.1.3.9, 2.3.6, 3.2.4	ExternalServiceResponse, APIGateway
External Services Orchestrator	serviceTarget	Variable	Identifies which external service the request targets and drives all routing logic within the orchestrator.	2.1.3.9, 2.3.6, 3.2.4	ExternalServiceRequest, routeRequest()

Structure Charts – API Gateway / Orchestrator

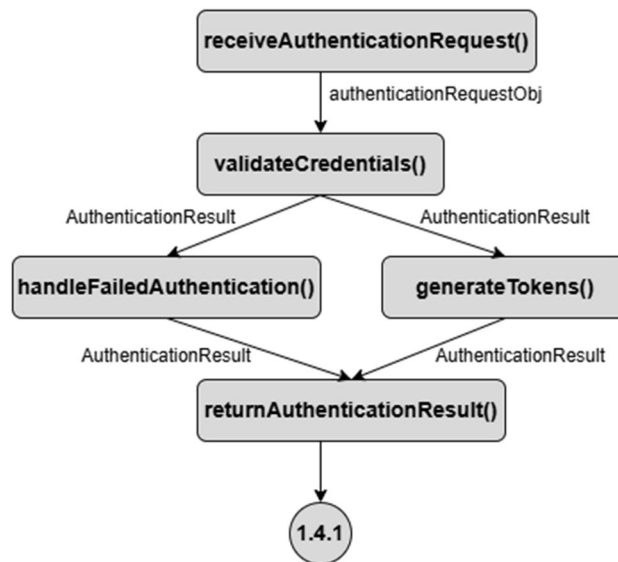
SC Figure 1.4.0.1 – recieveGatewayRequest()



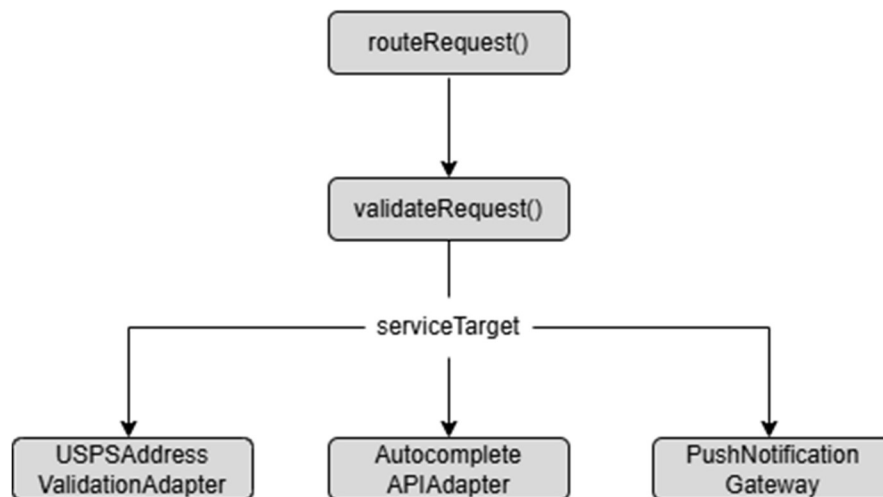
SC Figure 1.4.0.2 – recieveRouterRequest()



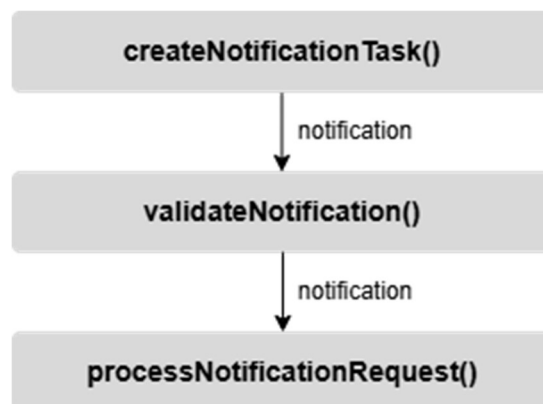
SC Figure 1.4.0.3 – receiveAuthenticationRequest()



SC Figure 1.4.0.11 – External Services Orchestrator

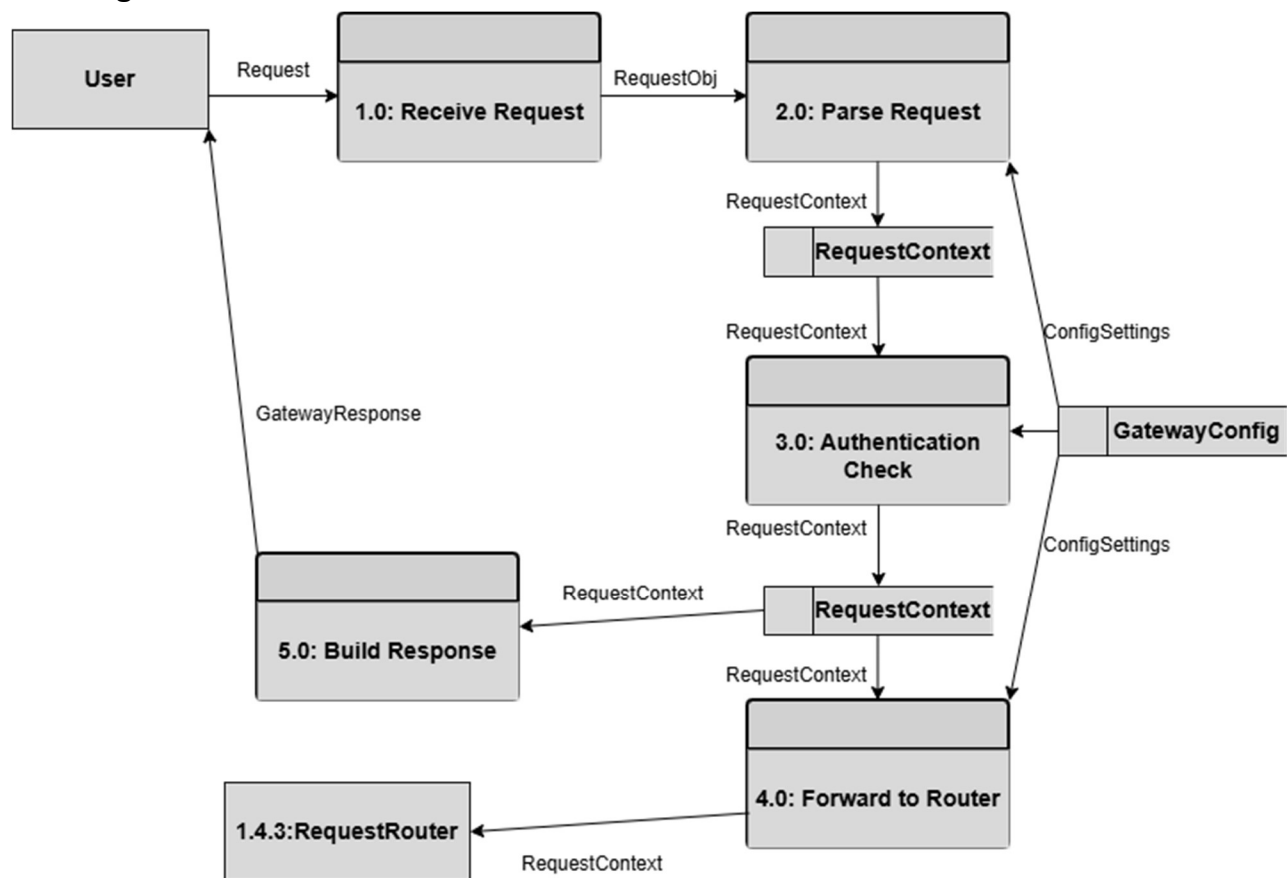


SC Figure 1.4.9.1 – createNotificationTask()

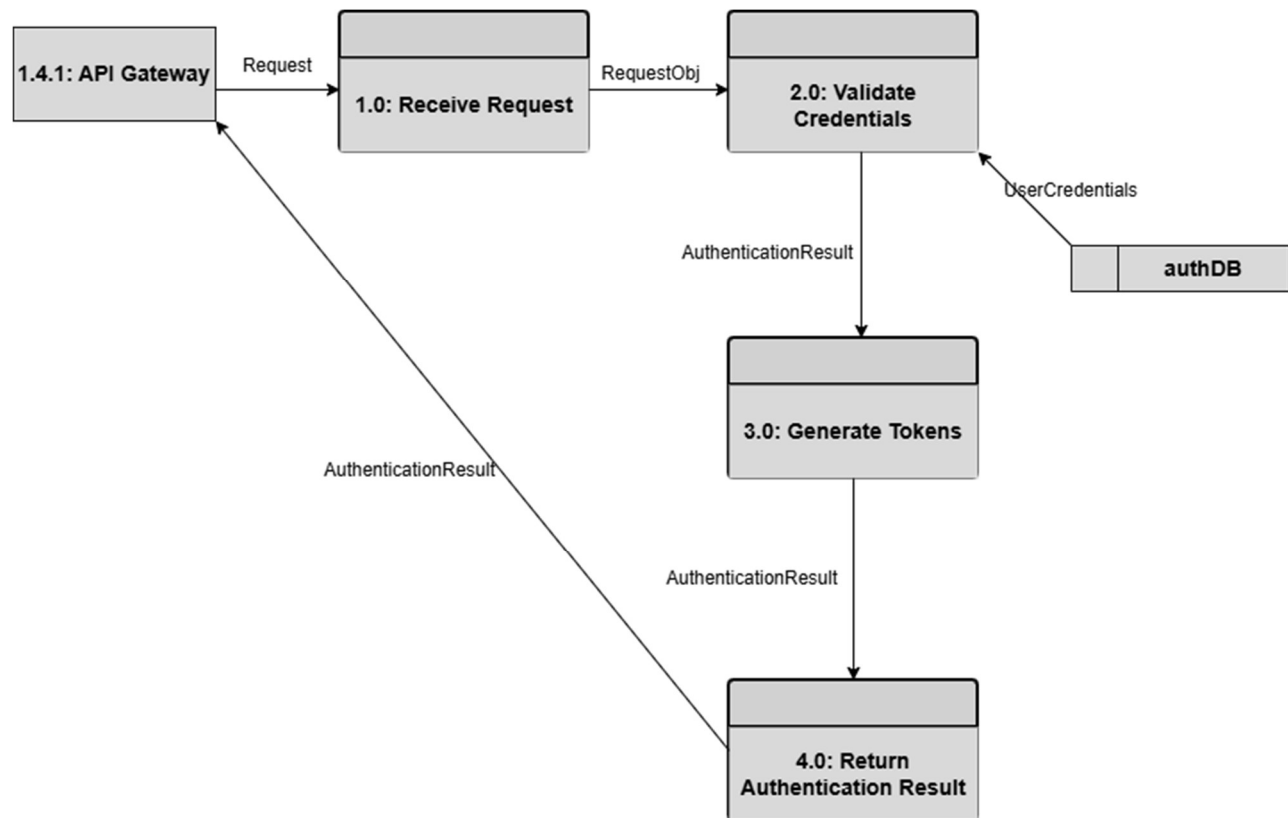


Data Flow Diagrams - API Gateway / Orchestrator

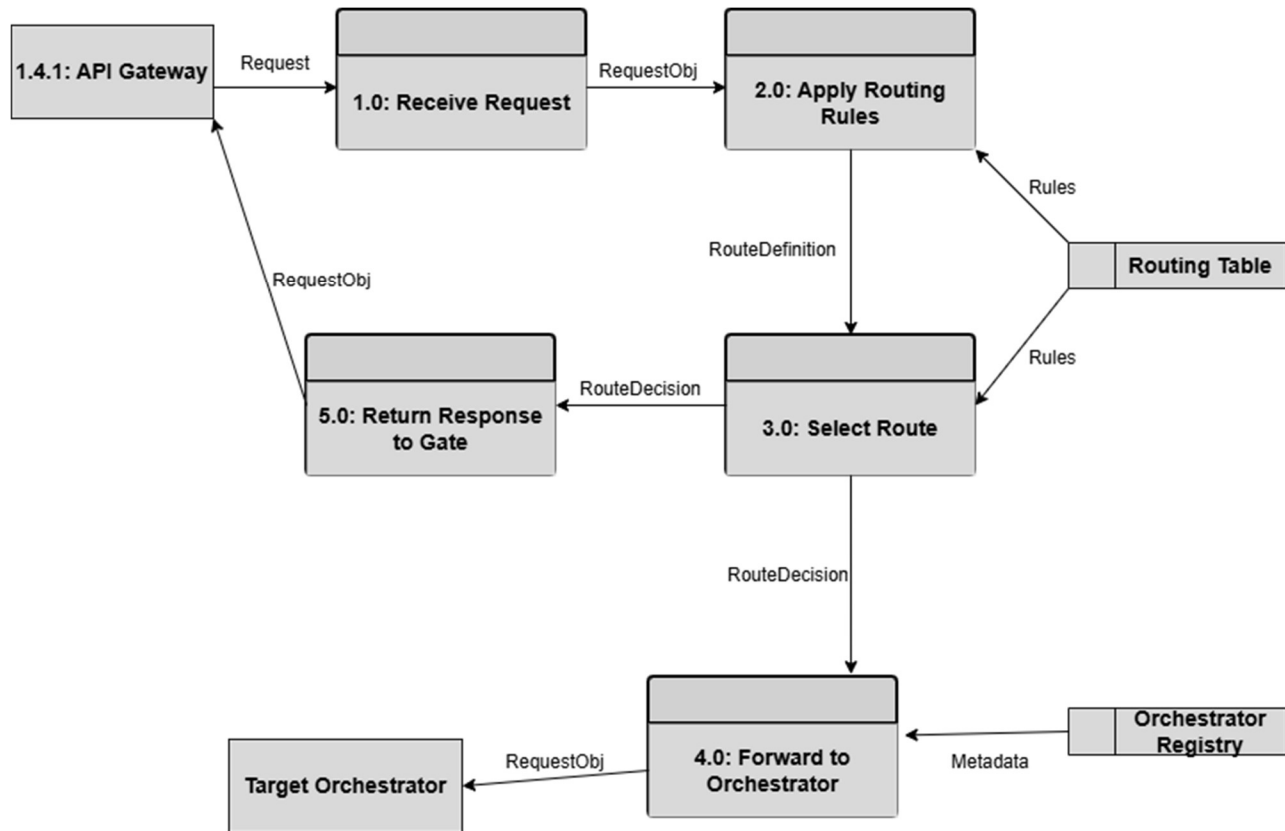
DFD Figure – 1.4.1.0.1 –



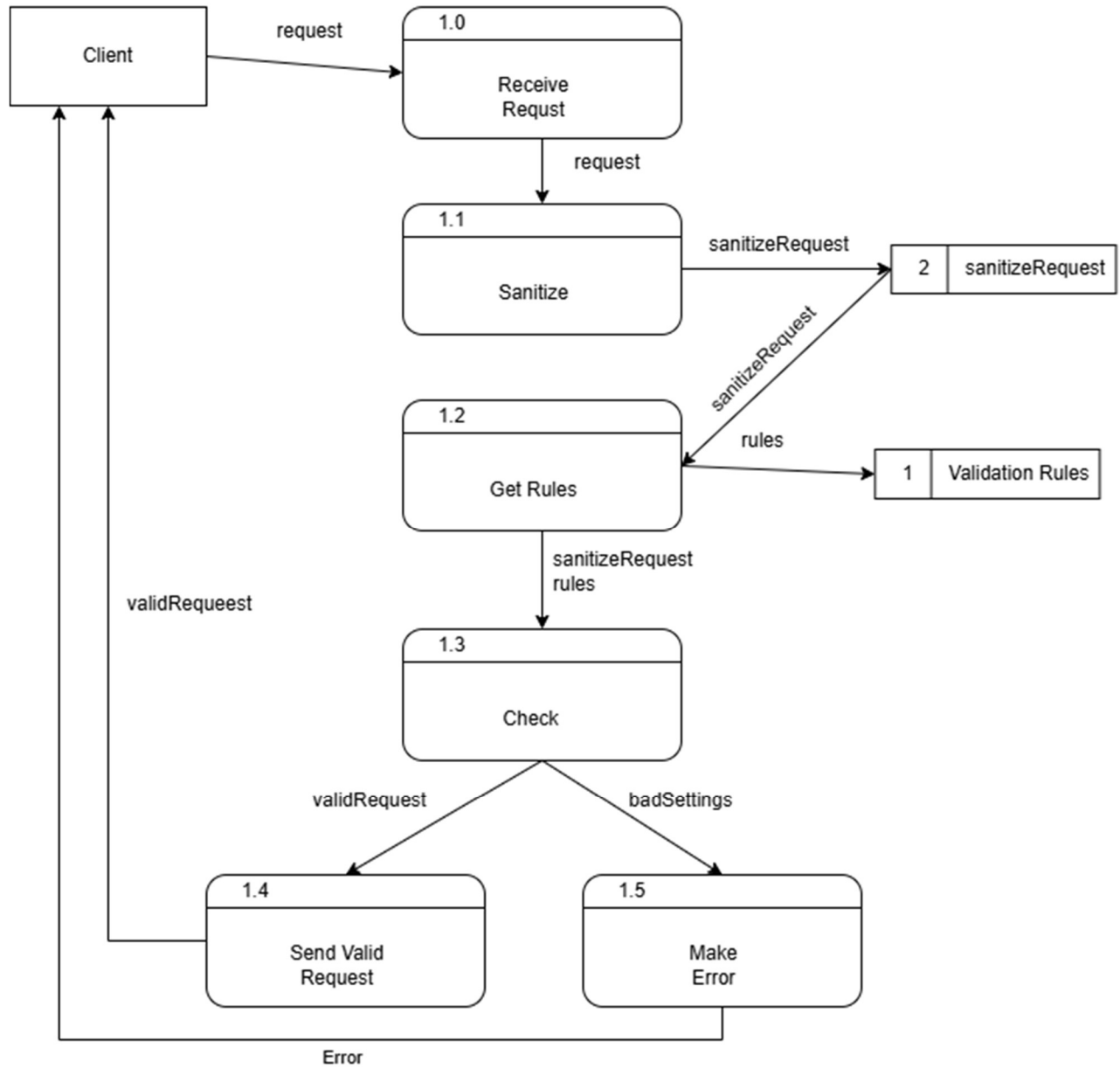
DFD Figure – 1.4.2.0.0.1 –



DFD Figure – 1.4.3.0.0.1 –



DFD Figure – 1.4.4.0.0.1 – Request Validator



DFD Figure – 1.4.5.0.0.1 – Service Orchestrator Pattern

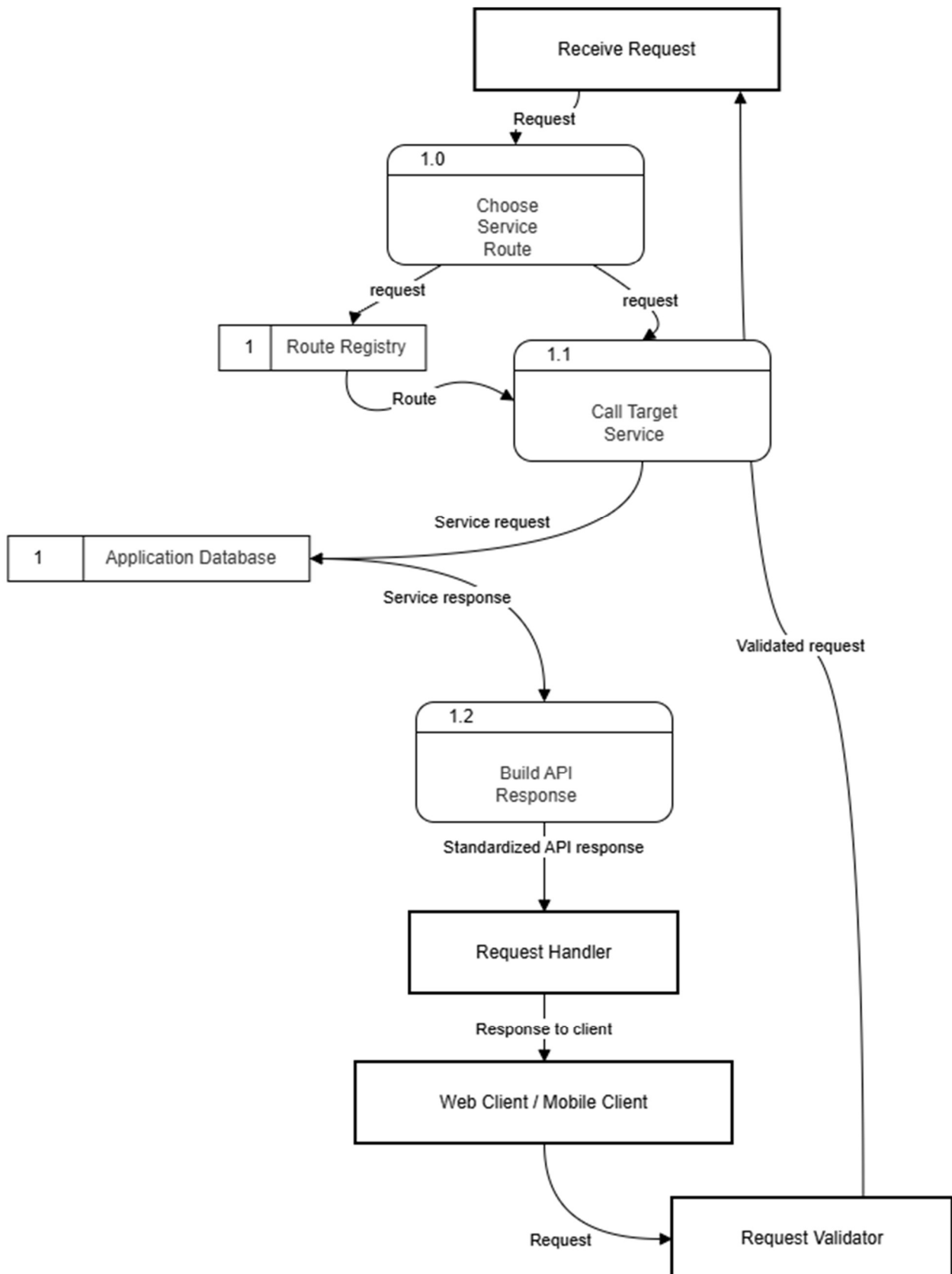
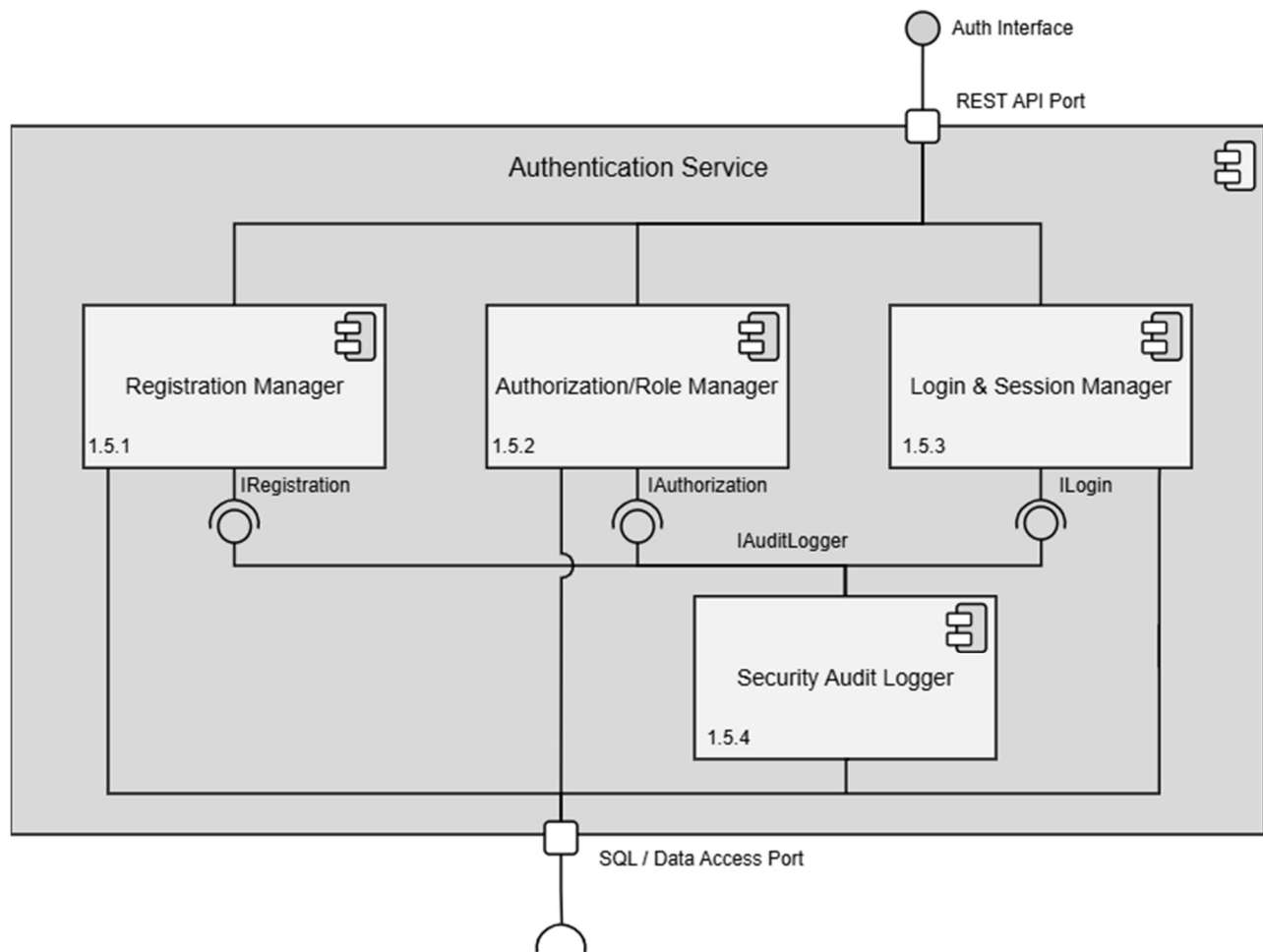


Figure 1.5 — Authentication Service Detail**Design Information – Authentication Service**

Field	Description / Purpose
View Name & Number	1.5 Authentication Service
Purpose	Describe the authentication system view.
Description	Basic service demonstrating the authentication service.
Requirements	Foundational requirement for all logged-in features — SRS 2.1.3.10 (Security), 3.1.1 (Student account), 2.1.1 (Company account)
Elements	1.5.1 Registration Manager
	1.5.2 Authorization/ Role Manager
	1.5.3 Login and Session Manager
	1.5.4 Security Audit Logger
Navigation / References	1.4 API Gateway / Orchestrator, 1.14 Database; Screen views 1.1 Login / Logout, 1.2 Registration
Viewpoint	Component Diagram

Entity Identification Tables – Authentication Service

El Table 1.5.1 – Registration Manager

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Registration Manager	UserAccount	Class	Represents a user's account which contains credentials and role information.	1.1, 1.2	UserData UserForm registerUser()
Registration Manager	UserForm	Class	A form which contains the user's information for registration.	1.1, 1.2	/api/auth/register/
Registration Manager	validateEmail()	Function	Checks if the email isn't already used and ensure it's in a valid format.	1.1, 1.2	UserForm
Registration Manager	validatePassword()	Function	Checks if the password contains any invalid characters and ensures it's in the valid format.	1.1, 1.2	UserForm
Registration Manager	hashPassword()	Function	Hashes the user's password to be stored securely.	1.1, 1.2	UserForm
Registration Manager	registerUser()	Function	Creates the user's account from the form data.	1.1, 1.2	validateEmail(), validatePassword(), hashPassword(), UserAccount
Registration Manager	User Data	Data Table	The database where all user information is held.	1.1, 1.2	UserAccount
Registration	/api/auth/register	API	Handles	1.1,	registerUser()

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Manager		Endpoint	incoming API requests for user registration.	1.2	UserForm

EI Table 1.5.2 – Authorization / Role Manager

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Authorization/ Role Manager	Authorization Manager	Class	Manages which users can access protected resources depending on their role.	1.1, 1.2, 2.1.1, 2.1.3.10, 3.1.2	checkPermissions()
Authorization/ Role Manager	UserAccount	Class	Represents a user's account which contains credentials and role information.	1.1, 1.2, 2.1.3.10, 2.5.6, 3.1.2, 3.1.3.4	User Data, role
Authorization/ Role Manager	role	Variable	The role the user has attached to their account.	1.1, 1.2	UserAccount, User Data
Authorization/ Role Manager	User Data	Data Table	The database where all user information is held.	1.1, 1.2	UserAccount, role
Authorization/ Role Manager	checkPermissions()	Function	Validates that a user has access to a protected resource.	1.1, 1.2, 2.1.1, 2.5.6,	UserAccount, User Data, role

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
				3.1.3.4	
Authorization/ Role Manager	/api/auth/authorize	API Endpoint	Handles incoming API requests for user authentication.	1.1, 1.2	checkPermissions()

EI Table 1.5.3 – Login & Session Manager

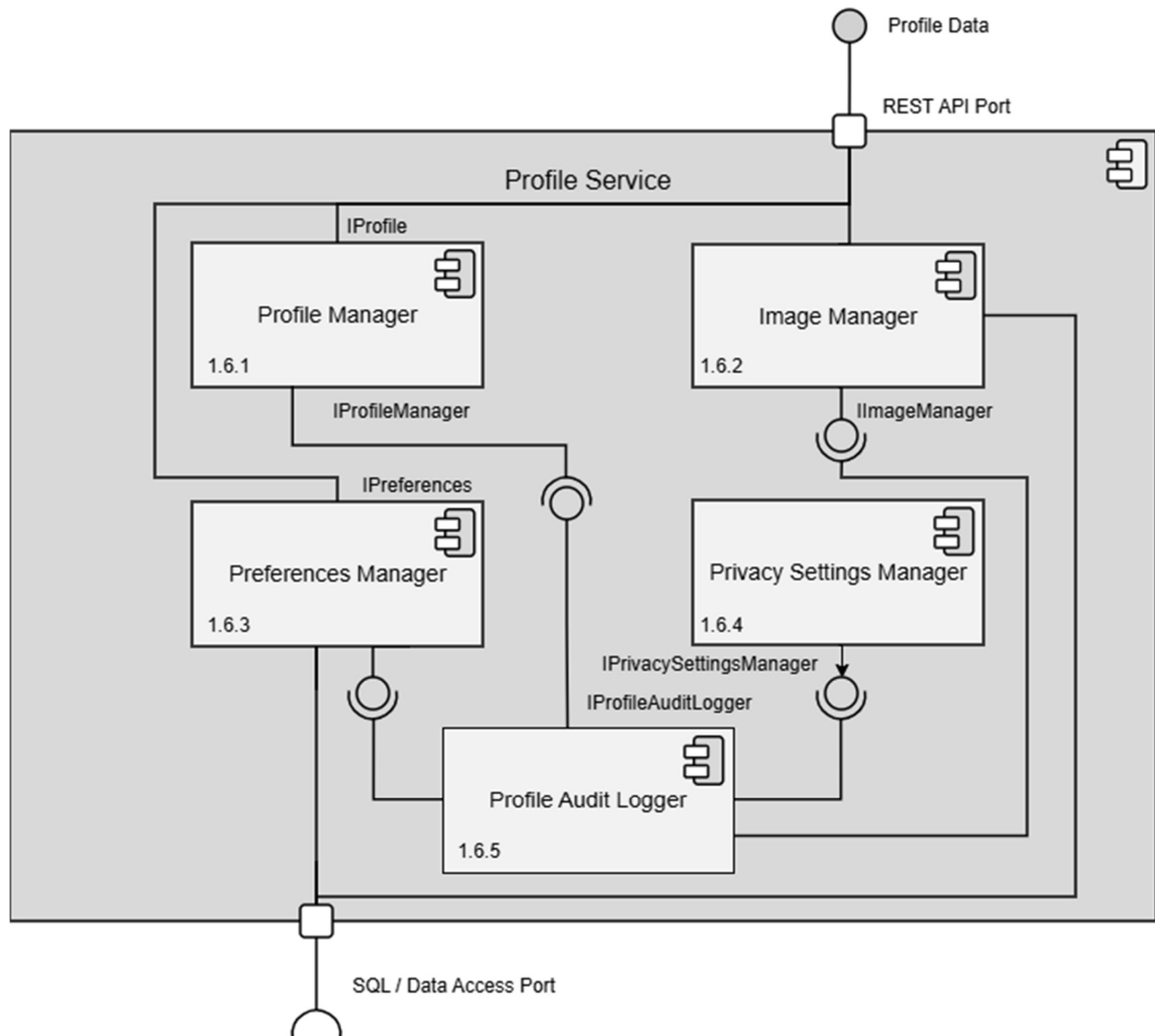
Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Login & Session Manager	UserAccount	Class	Represents a user's account which contains credentials and role information.	1.1, 1.2	loginUser(), Session
Login & Session Manager	UserForm	Class	A form which contains the user's email and password needed for logging in.	1.2	loginUser()
Login & Session Manager	Session	Class	Represents a user's active session for their device.	1.2	sessionToken
Login & Session Manager	loginUser()	Function	Reads the user's credentials and creates a new session if there's a match.	1.2	UserAccount, UserForm, Session

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Login & Session Manager	logoutUser()	Function	Deletes the user's session for their device on request.	1.2	Session, sessionToken
Login & Session Manager	sessionToken	Variable	Represents a user's session token for their device for later validation.	1.2	Session
Login & Session Manager	/api/auth/login	API Endpoint	Handles incoming API requests for user login.	1.2	loginUser()
Login & Session Manager	/api/auth/logout	API Endpoint	Handles incoming API requests for user logout.	1.2	logoutUser()

EI Table 1.5.4 – Security Audit Logger

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Security Audit Logger	buildAuditLog()	Method	Builds an audit log of different security metrics and displays it to the development team.	1.2	logId, loginAttempts, AuditLogException

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Security Audit Logger	loginAttempts	Variable	Represents how many times a failed login was attempted for a specific user.	1.2	logId, AuditLogException
Security Audit Logger	logId	Variable	Comes from user's input and gets packaged into the userData structure.	1.2	AuditLogException, buildAuditLog()
Security Audit Logger	AuditLogException	Class	Locks the user's account in the event that their account was potentially compromised.	1.2	buildAuditLog(), throwException()
Security Audit Logger	throwException()	Method	Throws an exception to the system when a security threat arises.	1.2	AuditLogException, submit()
Security Audit Logger	submit()	Method	Receives the user's data where the security risk was brought up to help in the threat assessment.	1.2	AuditLogException, throwException()
Security Audit Logger	/api/auth/log	API Endpoint	Handles incoming Security Audit Logger API requests.	1.2	buildAuditLog(), submit()

Figure 1.6 — Profile Management Service Detail**Design Information – Profile Management Service**

Field	Description / Purpose
View Name & Number	1.6 Profile Management Service
Purpose	Describe the profile management service.
Description	Basic service demonstrating the profile management service.
Requirements	SRS 2.1 Company Profile Page (2.1.1–2.1.3.10), 3.1 Student User Profile (3.1.1–3.1.3.3)
Elements	1.6.1 Profile Manager
	1.6.2 Image Manager
	1.6.3 Preferences Manager
	1.6.4 Privacy Settings Manager

Field	Description / Purpose
	1.6.5 Profile Audit Logger
Navigation / References	1.4 API Gateway / Orchestrator, 1.14 Database, 1.15 File / Blob Storage
Viewpoint	Component Diagram

Entity Identification Tables – Profile Management Service

El Table 1.6.1 – Profile Manager

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Profile Manager	StudentProfile	Class	Represents and organizes the main profile information for a student.	3.1.1, 3.1.2, 1.3.2	Accolades, addAccolade()
Profile Manager	Accolades	Class	Represents achievements, honors, certifications, awards, or recognition that belongs to a student profile.	3.1.1, 3.1.2, 1.3.2	StudentProfile
Profile Manager	addAccolade()	Method	Represents a method that adds an accolade.	3.1.1, 3.1.2, 1.3.2	StudentProfile
Profile Manager	updateAccolade()	Method	Represents a method that updates an accolade	3.1.1, 3.1.2, 1.3.2	StudentProfile
Profile Manager	deleteAccolade()	Method	Represents a method that deletes an accolade	3.1.1, 3.1.2, 1.3.2	StudentProfile

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Profile Manager	updateProfileImageReference()	Method	Connects the selected image to the student profile.	3.1.2, 1.3.2	StudentProfile
Profile Manager	ContactInformation	Data Store	Stores student contact details shown on the profile.	3.1.2, 1.3.2	StudentProfile
Profile Manager	EducationInformation	Data Store	Stores student education information shown on the profile	3.1.2, 1.3.2	StudentProfile
Profile Manager	Skill(s)	Data Store	Stores a list of one or more skills shown on the profile.	3.1.2, 1.3.2	StudentProfile
Profile Manager	AboutMe	Data Store	Stores a summary of content that describes the student, shown on the profile	3.1.2, 1.3.2	StudentProfile
Profile Manager	CompanyProfile	Class	Represents and organizes the main profile information for a company.	2.1.3.1, 2.1.2, 2.1.3.2	CompanyProfile, CompanyLocation, JobInterestForm
Profile Manager	CompanyLocation	Data Store	Represents a location where the company conducts business and may include a physical address	2.1.3.3, 2.1.3.4, 2.1.3.5, 2.1.3.6	CompanyProfile, PhysicalAddress
Profile Manager	PhysicalAddress	Data Store	Stores the address details for a company location.	2.1.3.5, 2.1.3.7	CompanyLocation

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Profile Manager	JobInterestForm	Data Store	Represents the form a student submits to express interest in a job	2.1.2	CompanyProfile, StudentUser, Qualification
Profile Manager	Qualification	Data Store	Represents a student's listed qualification for a job interest form	2.1.2	JobInterestForm, StudentUser
Profile Manager	StudentFollow	Data Store	Represents the relationship between a company representative and a followed student user.	2.1.2	CompanyRepresentative, StudentUser
Profile Manager	GPA	Variable	Represents the student's grade point average shown on the profile.	3.1.3.1	StudentProfile
Profile Manager	addProfileInformation	Method	Adds profile-owned data	3.1.2	StudentProfile, CompanyProfile
Profile Manager	updateProfileInformation	Method	Updates profile-owned data	3.1.2	StudentProfile, CompanyProfile
Profile Manager	deleteProfileInformation	Method	Deletes profile-owned data	3.1.2	StudentProfile, CompanyProfile
Profile Manager	addCompanyLocation	Method	Adds a new company location	2.1.3.3, 2.1.3.4	CompanyProfile

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Profile Manager	updateCompanyLocation	Method	Updates a company location from the location list.	2.1.3.3, 2.1.3.5, 2.1.3.7	CompanyProfile
Profile Manager	deleteCompanyLocation	Method	Deletes a company location from the location list.	2.1.3.3	CompanyProfile

EI Table 1.6.2 – Image Manager

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Image Manager	ImageHandler	Class	This oversees handling all image related methods	2.1, 2.1.3, 3.1, 3.1.3	UpdateImage(), ValidateImage(), ImageStore
Image Manager	updateImage()	Method	This handles updating images for users	2.1.3.10, 3.1.2	ImageManager, ImageValidator
Image Manager	processImage()	Method	This crops, resizes, and manipulates an image	2.1, 2.1.3, 3.1, 3.1.3	ImageManager, ImageValidator
Image Manager	retrieveImage()	Method	This handles retrieving image data from a database	2.1, 2.1.3, 3.1, 3.1.3	ImageManager, ImageStore

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Image Manager	ImageValidator	Class	This validates whether an image meets platform standards	2.1, 2.1.3, 3.1, 3.1.3	ImageManager
Image Manager	ImageStore	Data Store	This stores all profile images for various users	2.1, 2.1.3, 3.1, 3.1.3, 3.1.2, 2.1.3.10	ImageManager
Image Manager	/api/profile	API Endpoint	This is an API endpoint to the overall audit logger	2.1, 2.1.3, 3.1, 3.1.3	ImageManager

EI Table 1.6.3 – Preferences Manager

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Preferences Manager	updateNotification Preferences()	Method	Updates the user's notification delivery and filtering settings.	3.3.5, 3.3.6	NotificationPreferences, ProfileManager

Subcomponent	Entity Name	Entity Type	Description/ Purpose	Related Requirement(s)	Collaborators / Related Entities
Preferences Manager	NotificationPreferences	Data Table	Stores persistent user settings regarding which notifications to receive and how.	3.3.5, 3.3.6	updateNotificationPreferences(), getPreferences()
Preferences Manager	muteNotifications	Variable	Boolean flag representing whether a user has opted to silence all incoming notifications.	3.3.5	NotificationPreferences
Preferences Manager	getPreferences()	Method	Retrieves the user's saved preferences for system display and notification routing.	3.3.6	NotificationPreferences
Preferences Manager	/api/profile/preferences	API Endpoint	Handles incoming requests to view or modify user preference configurations.	3.3.5, 3.3.6	updateNotificationPreferences(), getPreferences()

EI Table 1.6.4 – Privacy Settings Manager

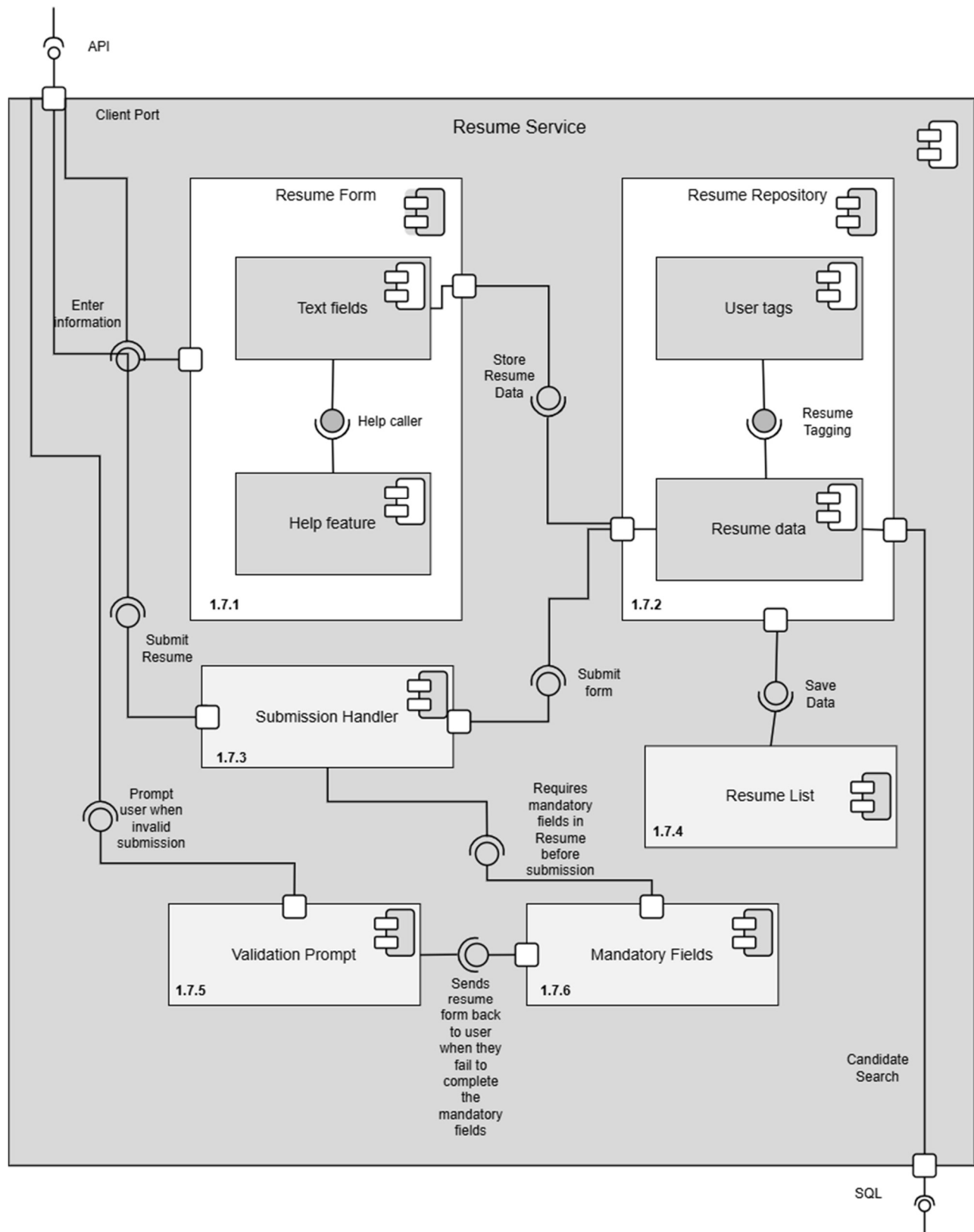
Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Privacy Settings Manager	UserPrivacySettings	Object	This contains variables controlling the privacy settings for the user	3.2	PrivacySettingsManager
Privacy Settings Manager	updateSettings()	Method	This edits settings for the user	3.2, 3.2.7	PrivacySettingsManager, SaveSettings()
Privacy Settings Manager	SettingsValidator	Class	This ensures that all settings for the user are all valid	3.2, 3.2.7	PrivacySettingsManager
Privacy Settings Manager	initializeSettings()	Method	This sets default settings for an account	3.2, 3.2.7	PrivacySettingsManager
Privacy Settings Manager	PrivacySettingsManager	Class	This handles all changes that occur for the privacy settings	3.2	UserPrivacySettings, SaveSettings, InitializeSettings(), SettingsValidator, UpdateSettings()
Privacy Settings Manager	saveSettings()	Method	Saves the settings associated with the account	3.2, 3.2.7	PrivacySettingsManager
Privacy	/api/profile/manage	API	This is an	3.2,	ImageManager

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Settings Manager		Endpoint	API endpoint to the overall audit logger	3.2.7	

EI Table 1.6.5 – Profile Audit Logger

Subcomponent	Entity Name	Entity Type	Description/ Purpose	Related Requirement(s)	Collaborators / Related Entities
Profile Audit Logger	logActivity()	Method	Records a specific user action or security event into the audit trail.	3.1.2, UI 4.4	AuditLog, ProfileManager
Profile Audit Logger	AuditLog	Data Store	Stores immutable records of account changes, logins, and security-related events.	UI 4.4	logActivity(), getLoginHistory()
Profile Audit Logger	getLoginHistory()	Method	Retrieves past login instances and locations for display on the security screen.	UI 4.4	AuditLog
Profile Audit Logger	SecurityEvent	Object	An instance of a tracked user action, such as a password change or profile update.	3.1.2, UI 4.4	logActivity(), AuditLog

Collaborators / Related Entities	Related Requirement(s)	Description/ Purpose	Entity Type	Entity Name	Subcomponent
getLoginHistory ()	UI 4.4	Handles requests to fetch the user's recent login and profile activity history.	API Endpoint	/api/profile/audit	Profile Audit Logger

Figure 1.7 — Resume Service Detail

Design Information – Resume Service

Field	Description / Purpose
View Name & Number	1.7 Resume Service
Purpose	Describe the resume service.
Description	Basic service demonstrating the resume service.
Requirements	SRS 3.5 Resume (3.5.1–3.5.5.2); referenced in 2.1.2, 2.2.5, 2.3.3, 2.3.5, 2.5.6
Elements	1.7.1 Resume Form
	1.7.2 Resume Repository
	1.7.3 Submission Handler
	1.7.4 Resume List
	1.7.5 Validation Prompt
	1.7.6 Mandatory Fields
Navigation / References	1.4 API Gateway / Orchestrator, 1.14 Database, 1.15 File / Blob Storage; Screen view 2.3 Resume Builder
Viewpoint	Component Diagram

Entity Identification Tables – Resume Service**EI Table 1.7.1 – Resume Form**

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Resume Service	ResumeInput	Object	Stores the information entered by the student in the resume form	3.5.1	ResumeSection, Submission Handler
Resume Service	ResumeSection	Class	Represents a section of the resume such as education, skills, or experience.	3.5.1	ResumeInput, Degree_Info, Experience, Contact_Info, Skill, Student, Resume, Submission Handler
Resume Service	ResumeFile	Object	Represents the resume file selected by the student for upload	3.5.2	Resume Repository, Database, Resume List

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Resume Service	FileType	Variable	Stores the type of file being uploaded	3.5.2	Resume Repository
Resume Service	FileSize	Variable	Stores the size of the uploaded resume file	3.5.2	Resume Repository
Resume Service	ResumeMetadata	Class	Stores information about the resume file such as name, type, size, and upload date	3.5.2	ResumeFile, Database
Resume Service	ResumeFiles	Data Store	Stores uploaded resume documents and related files	3.5.2	ResumeFile, File / Blob Storage
Resume Service	File / Blob Storage	Data Store	Stores the actual uploaded resume file	3.5.2	ResumeFiles, Resume Service
Resume Service	ResumeStatus	Variable	Tracks whether the resume is draft, uploaded, saved, or ready to submit	3.5.1, 3.5.2, 3.5.4.1	Resume, Student Dashboard
Resume Service	createResume()	Function	Starts the process of creating a new resume.	3.5.1	ResumeBuilder, Resume
Resume Service	collectResumeInput()	Function	Collects resume	3.5.1	ResumeForm, ResumeInput

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
			information entered by the student.		
Resume Service	saveResumeDraft()	Function	Saves the current resume as a draft.	3.5.1, 3.5.4.1	Resume, Database
Resume Service	uploadResume()	Method	Uploads a student resume file	3.5.2	ResumeFile, File / Blob Storage
Resume Service	saveResumeMetadata()	Function	Saves resume file information after upload	3.5.2	ResumeMetadata, Database
Resume Service	linkResumeToProfile()	Function	Connects the resume to the student profile	3.5.4.1	Resume, StudentProfile, Database

EI Table 1.7.2 – Resume Repository

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Resume Service	Resume	Class	Represents a student user resume	3.5.1	Contact_Info, Experience, Degree_Info, Skill_List, User_Tag, Resume Form, Database
Resume Service	Contact_Info	Class in a class	Represents the contact information within the	3.5.1	Resume, Full_Name, Email, Phone

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
			Resume class		
Resume Service	Experience	Class in a class	A professional or voluntary experience related to a company within specified dates	3.5.1	Resume, Company, Exp_Start_Date, Exp_End_Date, Position, Exp_Desc
Resume Service	Degree_Info	Class in a class	Information pertaining to a user's specific degree, including what type of degree, GPA, and expected graduation date	3.5.1	Resume, Degree_Title, School, GPA, Graduation_Date
Resume Service	Skill	Class	Represents a skill of a student or job requirement	3.5.1	Resume, Submission Handler
Resume Service	Exp_Start_Date	Member variable	The start date of an Experience object including month and year	3.5.1	Experience
Resume Service	Exp_End_Date	Member variable	The end date of an Experience object including month and year	3.5.1	Experience
Resume Service	Position	Member variable	The position title of an Experience object	3.5.1	Experience

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Resume Service	Degree_Title	Member variable	The degree name obtained from a school for a Degree_Info object	3.5.1	Degree_Info
Resume Service	GPA	Member variable	The numerical grade obtained from a degree for the Degree_Info object	3.5.1	Degree_Info
Resume Service	Graduation_Date	Member variable	The expected date of graduation for a Degree_Info object including month and year	3.5.1	Degree_Info
Resume Service	User_Tag	Member variable	A one-to-three-word long keyword that summarizes the outcome of a resume	3.5.1	Resume
Resume Service	Full_Name	Member variable	Represents a student's Full Name	3.5.1	Contact_Info
Resume Service	Email	Member variable	Represents a student's email address	3.5.1	Contact_Info
Resume Service	Phone	Member variable	Represents a student's contact number	3.5.1	Resume

EI Table 1.7.3 – Submission Handler

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Resume Service	saveResume()	Method	Saves a Resume object into the Database if validated	3.5.5.1	Resume, Database, Validation Prompt
Resume Service	saveContactInfo()	Function	Saves a Contact_Info to a Resume object	3.5.5.1	Resume, Contact_Info, saveResume()
Resume Service	saveExperience()	Function	Saves an Experience to a Resume object	3.5.5.1	Resume, saveResume(), Experience
Resume Service	saveDegree_Info()	Function	Saves a Degree_Info to a Resume object	3.5.5.1	Resume, saveResume(), Degree_Info
Resume Service	createSkill()	Function	Adds a new skill tagged from the Resume to the student submitter to the Skills Index	3.5.5.1	Resume Form, Database, Student
Resume Service	tagSkill()	Function	Tags a skill specified by the student submitter	3.5.5.1	Student, Database, Skill

EI Table 1.7.4 – Resume List

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Resume Service	listResumes()	Method	Creates a viewable list of resumes based on filters such as SearchFilter or tagged Student	3.5.3	SearchFilter, Student, Resume Repository, viewResume()
Resume Service	viewResume()	Method	Builds a view form of a Resume object	3.5.3	Resume, Database
Resume Service	listSkills()	Method	Creates a list of skills tagged to a student and resume	3.5.3	viewResume(), Skill, Database, Student, Resume

EI Table 1.7.5 – Validation Prompt

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Resume Service	checkResume()	Method	Validation check for Resume submission	3.5.5.2, 3.5.5.3	Submission Handler, Mandatory Fields
Resume Service	ResumeError	Class	Represents an alert for an invalid resume object submission	3.5.5.3	checkResume()
Resume Service	Error_ContactInfo	Member variable	Represents an error message for invalid contact information	3.5.5.3	ResumeError, Contact_Info, checkContactInfo()
Resume Service	Error_Experience	Member variable	Represents an error message for invalid Experience	3.5.5.3	ResumeError, Experience, checkExperience()
Resume Service	Error_Degree_Info	Member variable	Represents an error message for invalid Degree_Info	3.5.5.3	ResumeError, Degree_Info, checkDegreeInfo()
Resume Service	checkContactInfo()	Method	Validation checks for Contact_Info in a Resume object	3.5.5.3	Contact_Info, Error_ContactInfo, Mandatory Fields
Resume Service	checkExperience()	Method	Validation checks for Experience in a Resume object	3.5.5.3	Experience, Error_Experience, Mandatory Fields
Resume Service	checkDegreeInfo()	Method	Validation checks for Degree_Info in a Resume object	3.5.5.3	Degree_Info, Error_Degree_Info, Mandatory Fields

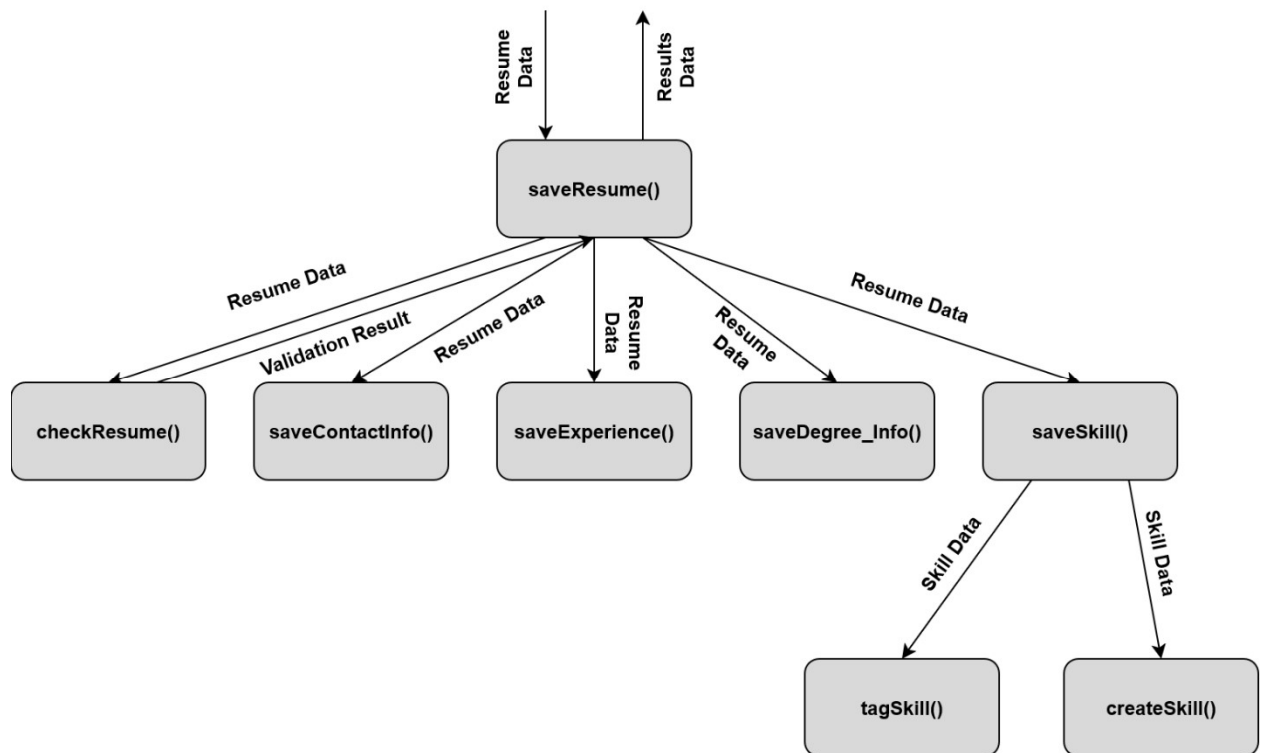
EI Table 1.7.6 – Mandatory Fields

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Resume Service	requireStartDate()	Function	Checks that Experience	3.5.5.2, 3.5.5.3	Experience, checkExperience()

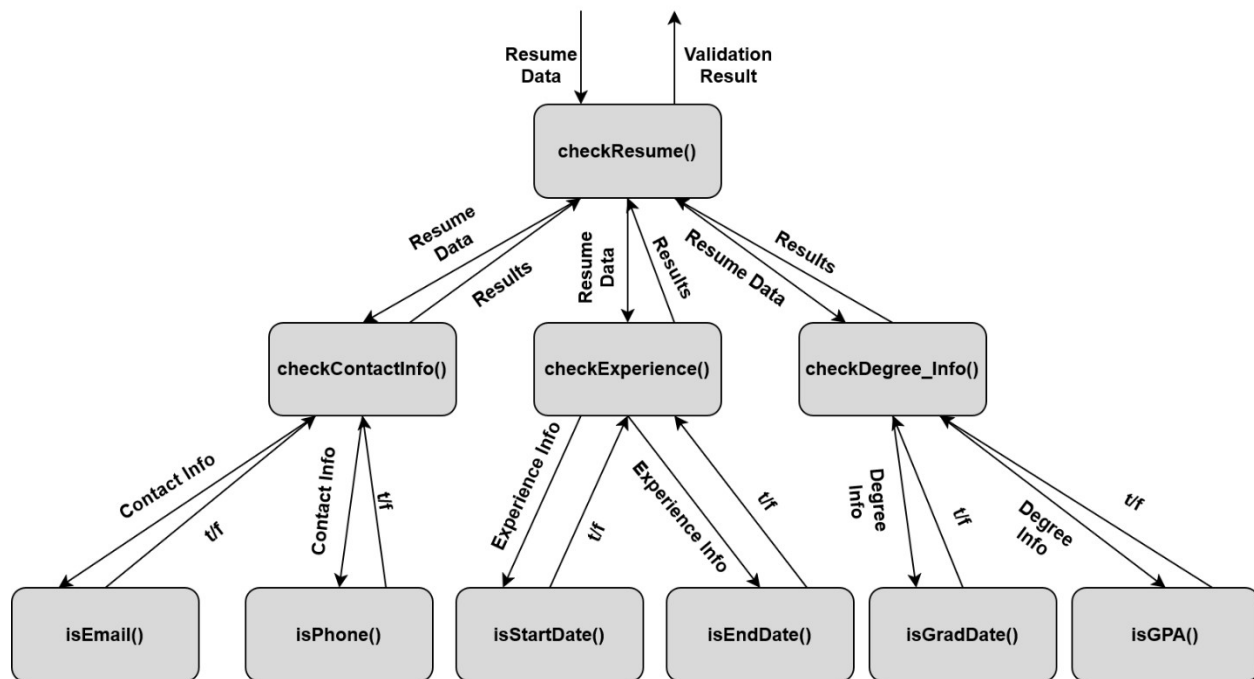
Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
			Start Date is specified		
Resume Service	requireGradDate()	Function	Checks that graduation date is specified	3.5.5.2, 3.5.5.3	Degree_Info, checkDegreeInfo()
Resume Service	requireEmail()	Function	Checks that email is specified	3.5.5.2, 3.5.5.3	Contact_Info, checkContactInfo()
Resume Service	requirePhone()	Function	Checks that phone is specified	3.5.5.2, 3.5.5.3	Contact_Info, checkContactInfo()
Resume Service	requireDegName()	Function	Checks that Degree name is specified	3.5.5.2, 3.5.5.3	Degree_Info, checkDegreeInfo()
Resume Service	requirePosition()	Function	Checks that position title is specified	3.5.5.2, 3.5.5.3	Experience, checkExperience()

Structure Charts – Resume Service

SC Figure 1.7.0.3 — saveResume()



SC Figure 1.7.0.4 — checkResume()



Data Flow Diagrams – Resume Service

DFD Figure 1.7.0.0.2 — Submission and Storage Workflow

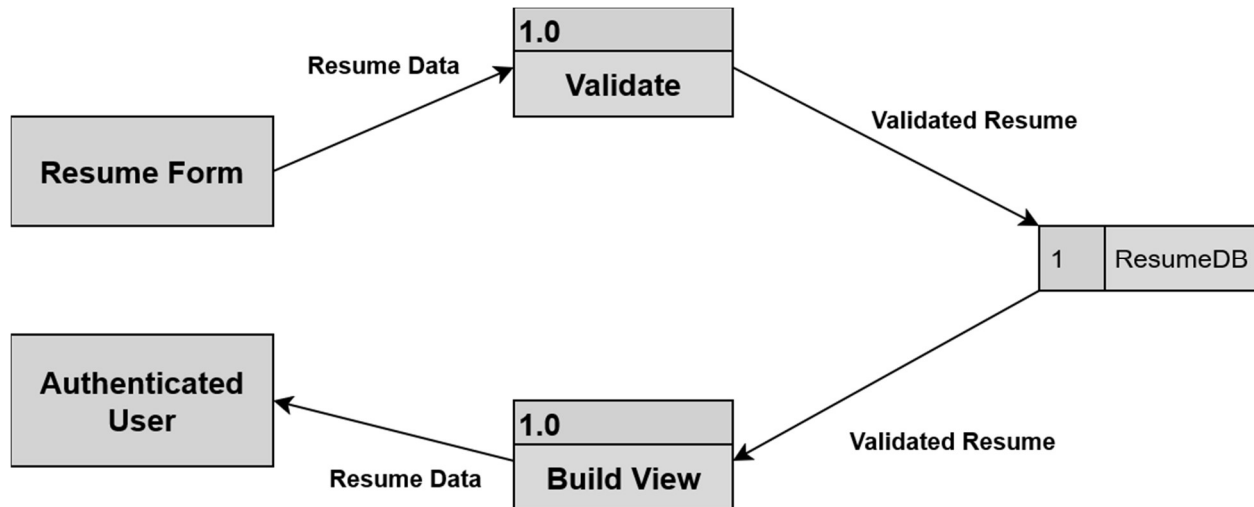
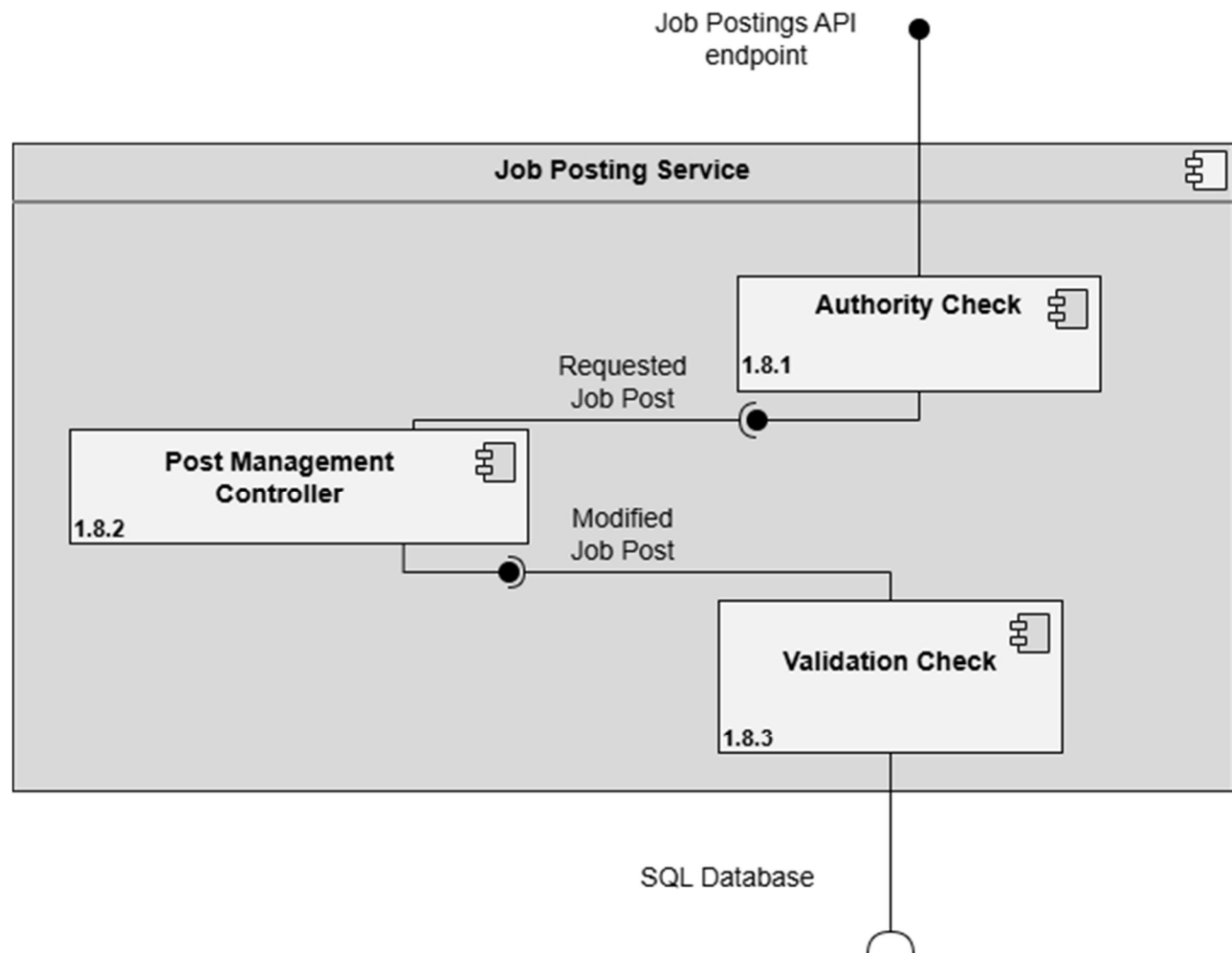


Figure 1.8 — Job Posting Service Detail**Design Information – Job Posting Service**

Field	Description / Purpose
View Name & Number	1.8 Job Postings Service
Purpose	Describe the jobs posting service.
Description	Basic service demonstrating the jobs posting service.
Requirements	SRS 2.5 Job Postings (2.5.1–2.5.7); referenced in 3.3.1, 3.4.1, 2.2.5
Elements	1.8.1 Authority Check
	1.8.2 Post Management Control
	1.8.3 Validation Check
Navigation / References	1.4 API Gateway / Orchestrator, 1.14 Database; Screen views 3.3 Job Posting Creation, 2.4 Job Search, 2.5 Job Detail
Viewpoint	Component Diagram

Entity Identification Tables – Job Posting Service

El Table 1.8.1 – Authority Check

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Authority Check	AuthorizationService	Class	Main container for authorization logic	2.5.1-2.5.4	
Authority Check	PermissionManager	Class	Manages permission rules	2.5.1-2.5.4	AuthorizationService
Authority Check	RoleManager	Class	Determines user roles	2.5.1-2.5.4	AuthorizationService
Authority Check	verifyOwner()	Method	Verifies that a user owns a given job listing	2.5.1-2.5.4	AuthorizationService, PermissionManager
Authority Check	canCreateJob()	Method	Checks if a user is allowed to post a job	2.5.1-2.5.4	AuthorizationService, PermissionManager
Authority Check	canEditJob()	Method	Checks if a user is allowed to edit a job	2.5.1-2.5.4	AuthorizationService, PermissionManager

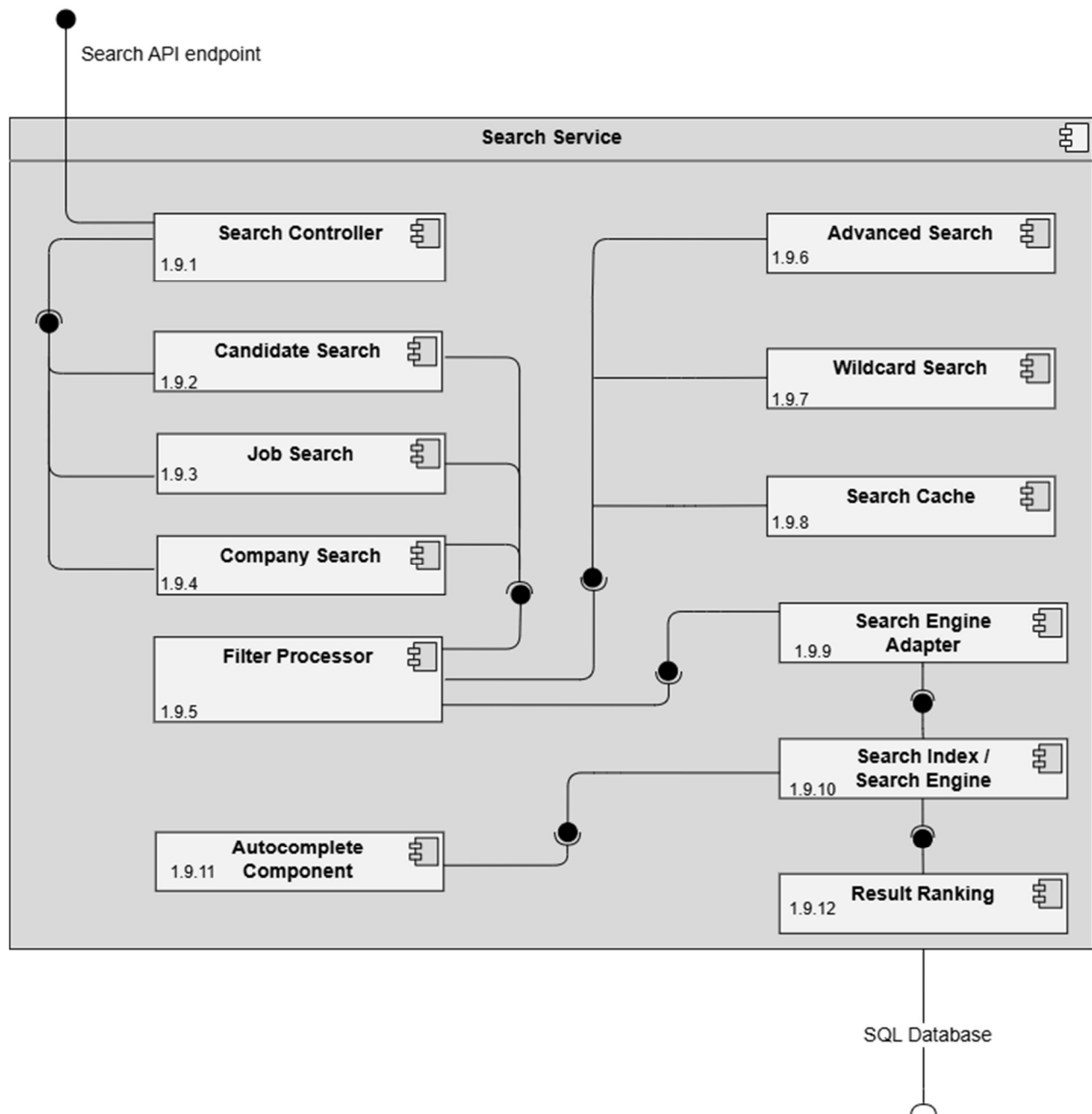
EI Table 1.8.2 – Post Management Controller

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Post Management Controller	JobPosting	Class	Job data	2.5.1-2.5.4	currentPost, JobPostingsTable
Post Management Controller	currentPost	Object	The active job post is being modified by the recruiter	2.5.1-2.5.4	JobPosting, updatePostDetails
Post Management Controller	createPost	Method	Initiate a new JobPosting object	2.5.1-2.5.4	JobPosting, JobPostingsTable
Post Management Controller	updatePostDetails	Method	Modifies the job data	2.5.1-2.5.4	currentPost, JobPostingTable
Post Management Controller	archivePost	Method	Hide the post from student searches	2.5.1-2.5.4	currentPost, JobPostingsTable
Post Management Controller	isDraft	Variable	A boolean indicating if the job post is still a hidden draft	2.5.1-2.5.4	JobPosting, createPost
Post Management Controller	authorId	Variable	Identifier of the recruiter	2.5.1-2.5.4	JobPosting
Post Management Controller	JobPostingsTable	Data Store	The database table that stores all job records	2.5.1-2.5.4	JobPosting, createPost, updatePostDetails

EI Table 1.8.3 – Validation Check

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Validation Check	PostValidator	Class	Contains the core logic algorithms used to validate the job post	2.5.1-2.5.4	ValidationResult, validateRequiredFields
Validation Check	ValidationResult	Object	Stores the pass/fail breakdown of a completed validation check	2.5.1-2.5.4	PostValidator, isValid, errorMessage
Validation Check	validateRequiredFields	Method	Check if any required fields are empty	2.5.1-2.5.4	PostValidator, isValid
Validation Check	verifyExpirationDate	Method	Confirms that the job post is still in a valid posting timeline	2.5.1-2.5.4	PostValidator, isValid
Validation Check	checkProfanity	Method	Searches text strings for forbidden terms	2.5.1-2.5.4	PostValidator, SystemRulesConfig, errorMessage
Validation Check	isValid	Variable	A boolean tracking if the job post is valid to be posted	2.5.1-2.5.4	ValidationResult
Validation Check	errorMessage	Variable	Notification to user when the validation check failed	2.5.1-2.5.4	ValidationResult
Validation Check	SystemRulesConfig	Data Store	A reference configuration file or directory	2.5.1-2.5.4	checkProfanity

Collaborators / Related Entities	Related Requirement(s)	Description / Purpose	Entity Type	Entity Name	Subcomponent
		tracking system constants and forbidden word lists			

Figure 1.9 — Search Service Detail**Design Information – Search Service**

Field	Description / Purpose
View Name & Number	1.9 Search Service
Purpose	Describe the search service.
Description	Basic service demonstrating the search service.
Requirements	SRS 2.3 Search Features (2.3.1–2.3.8), 3.3 Student Search Features (3.3.1–3.3.2)

Field	Description / Purpose
Elements	1.9.1 Search Controller
	1.9.2 Candidate Search
	1.9.3 Job Search
	1.9.4 Company Search
	1.9.5 Filter Processor
	1.9.6 Advanced Search
	1.9.7 Wildcard Search
	1.9.8 Search Cache
	1.9.9 Search Engine Adapter
	1.9.10 Search Index / Search Engine
	1.9.11 Autocomplete Component
	1.9.12 Result Ranking
Navigation / References	1.4 API Gateway / Orchestrator, 1.14 Database; Screen views 2.4 Job Search, 3.2 Candidate Search
Viewpoint	Component Diagram

Entity Identification Tables – Search Service

El Table 1.9.1 — Search Controller

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Search Controller	SearchController	Class	Coordinates all search requests and routes them to the appropriate search module.	2.3.6, 2.3.7, 2.3.8, 3.3.1	CandidateSearch, JobSearch, AutofillAPI
Search Controller	searchQuery	Function	Stores the text entered into the search bar.	2.3.6, 3.3.1	SearchController
Search Controller	searchFilters	Function	Stores advanced search filter criteria.	2.3.8, 3.3.3, 3.3.4	CandidateSearch, JobSearch
Search Controller	submitSearch()	Method	Processes a search request submitted by a user.	2.3.5, 3.3.2	CandidateSearch, JobSearch
Search Controller	autocompleteSearch()	Method	Retrieves autocomplete suggestions	2.3.6	AutofillAPI

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
			from the API.		
Search Controller	wildcardSearch()	Method	Handles wildcard search patterns entered by users.	2.3.7	CandidateSearch, JobSearch
Search Controller	advancedSearch()	Method	Executes searches using advanced filtering options.	2.3.8, 3.3.4	CandidateSearch, JobSearch

EI Table 1.9.2 – Candidate Search

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Candidate Search	CandidateSearch	Class	Performs searches on student users for company representatives.	2.3.1 – 2.3.8	CompanyRepresentative, StudentUserCollections
Candidate Search	CompanyRepresentative	Object	User performing candidate searches.	2.3.1, 2.3.4, 2.3.5	CandidateSearch
Candidate Search	StudentUserCollection	Class	Collection containing all student user profiles.	2.3.1, 2.3.2	StudentUser
Candidate Search	candidateFilters	Function	Stores filter criteria entered by the company	2.3.2, 2.3.4	CompanyRepresentative

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
			representative .		
Candidate Search	searchResults	Function	Stores matching student profiles and resumes.	2.3.5	StudentUser
Candidate Search	inputFilters()	Method	Prompts the company representative to enter search filters.	2.3.4	CompanyRepresentative
Candidate Search	filterCandidates()	Method	Selects student users matching the specified criteria.	2.3.2, 2.3.3	StudentUserCollection
Candidate Search	displayResults()	Method	Displays resumes and profiles matching the search criteria.	2.3.5	SearchResults

EI Table 1.9.3 – Job Search

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Job Search	JobSearch	Class	Performs searches for job postings	3.3.1 – 3.3.4	StudentUser, JobPosting, CompanyProfile

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
			and company profiles.		
Job Search	StudentUser	Object	User searching for jobs and companies.	3.3.1, 3.3.3, 3.3.4	JobSearch
Job Search	JobPostingCollection	Class	Collection of available job postings.	3.3.1, 3.3.3	JobPosting
Job Search	CompanyProfileCollection	Class	Collection of company profile pages.	3.3.1	CompanyProfile
Job Search	selectedFilters	Function	Stores search filters chosen by the student user.	3.3.3, 3.3.4	StudentUser
Job Search	searchResults	Function	Stores matching jobs and company profiles.	3.3.2, 3.3.3	JobPosting, CompanyProfile
Job Search	searchJobs()	Method	Searches job postings using keywords and filters.	3.3.1, 3.3.3	JobPostingCollection
Job Search	searchCompanies()	Method	Searches company profile	3.3.1	CompanyProfileCollection

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
			pages.		
Job Search	filterResults()	Method	Narrows search results according to selected filters.	3.3.3, 3.3.4	StudentUser
Job Search	displayResults()	Method	Displays matching jobs and company profiles to the student user.	3.3.2	SearchResults

EI Table 1.9.4 — Company Search

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Company Search	CompanySearch	Class	Represents the class containing the methods and variables that handle company searches	3.3.1, 3.3.4	Search Controller, Wildcard Search
Company Search	searchCompanies()	Function	Processes company search requests submitted by the user	3.3.1, 3.3.4	CompanySearch, Filter Processor, searchQuery

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Company Search	searchQuery	Variable	Stores the keyword or phrase entered into the input field	3.3.4	searchCompanies()

EI Table 1.9.5 — Filter Processor

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Filter Processor	SearchFilter	Class	Represents the filters selected by the user for company searching	3.3.1, 3.3.4	Company Search, Job Search, Candidate Search, Advanced Search, Wildcard Search
Filter Processor	validateFilters()	Method	Validates user-selected search filters before processing the search	3.3.4	SearchFilter

EI Table 1.9.6 — Advanced Search

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Advanced	Advanced	Class	Represents the	3.3.1,	Search Controller,

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Search	CompanySearch		class that inherits from CompanySearch class and contains the overridden methods, additional methods if necessary and additional variables that handle advanced company searches	3.3.4	Wildcard Search
Advanced Search	searchCompanies()	Function	Processes company search requests submitted by the user for advanced search	3.3.1, 3.3.4	Advanced CompanySearch, FilterProcessor, searchQuery
Advanced Search	Advanced JobSearch	Class	Represents the class that inherits from JobSearch class and contains the overridden methods, additional methods if necessary and additional variables that handle advanced job searches	3.3.1, 3.3.4	Search Controller, Wildcard Search
Advanced Search	searchJobs()	Function	Processes job search requests submitted by the user for advanced search	3.3.1, 3.3.4	Advanced JobSearch, FilterProcessor, searchQuery

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Advanced Search	Advanced CandidateSearch	Class	Represents the class that inherits from CandidateSearch class and contains the overridden methods, additional methods if necessary and additional variables that handle advanced candidate searches	3.3.1, 3.3.4	Search Controller, Wildcard Search
Advanced Search	searchCandidates()	Function	Processes candidate search requests submitted by the user for advanced search	3.3.1, 3.3.4	Advanced CandidateSearch, FilterProcessor, searchQuery
Advanced Search	searchQuery	Variable	Stores the keyword or phrase entered into the search bar for advanced search	3.3.4	searchCompanies(), searchJobs(), searchCandidates()

EI Table 1.9.7 – Wildcard Search

Subcomponent	Entity Name	Entity Type	Description/ Purpose	Related Requirement(s)	Collaborators / Related Entities
Wildcard Search	WildcardSearch	Class	Represents the logic used to process wildcard search queries and match pattern-based input.	2.3.7, 3.3.1, 3.3.2	SearchEngineAdapter, SearchIndex, searchQuery
Wildcard Search	processWildcardSearch()	Function	Processes a search request using wildcard matching rules and returns matching results.	2.3.7, 3.3.1, 3.3.2	WildcardSearch, SearchIndex
Wildcard Search	wildcardPattern	Variable	Stores the pattern entered by the user for wildcard matching.	2.3.7	processWildcardSearch()

EI Table 1.9.8 – Search Cache

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Search Cache	SearchCache	Class	Stores recently used search queries and results to improve repeated search performance.	2.3.5, 3.3.2	SearchEngineAdapter, SearchIndex, cachedResults
Search Cache	cachedResults	Member Variable	Holds cached search result sets for faster retrieval.	2.3.5, 3.3.2	SearchCache, storeResults(), retrieveResults()

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Search Cache	storeResults()	Method	Saves search results into the cache for later reuse.	2.3.5	SearchCache, cachedResults
Search Cache	retrieveResults()	Method	Retrieves previously cached results for a repeated search request.	2.3.5, 3.3.2	SearchCache, cachedResults

EI Table 1.9.9 – Search Engine Adapter

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Search Engine Adapter	SearchEngineAdapter	Class	Acts as an interface between the search UI and the internal search engine logic.	2.3.1, 2.3.2, 2.3.7, 3.3.1	SearchCache, WildcardSearch, SearchIndex
Search Engine Adapter	executeSearch()	Method	Sends search requests to the search engine and returns processed results to the caller.	2.3.1, 2.3.5, 3.3.1, 3.3.2	SearchEngineAdapter, SearchIndex, SearchCache

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Search Engine Adapter	searchRequest	Variable	Stores the incoming search query and filter data before execution.	2.3.1, 2.3.2	executeSearch(), SearchCache

EI Table 1.9.10 — Search Index / Search Engine

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Search Index / Search Engine	SearchEngine	Class	Class responsible for managing the Search Index, processing search queries, handling wildcard logic, and executing filtering criteria	2.3.1, 3.3.1	searchIndex, searchFilters, executeSearch()
Search Index / Search Engine	searchIndex	Member Variable	Data Structure responsible for storing all search data including profiles, resumes, listings, etc.	2.3.1, 3.3.1	SearchEngine, executeSearch()
Search Index / Search Engine	searchFilters	Member Variable	Key Value data related to the searchIndex used to “filter” results from the search queries	2.3.2, 2.3.3, 2.3.4	SearchEngine, executeSearch()
Search Index / Search Engine	executeSearch()	Method	Method responsible for executing the search and returning results according to the search query and searchFilters	3.3.1, 2.3.5, 3.3.2	SearchEngine, searchIndex, searchFilters

EI Table 1.9.11 — Autocomplete Component

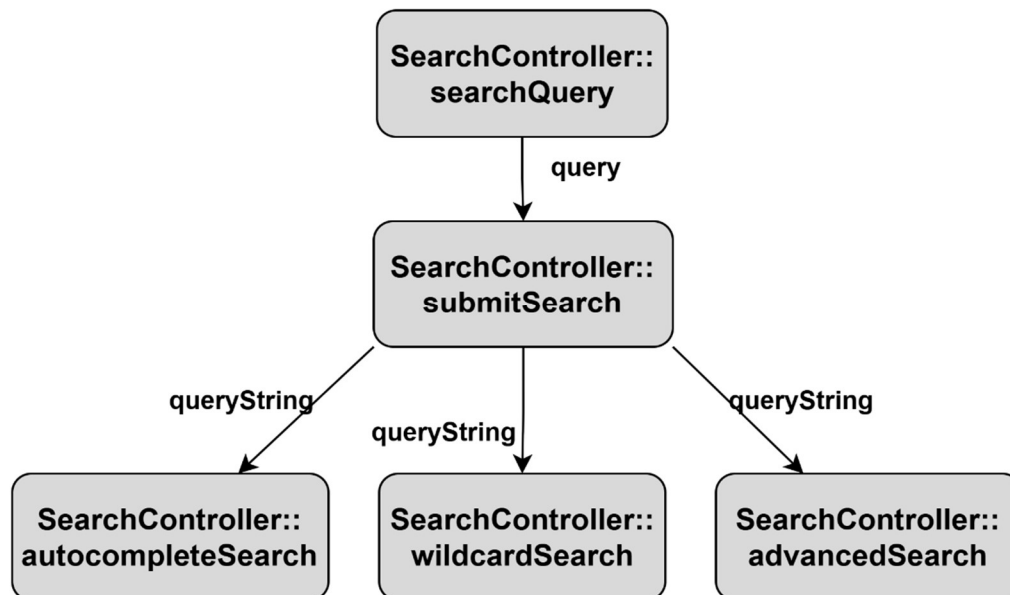
Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Autocomplete Component	AutocompleteComponent	Class	Frontend Class managing the Autocomplete API interactions	2.3.6	apiEndpoints, suggestions, fetchAutocomplete(), Autocomplete API
Autocomplete Component	apiEndpoints	Member Variable	String storing URL to the Autocomplete API	2.3.6	Autocomplete Component, fetchAutocomplete()
Autocomplete Component	suggestions	Member Variable	List of String suggestions returned from the Autocomplete API	2.3.6	Autocomplete Component, fetchAutocomplete()
Autocomplete Component	fetchAutocomplete()	Method	Calls API using apiEndpoints URL and returns list of suggestions	2.3.6	Autocomplete Component, apiEndpoints, suggestions
Autocomplete Component	Autocomplete API	Object	External API handling the autocomplete queries	2.3.6	Autocomplete Component, fetchAutocomplete()

EI Table 1.9.12 — Result Ranking

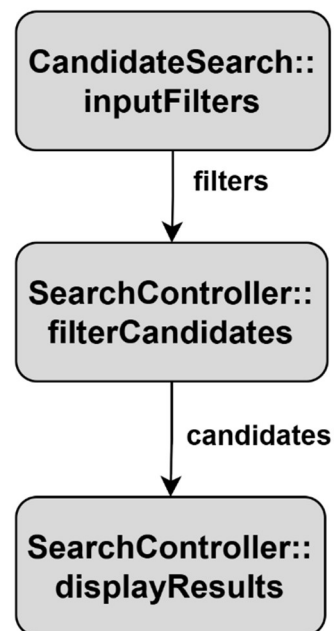
Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Result Ranking	ResultsRanker	Class	Class responsible for ordering or “ranking” results to be provided to the user	2.5.8, 3.3.4	rankingCriteria, rankResults()
Results Ranking	rankingCriteria	Member Variable	Attribute weighted to evaluate result priority (this could be GPA, Skills, Exp Graduation, etc.)	2.5.8, 3.3.4	ResultsRanker, rankResults()
Results Ranking	rankResults()	Method	Method responsible for prioritizing search results according to the ranking criteria	2.5.8, 3.3.4	ResultsRanker, rankingCriteria

Structure Charts – Search Service

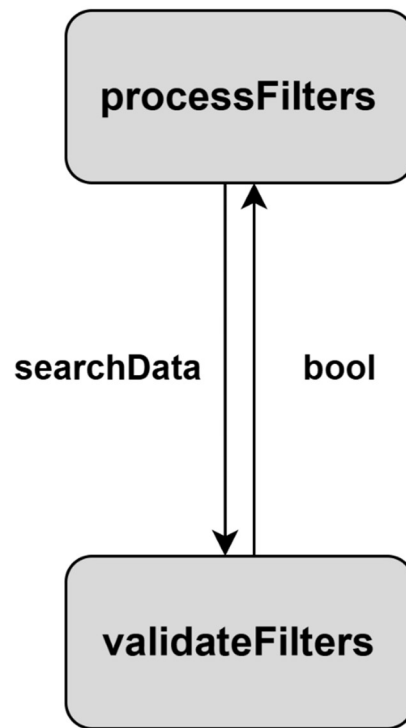
SC Figure 1.9.0.1 – SearchQuery()



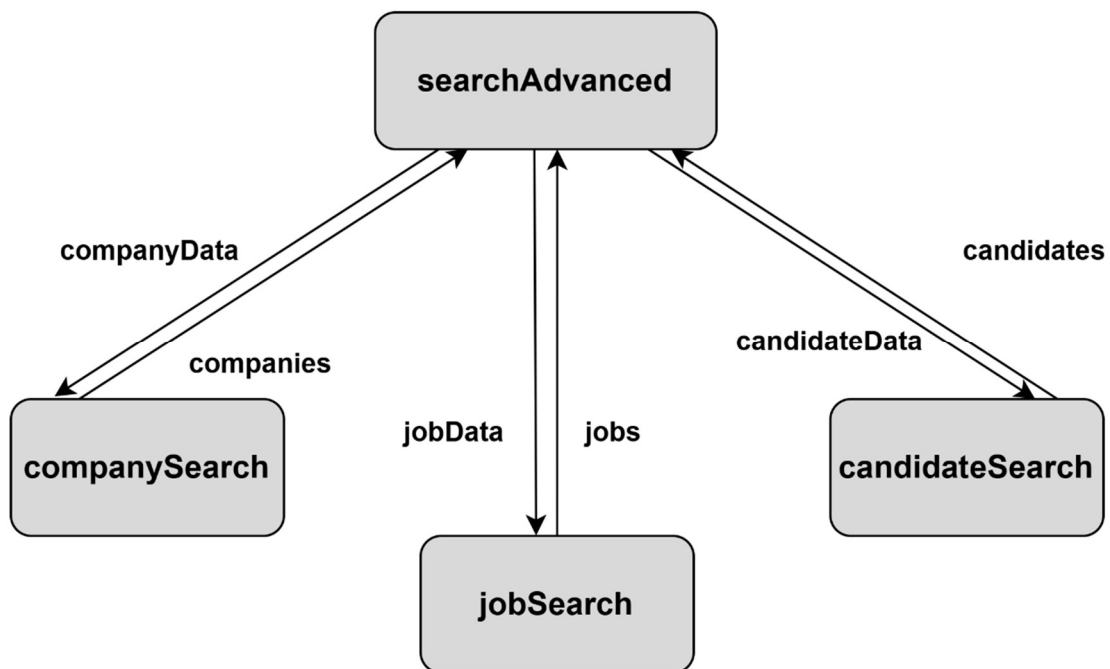
SC Figure 1.9.0.2 – InputFilters()



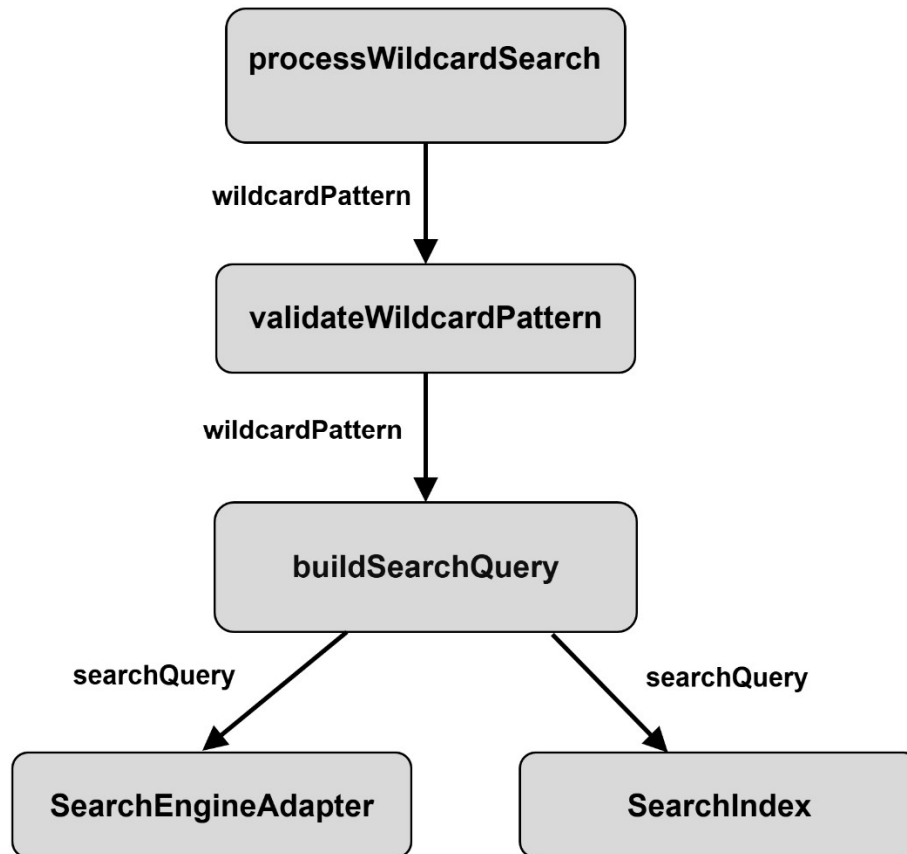
SC Figure 1.9.0.3 – ProcessFilters()



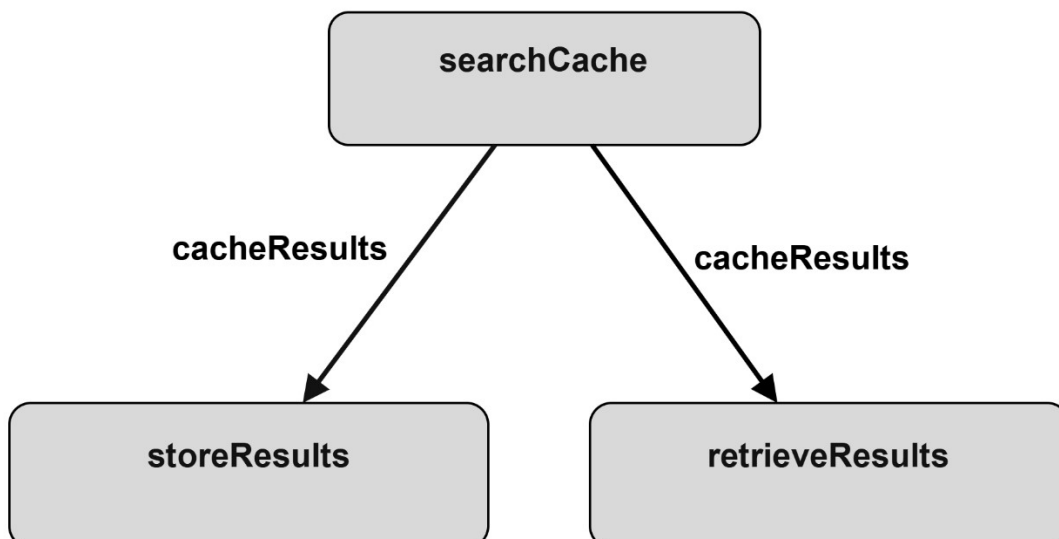
SC Figure 1.9.0.4 – Search()



SC Figure 1.9.0.5 – processWildcardSearch()

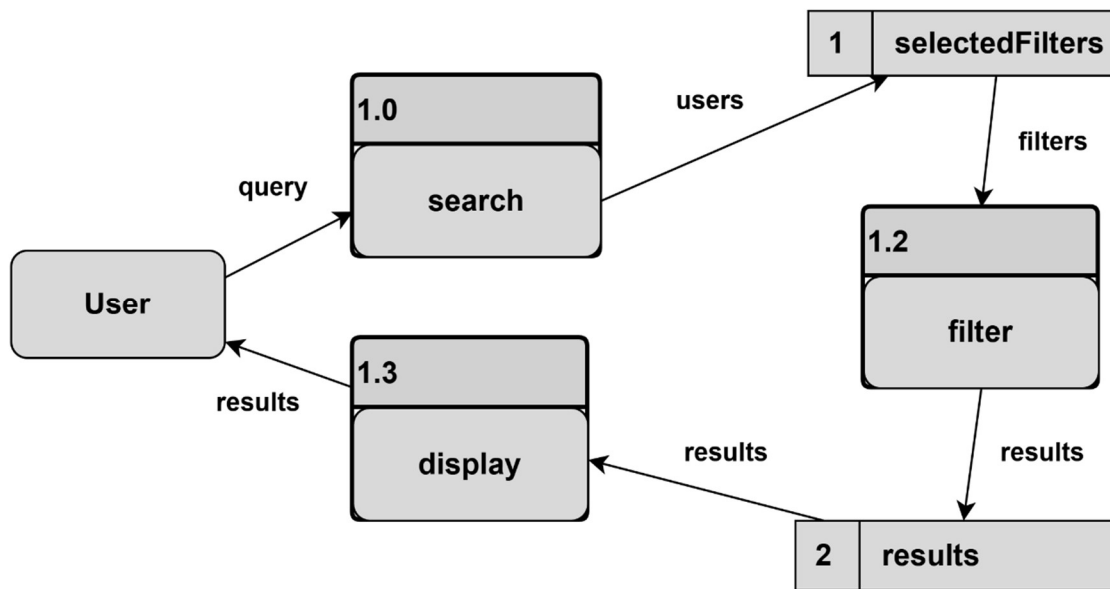


SC Figure 1.9.0.6 – SearchCache()

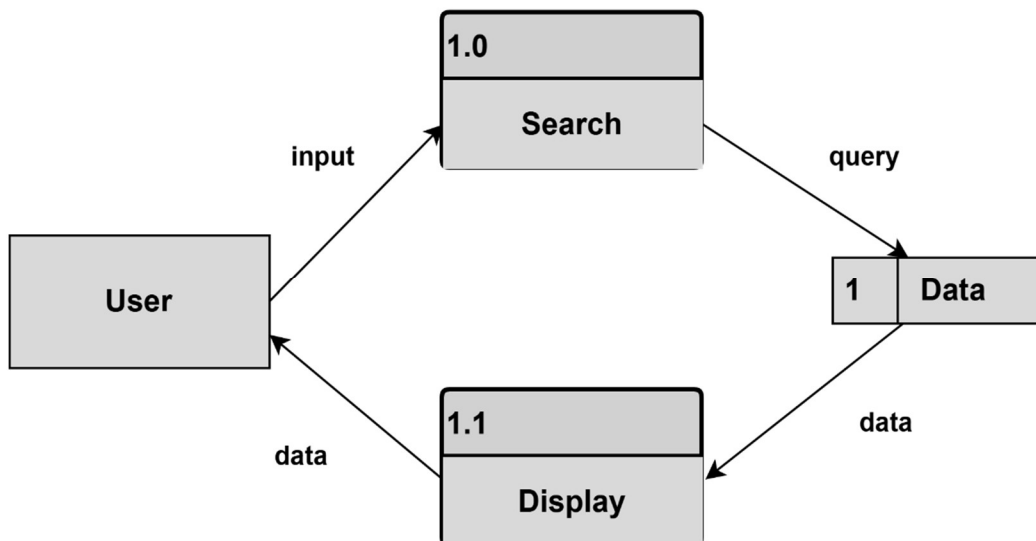


Data Flow Diagrams – Search Service

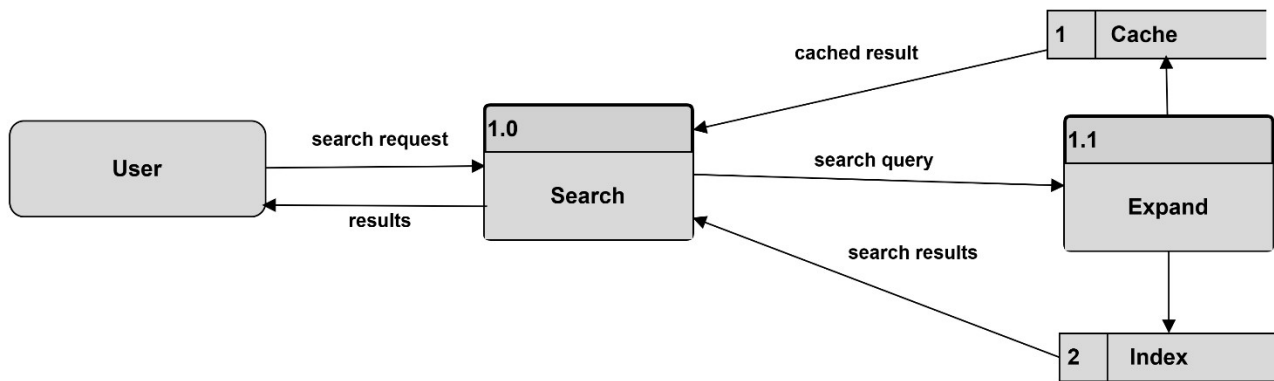
DFD 1.9.0.0.1 – Job Search



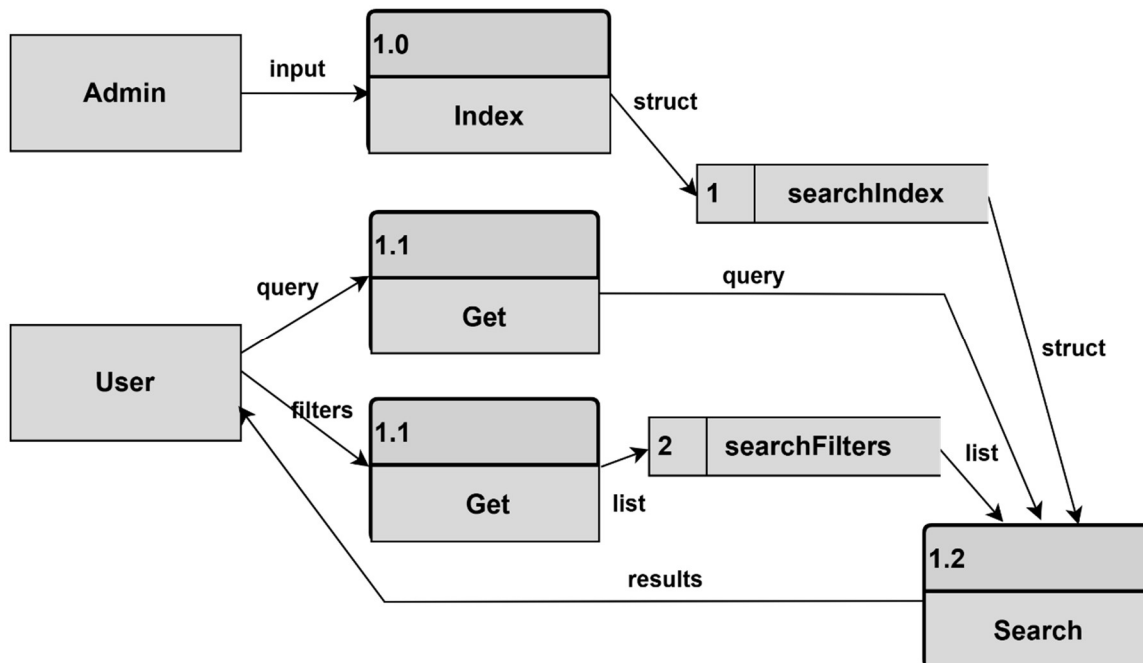
DFD Figure 1.9.0.0.2 – Company Search



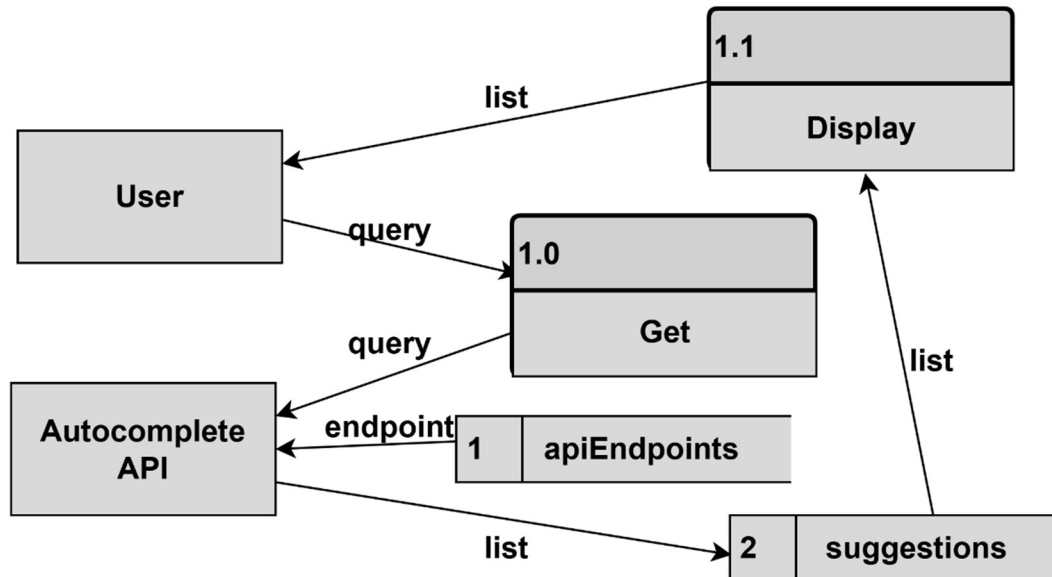
DFD Figure 1.9.0.0.3 – Search Engine Adapter



DFD Figure 1.9.0.0.4 – Search Index / Search Engine



DFD Figure 1.9.0.0.5 – Autocomplete



DFD Figure 1.9.0.0.6 – Ranking Results

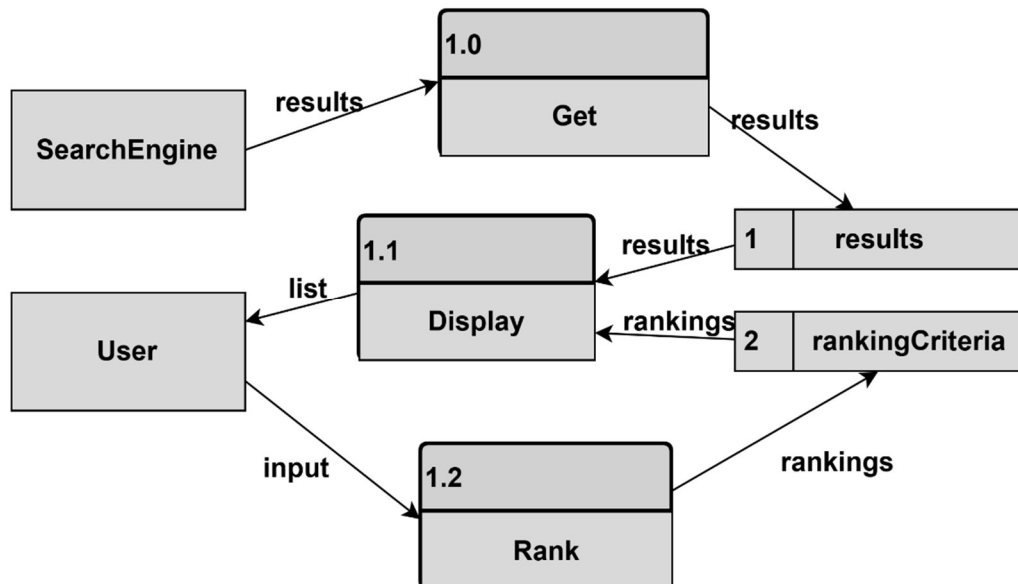
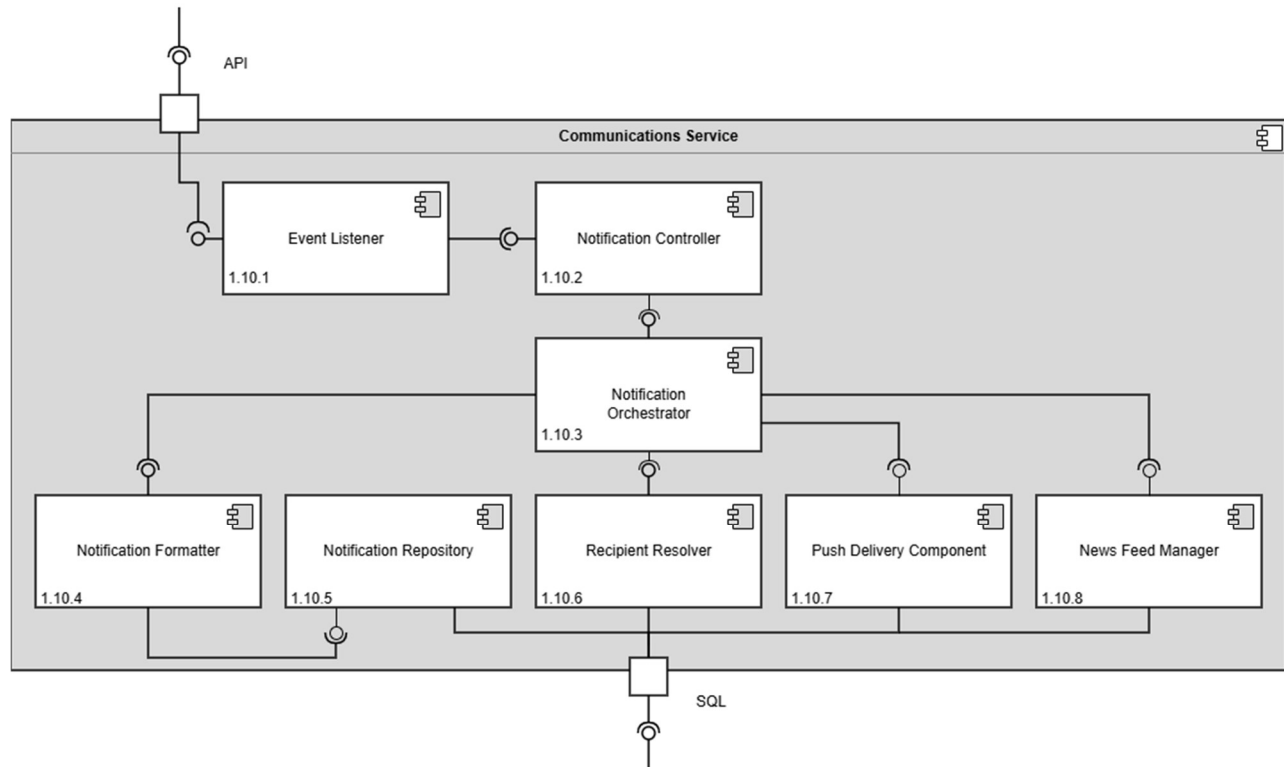


Figure 1.10 — Communication Services (Notifications + News Feed) Detail**Design Information – Communication Services (Notifications + News Feed)**

Field	Description / Purpose
View Name & Number	1.10 Communication Service
Purpose	Describe the communication service.
Description	Basic service demonstrating the communication service.
Requirements	SRS 2.2 Notifications (2.2.1–2.2.5), 2.4 News Feed (2.4.1–2.4.6), 3.2 Notifications (3.2.1–3.2.2), 3.4 News Feed (3.4.1–3.4.2)
Elements	1.10.1 Event Listener
	1.10.2 Notification Controller
	1.10.3 Notification Orchestrator
	1.10.4 Notification Formatter
	1.10.5 Notification Repository
	1.10.6 Recipient Resolver
	1.10.7 Push Delivery
	1.10.8 News Feed Manager
Navigation / References	1.4 API Gateway / Orchestrator, 1.14 Database; Screen views 4.1 Notifications, 4.2 News Feed
Viewpoint	Component Diagram

Entity Identification Tables – Communication Service

El Table 1.10.1 – Event Listener

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Event Listener	listenForEvents()	Function	Continuously monitors incoming system events	1.4	EventQueue
Event Listener	validateEvent()	Method	Verifies incoming event data before forwarding	1.10.2	Event
Event Listener	ForwardEvent()	Function	Stores the list of matching job postings returned from a search	1.10.2	Event
Event Listener	EventQueue	Data Structure	Temporary storage for received events awaiting processing	1.4	Event, listenForEvents()
Event Listener	Event	Object	Represents an incoming event requiring notification processing	1.10.2	listenForEvents()

El Table 1.10.2 – Notification Controller

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Notification Controller	processNotificationRequest()	Function	Processes incoming notification requests from Event Listener	1.10.1	NotificationRequest
Notification	NotificationRequest	Object	Encapsulat	1.10.	createNotificationTask()

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
on Controller			es notification processing information	2	
Notification Controller	validateNotification()	Method	Ensures notification request data is complete and valid	1.10.1	processNotificationRequest()
Notification Controller	NotificationPolicy	Class	Contains rules governing notification generation	1.10.4	validateNotification()
Notification Controller	createNotificationTask()	Function	Creates processing tasks for the orchestrator	1.10.1	validateNotification()
Notification Controller	NotificationTask	Object	Represents a pending notification workflow	1.10.1	createNotificationTask()

EI Table 1.10.3 – Notification Orchestrator

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Notification Orchestrator	orchestrateNotification()	Function	Coordinates the	2.2.1 –	Formatter, Repository, Resolver,

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
or			complete notification workflow	2.2.5, 2.4, 2.4.1 – 2.4.6, 3.2, 3.2.1 – 3.2.2, 3.4, 3.4.1 – 3.4.2	PushDeliveryComponent
Notification Orchestrator	NotificationJob	Object	Represents a notification being processed	1.10.2	orchestrateNotification()
Notification Orchestrator	buildNotification()	Method	Constructs notification content from event data	1.10.4	NotificationFormatter
Notification Orchestrator	resolveRecipients()	Method	Determines intended notification recipients	1.10.6	RecipientResolver
Notification Orchestrator	deliverNotification()	Method	Sends notifications through delivery channels	1.10.7	PushDeliveryComponent
Notification Orchestrator	archiveNotification()	Method	Stores notification records for auditing and retrieval	1.10.5	NotificationRepository
Notification Orchestrator	updateNewsFeed()	Method	Publishes notification	1.10.8	NewsFeedManager

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
or			content to the news feed system		
Search Query Manager	processingStatus	Variable	Tracks workflow progress and state	1.10.2 – 1.10.8	NotificationJob

EI Table 1.10.4 – Notification Formatter

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Notification Formatter	NotificationFormatter	Class	Defines formatting behavior for user notifications.	2.2.1 - 2.2.5, 3.2.1 - 3.2.7	Notification Orchestrator, NotificationContent
Notification Formatter	formatNotification()	Function	Converts event data into a standard notification message.	2.2.1, 2.2.2, 3.2.1, 3.2.2	NotificationJob, NotificationTemplate, NotificationContent

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Notification Formatter	NotificationTemplate	Class	Represents reusable layouts for notification messages.	2.2.1 - 2.2.5, 3.2.1 - 3.2.7	formatNotification(), NotificationContent
Notification Formatter	NotificationContent	Object	Stores the formatted title, body, link, and display data.	2.2.1 - 2.2.5, 3.2.1 - 3.2.7	NotificationTemplate, NotificationRepository
Notification Formatter	formatPushMessage()	Method	Formats content for mobile push delivery.	2.2.3, 2.2.4, 3.2.3, 3.2.4	NotificationContent, Push Delivery Component
Notification Formatter	formatInAppMessage()	Method	Formats content for in-app display.	2.2.1, 2.2.2, 3.2.1, 3.2.2	NotificationContent, Notification Controller
Notification Formatter	messageBody	Variable	Stores the final notification body text.	2.2.1, 2.2.2, 3.2.1, 3.2.2	NotificationContent

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Notification Formatter	notificationLink	Variable	Stores the destination link opened from the notification.	2.2.1 - 2.2.5, 3.2.1 - 3.2.7	NotificationContent, News Feed Manager

EI Table 1.10.5 – Notification Repository

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Notification Repository	NotificationRepository	Class	Manages stored notification records.	2.2.1-2.2.5, 3.2.1-3.2.7	Notification Orchestrator, Notification Record, Notification DB
Notification Repository	saveNotification()	Function	Stores a completed notification record.	2.2.1-2.2.5, 3.2.1-3.2.7	Notification Content, Notification Record, Notification DB
Notification Repository	getNotificationHistory()	Function	Retrieves notifications for the notification screen.	2.2.1, 2.2.2, 3.2.1, 3.2.7	Notification Record, Notification DB

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Notification Repository	NotificationRecord	Object	Represents one stored notification item.	2.2.1-2.2.5, 3.2.1-3.2.7	Notification Content, RecipientList
Notification Repository	markNotificationRead()	Method	Updates a notification when the user views it.	2.2.1, 3.2.1, 3.2.7	Notification Record, readStatus
Notification Repository	archiveNotification()	Method	Stores notification records for audit and history.	2.2.1-2.2.5, 3.2.7	Notification Record, Notification DB
Notification Repository	readStatus	Variable	Tracks whether a notification has been read.	2.2.1, 3.2.1, 3.2.7	Notification Record, markNotificationRead()
Notification Repository	NotificationDB	Data Store	Stores notification records, delivery status, and history data.	2.2.1-2.2.5, 3.2.1-3.2.7	Notification Repository, Database
Notification Repository	/notifications/history	API Endpoint	Handles requests for notification history.	2.2.1, 2.2.2, 3.2.1, 3.2.7	getNotificationHistory(), API Gateway

EI Table 1.10.6 – Recipient Resolver

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Recipient Resolver	RecipientResolver	Class	Determine s which users should receive a notification .	2.2.1 - 2.2.5 , 3.2.1 - 3.2.7	Notification Orchestrator, RecipientList
Recipient Resolver	resolveRecipients()	Function	Builds the recipient list for a workflow.	2.2.1 , 2.2.2 , 3.2.1 , 3.2.2	NotificationJob, RecipientList
Recipient Resolver	RecipientList	Object	Stores the selected notification recipients.	2.2.1 - 2.2.5 , 3.2.1 - 3.2.7	resolveRecipients(), NotificationRepository
Recipient Resolver	NotificationPreference	Class	Represent s user notification settings and filters.	2.2.5 , 3.2.6 , 3.2.7	RecipientList, filterMutedUsers()
Recipient Resolver	followedCompanyList	Variable	Stores followed companies used for recipient matching.	2.2.2 , 3.2.2 , 3.2.5	NotificationPreference, Profile Management Service

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Recipient Resolver	filterMutedUsers()	Method	Removes users who muted or filtered the notification type.	2.2.5 , 3.2.6 , 3.2.7	NotificationPreference, RecipientList
Recipient Resolver	applyRecipientRules()	Method	Applies follow, message, profile-view, and preference rules.	2.2.1 - 2.2.5 , 3.2.1 - 3.2.7	recipientRules, RecipientList
Recipient Resolver	recipientRules	Variable	Stores conditions used to match events to recipients.	2.2.1 - 2.2.5 , 3.2.1 - 3.2.7	applyRecipientRules(), NotificationPreference
Recipient Resolver	UserPreferences	Data Table	Stores notification preferences and filter settings.	2.2.5 , 3.2.6 , 3.2.7	NotificationPreference, Database

EI Table 1.10.7 — Push Delivery Component

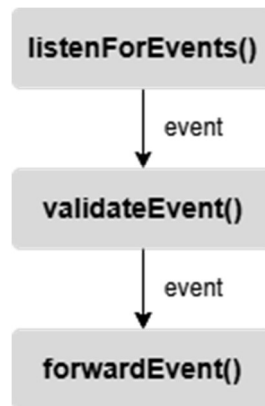
Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Push Delivery Component	PushNotification	Class	Stores and manages push notifications	2.2.1-2.2.5, 3.2.1-3.2.2	IsDelivered(), SendNotification(), ToQueue(), NotificationsAvailable(), RetrieveNotification()
Push Delivery Component	Isdelivered()	Function	Changes the attribute of delivered to true	2.2.1-2.2.5, 3.2.1-3.2.2	PushNotification, Notification Repository
Push Delivery Component	SendNotification()	Function	Sends notification	2.2.1-2.2.5, 3.2.1-3.2.2	NotificationOrchestrator, ToQueue()
Push Delivery Component	ToQueue()	Function	Queues for notifications to be sent	2.2.1-2.2.5, 3.2.1-3.2.2	SendNotification(), RetrieveNotification(), NotificationOrchestrator
Push Delivery Component	NotificationsAvailable()	Function	Make sure that notifications are enabled	2.2.1-2.2.5, 3.2.1-3.2.2	NotificationOrchestrator
Push Delivery Component	RetrieveNotification()	Function	Retrieves the notification to queue it up	2.2.1-2.2.5, 3.2.1-3.2.2	ToQueue(), NotificationRepository

EI Table 1.10.8 — News Feed Manager

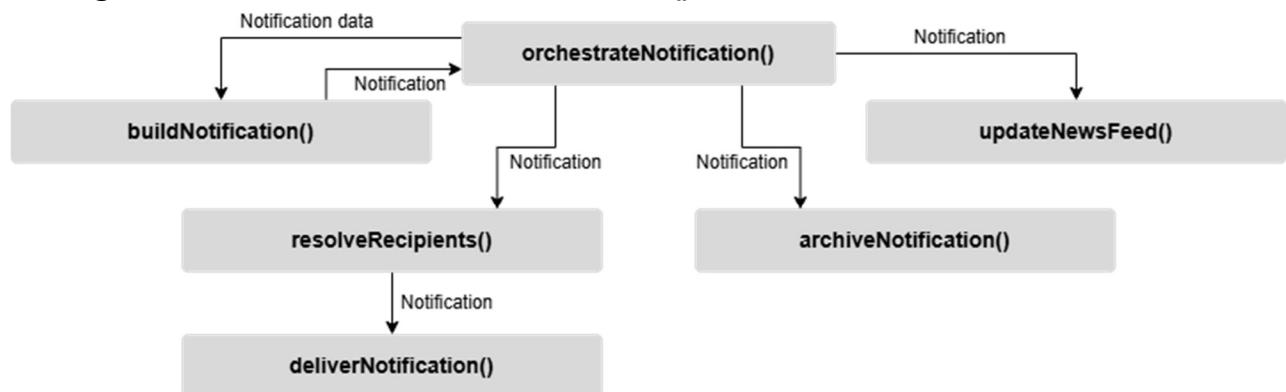
Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Profile Orchestrator	CheckForNotifications()	Function	Checks for anything to display	2.2 , 2.4 , 3.2 , 3.4	Notification Orchestrator, Notification Repository
Resume Orchestrator	FilterBy()	Function	Filters Newsfeed	2.2 , 2.4 , 3.2 , 3.4	Notification Repository, Notification Orchestrator
Job Orchestrator	Display()	Function	Displays Notifications/postings	2.2 , 2.4 , 3.2 , 3.4	Notification Orchestrator, Notification Repository
Search Orchestrator	HasBeenRead()	Function	Marks if a posting has been seen by the user	2.2 , 2.4 , 3.2 , 3.4	Display(), Notification Orchestrator, Notification Repository

Structure Charts - Communication Service

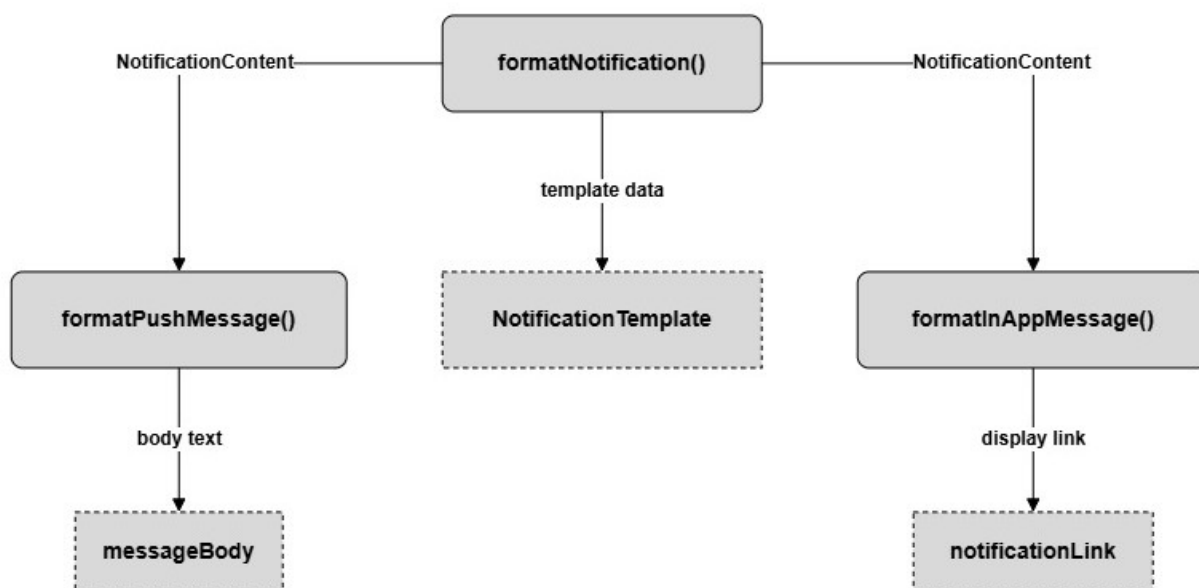
SC Figure 1.10.0.1 – listenForEvents()



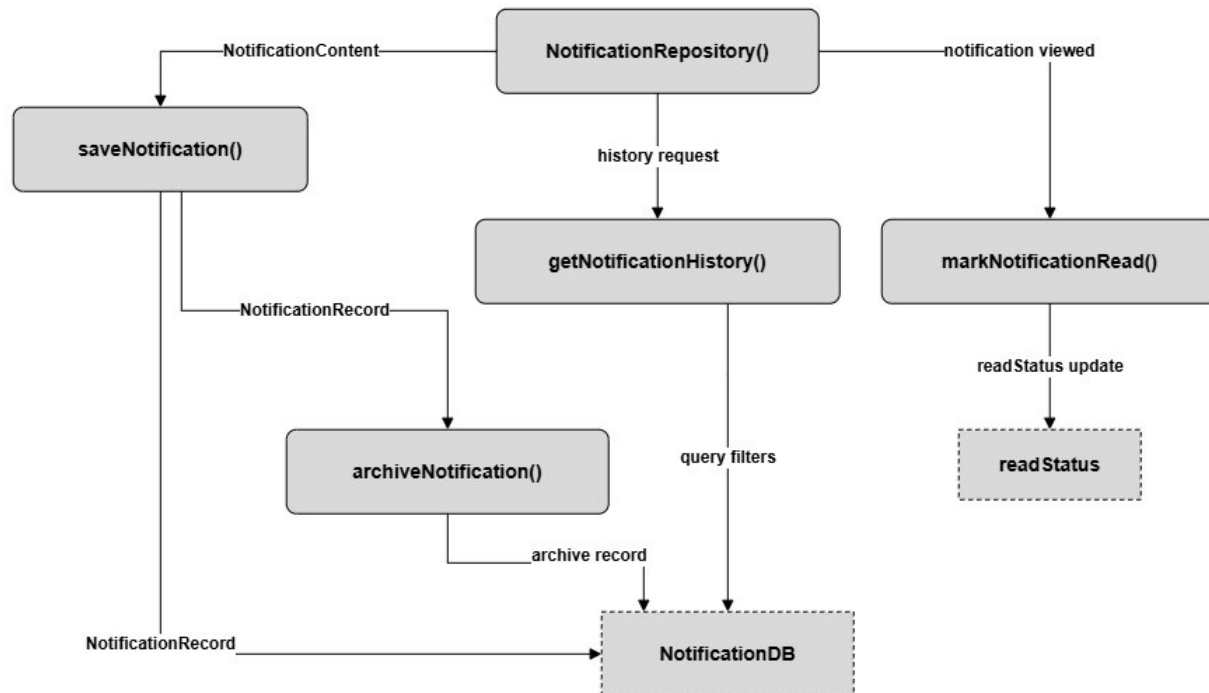
SC Figure 1.10.3.1 – orchestrateNotification()



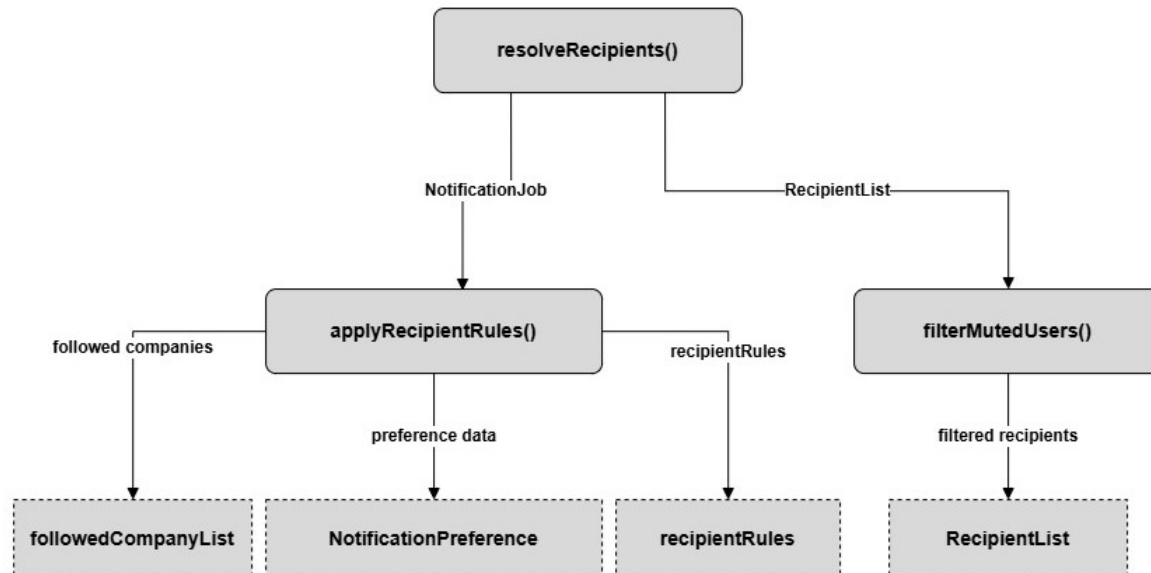
SC Figure 1.10.4.1 – formatNotification()



SC Figure 1.10.5.1 – notificationRepository()



SC Figure 1.10.6.1 – resolveRecipient()



Data Flow Diagrams - Communication Service

DFD Figure 1.10.0.0.1 – Event Listener



DFD Figure 1.10.0.0.2 – Event Source

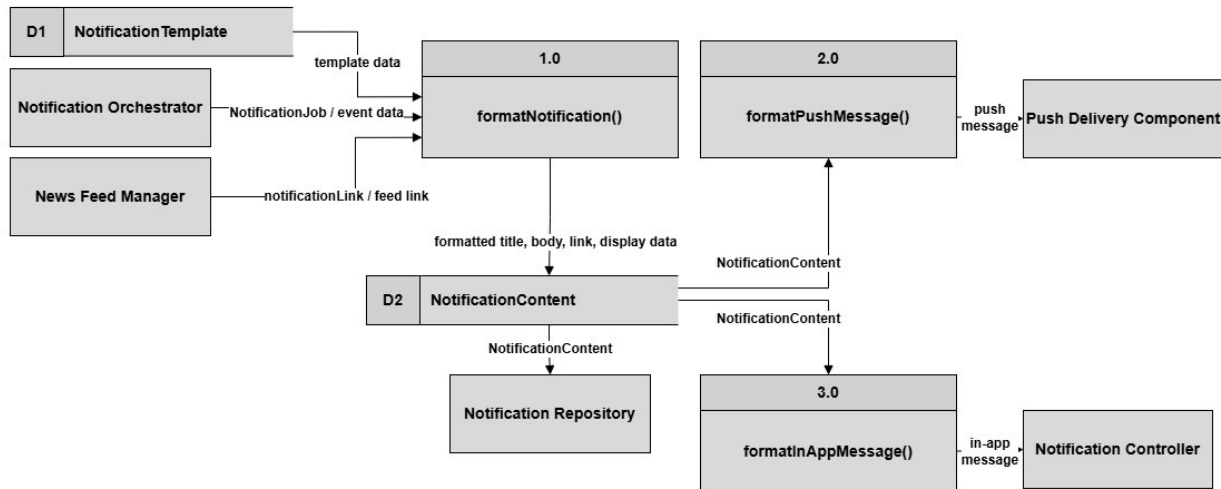


DFD Figure 1.10.0.0.3 – Notification Controller

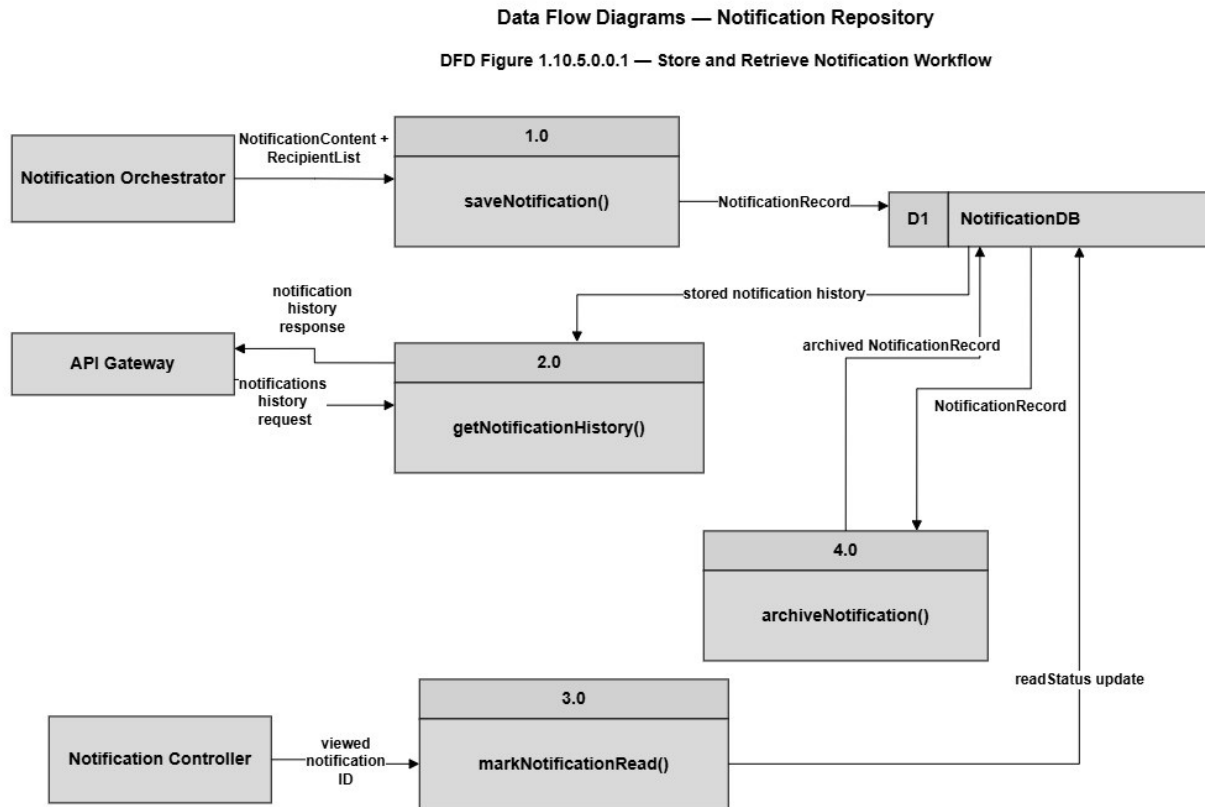


DFD Figure 1.10.4.0.0.1

Data Flow Diagrams — Notification Formatter
DFD Figure 1.10.4.0.0.1 — Format Notification Workflow



DFD Figure 1.10.5.0.0.1



DFD Figure 1.10.6.0.0.1

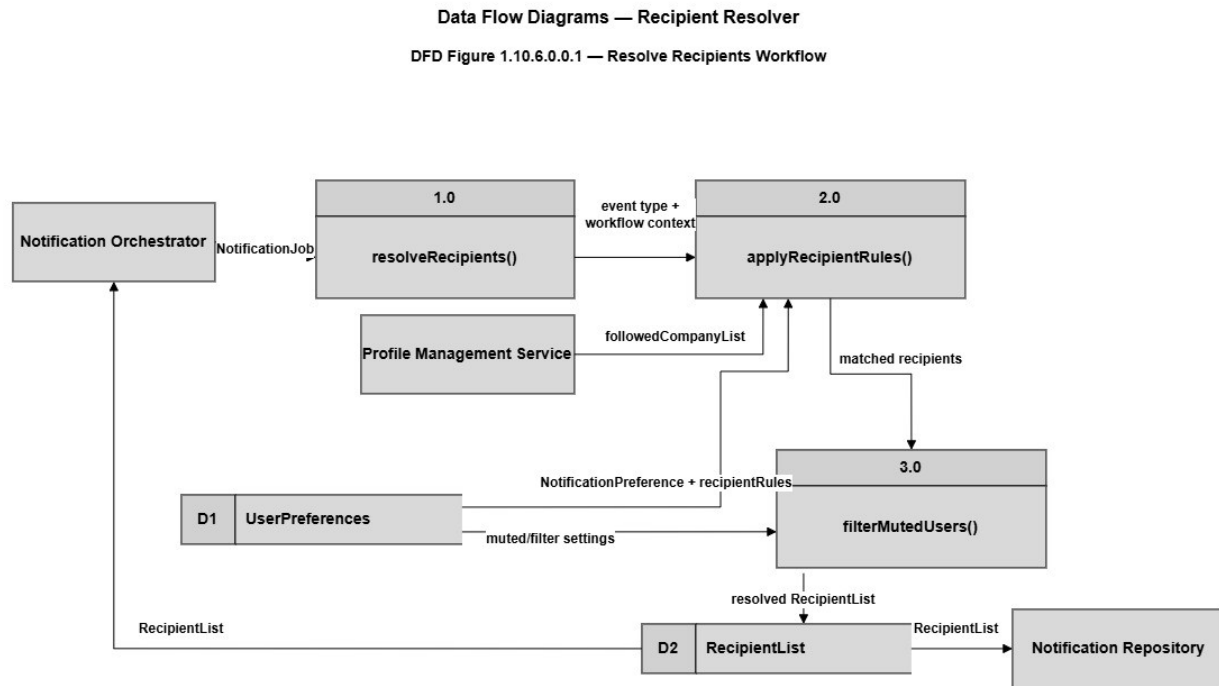
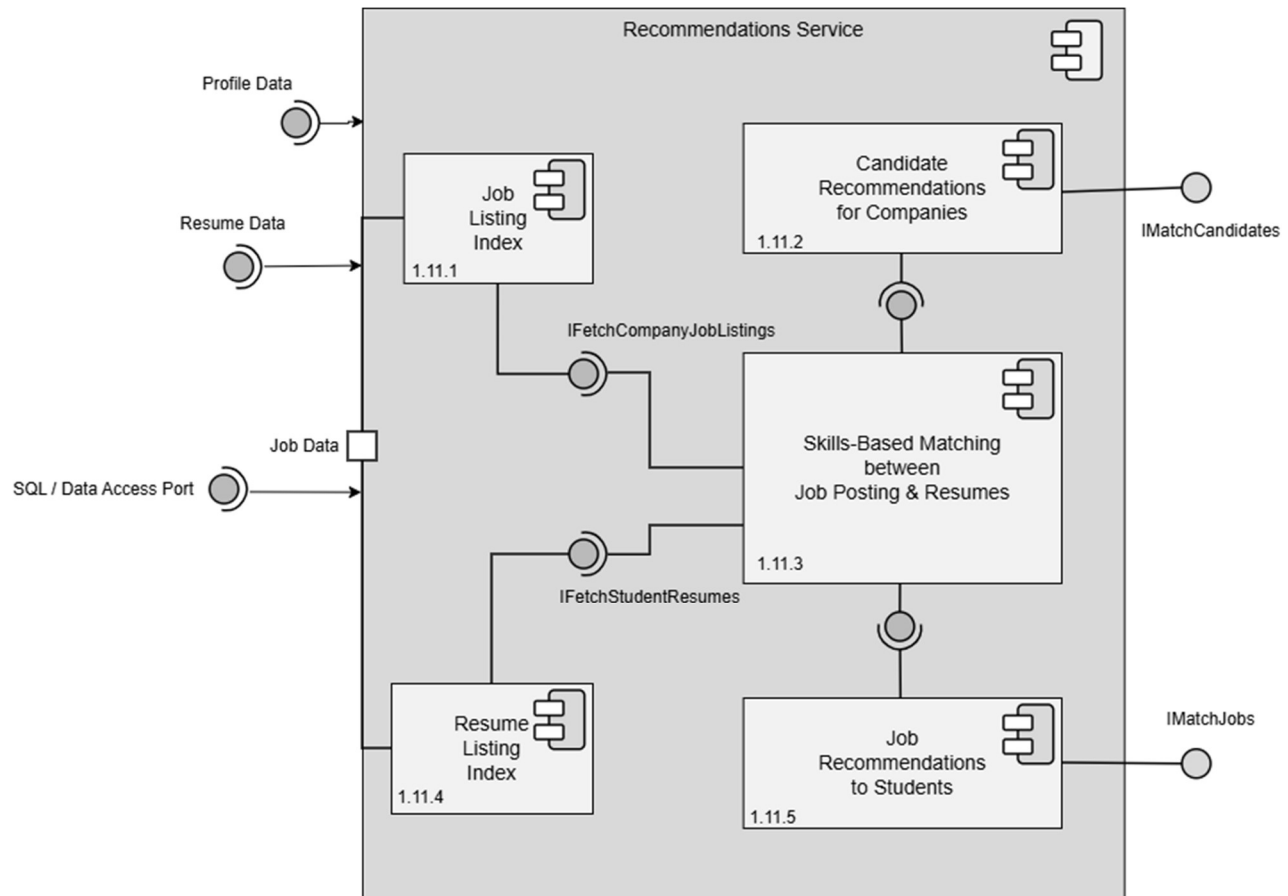


Figure 1.11 — Recommendation Service Detail**Design Information – Recommendation Service**

Field	Description / Purpose
View Name & Number	1.11 Recommendation Service
Purpose	Describe the recommendation service.
Description	Basic service demonstrating the recommendation service.
Requirements	Implicit in SRS 2.3 Search (2.3.7 Ranking), 2.1.3.8 (Search), 3.3 (Student Search); supports job and candidate recommendations
Elements	1.11.1 Job Posting Index
	1.11.2 Candidate Recommendations
	1.11.3 Skills-Based Matching
	1.11.4 Resume Listing Index
	1.11.5 Job Recommendations to Students
Navigation /	1.4 API Gateway / Orchestrator, 1.6 Profile Management Service, 1.7

Field	Description / Purpose
References	Resume Service, 1.8 Job Posting Service, 1.14 Database
Viewpoint	Component Diagram

Entity Identification Tables – Recommendation Service

El Table 1.11.1 – Job Listing Index

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Recommendation Service	JobIndex	Data Store	Stores the indexed list of all active job postings for fast recommendation retrieval	2.5.9	IndexJob()
Recommendation Service	IndexJob()	Function	Adds a newly created or updated job posting into the searchable index	2.5.1, 2.5.9	JobIndex

El Table 1.11.2 – Candidate Recommendations for Companies

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Recommendation Service	CandidateRecommender	Module	Generates a ranked list of student profiles for a specific company based on requirement	2.3.1, 2.3.3	recommendCandidates()
Recommendation Service	recommendCandidates()	Function	Processes company search filters to suggest matching student resumes	2.3.2, 2.3.3	CandidateRecommender, ResumeIndex

EI Table 1.11.3 – Skills-Based Matching Between Job Posting & Resumes

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Recommendation Service	SkillMatcher	Class	Compares the requires skills of a job to the skills listed on a student's resume	2.5.2, 3.1.3.1	calculateMatchScore()
Recommendation Service	calculateMatchScore()	Method	Computes a compatibility percentage based on shared skills	2.3.3, 3.3.3	Skill Matcher

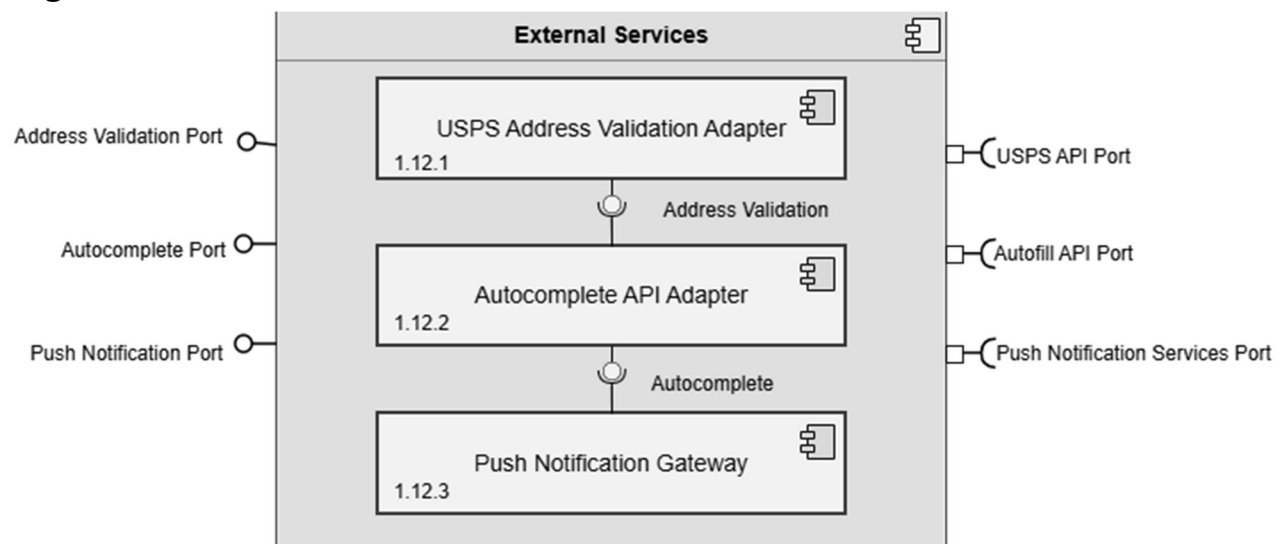
Collaborators / Related Entities	Related Requirement(s)	Description / Purpose	Entity Type	Entity Name	Subcomponent
		between the job and resume			

EI Table 1.11.4 – Resume Listing Index

Collaborators / Related Entities	Related Requirement(s)	Description / Purpose	Entity Type	Entity Name	Subcomponent
indexResume()	3.5.1, 3.5.3	Maintains a highly optimized database of student resumes for recommendation operations	Data Store	ResumeIndex	Recommendation Service
ResumeIndex	3.5.2, 3.5.5.1	Updates the searchable index whenever a student modifies or uploads their resume.	Function	indexResume()	Recommendation Service

EI Table 1.11.5 – Job Recommendations to Students

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Recommendation Service	JobRecommender	Module	Generates a list of suggested job postings tailored to a student's specific profile criteria	3.3.3	recommendJobs()
Recommendation Service	recommendJobs()	Function	Analyzes a student's skills and degree to recommend relevant job postings to their dashboard	3.3.3, 3.3.4	JobRecommender, JobIndex

Figure 1.12 — External Services Detail**Design Information – External Services**

Field	Description / Purpose
View Name & Number	1.12 External Services
Purpose	Describe the external services.
Description	Basic service demonstrating the external services.
Requirements	SRS 2.1.3.9 (USPS address validation), 2.3.6 (Autocomplete API), 2.2.3–2.2.4 (Push notification delivery)
Elements	1.12.1 USPS Validation
	1.12.2 Autocomplete API Adapter
	1.12.3 Push Notifications Gateway
Navigation / References	1.4 API Gateway / Orchestrator
Viewpoint	Component Diagram

Entity Identification Tables – External Services**EI Table 1.12.1 – USPS Validation Adapter**

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
USPS Address Validation Adapter	validateAddress()	Function	Validates a company location against USPS address standards before saving.	2.1.3.7 , 2.1.3.9	USPS API, CompanyLocation
USPS Address Validation Adapter	AddressRequest	Object	Stores address information submitted by a company representative.	2.1.3.5 , 2.1.3.7	CompanyLocationForm, validateAddress()
USPS Address Validation Adapter	ValidatedAddress	Object	Stores the USPS-standardized address returned from validation.	2.1.3.9	USPS API, CompanyLocation
USPS Address Validation Adapter	addressFields	Variable	Stores street, suite/building number, city, state, country, and ZIP code values.	2.1.3.7	AddressRequest
USPS Address Validation Adapter	CompanyLocationData	Data Table	Stores validated company location records.	2.1.3.3 , 2.1.3.4 , 2.1.3.5	Database, CompanyLocation
USPS Address Validation Adapter	/api/address/validate	API Endpoint	Receives address validation requests and returns USPS-standardized addresses.	2.1.3.7 , 2.1.3.9	validateAddress(), ValidatedAddress

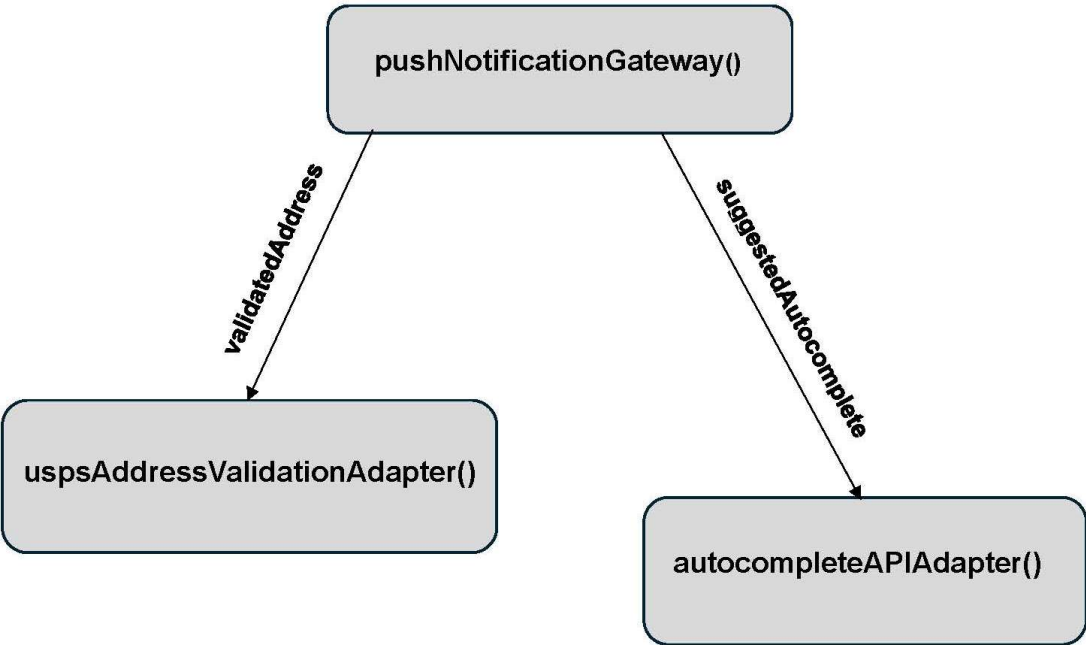
EI Table 1.12.3 – Push Notification Gateway

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
External Services Orchestrator	pushNotificationGateway()	Function	Manages and securely delivers app notifications by standardizing data, routing messages to each user's device, and temporarily storing notifications for offline users.	1.12.1–1.12.3	USPSAddressValidationAdapter, AutocompleteAPIAdapter
External Services Orchestrator	USPSAddressValidationAdapter	Module	Validates and standardizes user addresses through USPS services.	1.12.1–1.12.3	pushNotificationGateway()
External Services Orchestrator	AutocompleteAPIAdapter	Module	Provides address and location autocomplete suggestions	1.12.1–1.12.3	pushNotificationGateway()

			through external APIs.		
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Structure Charts – External Services

SC Figure 1.12.0.3 — pushNotificationGateway()



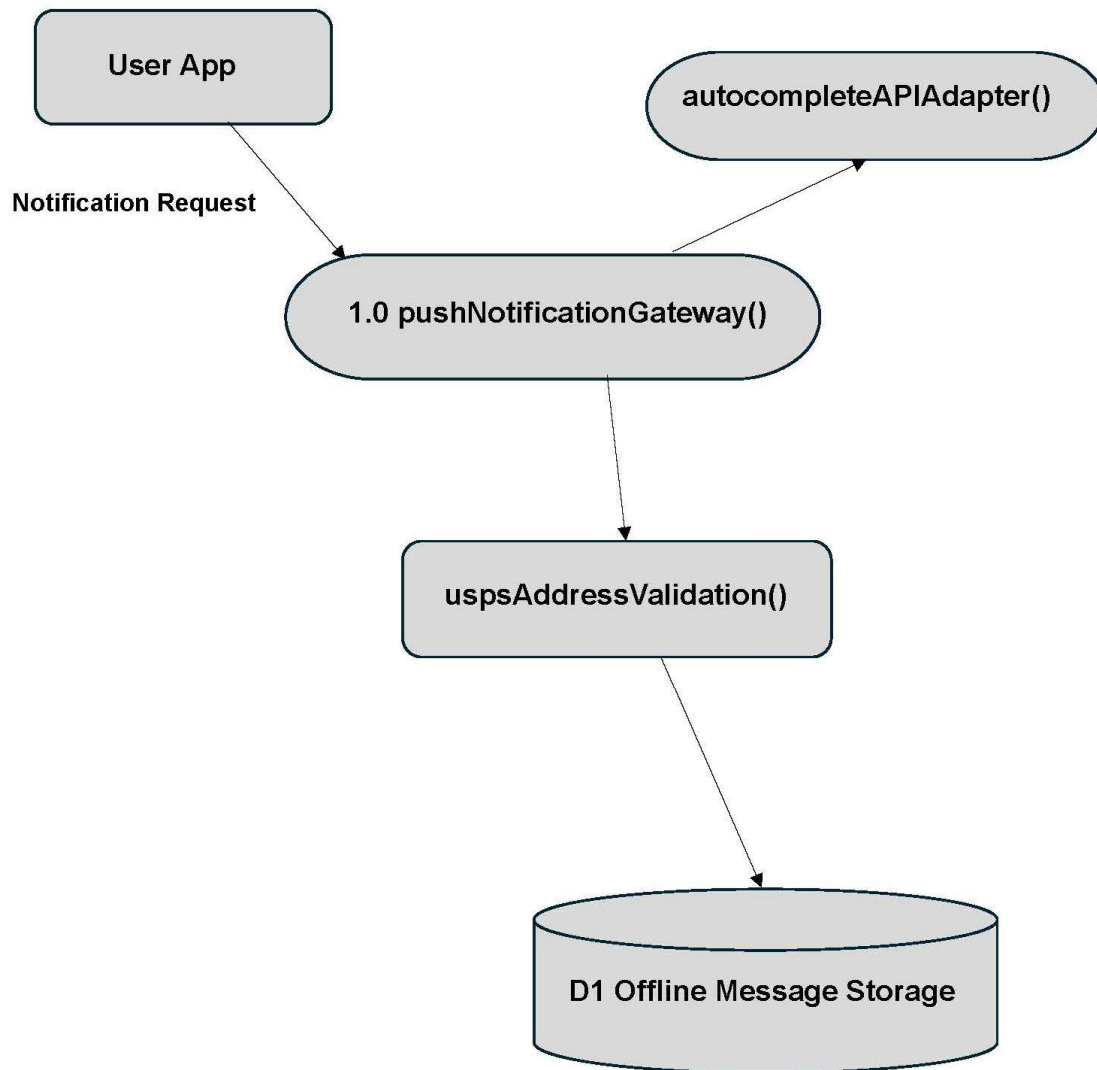
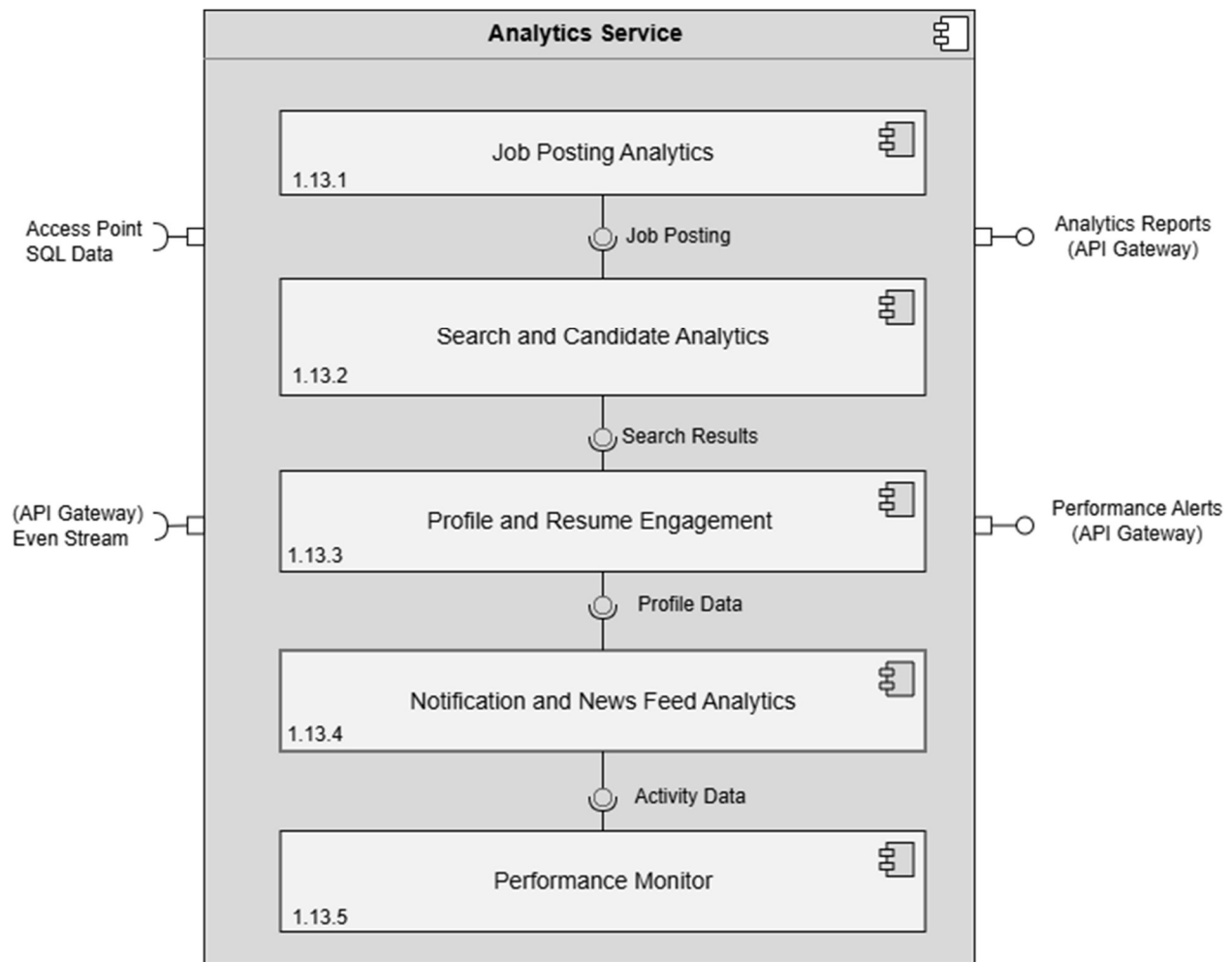
Data Flow Diagram – External Services**DFD Figure 1.12.0.0.3 — Push Notification Gateway****DFD Figure 1.12.0.0.3 — Push Notification Gateway**

Figure 1.13 — Analytics Service Detail**Design Information – Analytics Service**

Field	Description / Purpose
View Name & Number	1.13 Analytics Service
Purpose	Describe the analytics service.
Description	Basic service demonstrating the analytics service.
Requirements	SRS 2.1.3.6 (Company analytics), 2.5.7 (Job posting status/performance tracking)
Elements	1.13.1 Job Posting Analytics
	1.13.2 Search and Candidate Analytics
	1.13.3 Profile and Resume Engagement
	1.13.4 Notifications and News Feed Analytics
	1.13.5 Performance Monitor
Navigation / References	1.4 API Gateway / Orchestrator, 1.14 Database
Viewpoint	Component Diagram

Entity Identification Tables – Analytics Service Detail

El Table 1.13.1 – Job Posting Analytics

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Related Collaborators / Entities
Job Posting Manager	manageJobPosting()	Function	Used so a company representative can add, edit, or delete a job posting.	2.5.1	addJobPosting(), editJobPosting(), deleteJobPosting(), jobPostingId, JobPosting, /api/job-postings
Job Posting Manager	addJobPosting()	Function	Used so a company representative can add/save a job posting to the database, including adding a list of required skills and free-form job descriptions	2.5.1, 2.5.7	manageJobPosting(), jobPostingData, JobPosting, /api/job-postings
Job Posting Manager	editJobPosting()	Function	Used so a company representative can edit/save a job posting to the database, including editing a list of required skills and free-form job descriptions	2.5.1, 2.5.7	manageJobPosting(), jobPostingData, jobPostingId, JobPosting, /api/job-postings
Job Posting Manager	deleteJobPosting()	Function	Used so a company representative can delete a job posting from the	2.5.1, 2.5.7	manageJobPosting(), jobPostingId, JobPosting, /api/job-postings

			database, including deleting a list of required skills and free-form job descriptions		
Job Posting Manager	jobPostingData	Variable	Stores the job posting data to be stored in the JobPosting data table	2.5.1, 2.5.7, 2.5.9	manageJobPosting(), JobPosting, addJobPosting(), editJobPosting()
Job Posting Manager	jobPostingId	Variable	Stores the job posting ID to be used to edit or delete a job posting	2.5.1	manageJobPosting(), editJobPosting(), deleteJobPosting(), JobPosting
Job Posting Manager	JobPosting	Data Table	Used so a company representative can store a job posting to the database, including a list of required skills and free-form job descriptions	2.5.1-2.5.2, 2.5.7, 2.5.9	addJobPosting(), editJobPosting(), deleteJobPosting(), jobPostingData, jobPostingId, /api/job-postings
Job Posting Manager	job_posting_skill_ids	Variable	Stores a list of skill IDs for a job posting	2.5.2	RequiredJobSkills Dictionary, saveRequiredSkills(), addRequiredSkill(), editRequiredSkill(), deleteRequiredSkill(), /api/job-postings/required-skills
Job Posting Manager	RequiredJobSkillsDictionary	Data Table	A normalized data table dictionary used to retrieve the text for required skills for a job posting by skill ID and allows a skill to	2.5.2, 2.5.5	job_posting_skill_ids, saveRequiredSkills(), addRequiredSkill(), editRequiredSkill(), deleteRequiredSki

			accommodate any text length		ll(), /api/job-postings/required-skills
Job Posting Manager	saveRequiredSkills()	Function	Used so a company representative may save required job skills to a job posting by checking checkboxes	2.5.3	job_posting_skill_ids, RequiredJobSkills Dictionary, /api/job-postings/required-skills
Job Posting Manager	addRequiredSkill()	Function	Used so a company representative can add/save a new required skill to a job posting	2.5.4	job_posting_skill_ids, RequiredJobSkills Dictionary, /api/job-postings/required-skills
Job Posting Manager	editRequiredSkill()	Function	Used so a company representative can edit an existing required skill on a job posting	2.5.4	job_posting_skill_ids, RequiredJobSkills Dictionary, /api/job-postings/required-skills
Job Posting Manager	deleteRequiredSkill()	Function	Used so a company representative can delete a required skill from a job posting	2.5.4	job_posting_skill_ids, RequiredJobSkills Dictionary, /api/job-postings/required-skills
Job Posting Manager	ResumeSubmissions	Data Table	Stores all resume form submissions for a job posting	2.5.6	getResumeSubmissions(), DisplayResumeSubmissions
Job Posting Manager	getResumeSubmissions()	Function	Fetches all resume form submissions for a job posting and accepts parameters to sort and filter candidates	2.5.6, 2.5.8	ResumeSubmissions, DisplayResumeSubmissions, SortCandidates
Job	DisplayResume	Class	Displays all	2.5.6	ResumeSubmissions

Posting Manager	Submissions		resume form submissions for a job posting		ons, getResumeSubmissions()
Job Posting Manager	SortCandidates	Class	Used so a company representative may sort candidates using filters	2.5.8	getResumeSubmissions()
Job Posting Manager	/api/job-postings/required-skills	API Endpoint	Handles incoming API HTTP requests POST, PUT, DELETE for required skills	2.5.2-2.5.5	job_posting_skill_ids, RequiredJobSkills Dictionary, saveRequiredSkills(), addRequiredSkill(), editRequiredSkill(), deleteRequiredSkill()
Job Posting Manager	/api/job-postings	API Endpoint	Handles incoming API HTTP requests: POST, PUT, DELETE for job postings	2.5.1, 2.5.7, 2.5.9	JobPosting, addJobPosting(), editJobPosting(), deleteJobPosting()

EI Table 1.13.2 – Search and Candidate Analytics

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Related Collaborators / Entities
Candidate Search Manager	searchCandidates()	Function	Pulls a list of all the candidates matching a user's selected criteria	2.3.1-2.3.8	activeSearchFilters, candidateSearchResults, searchQuery,

					parsedQuery, /api/search/candidates
Candidate Search Manager	CandidateSearchFilter	Class	Filters candidates by the filter the user selects (skills, location, work preference, willingness to relocate, position type, start date availability, years of experience, degree level, graduation date, application source, GPA)	2.3.2-2.3.4, 2.3.8	activeSearchFilters
Candidate Search Manager	activeSearchFilters	Variable	Stores all of the active filters the user selects	2.3.2-2.3.4, 2.3.8	searchCandidates(), CandidateSearchFilter, /api/search/candidates
Candidate Search Manager	candidateSearchResults	Variable	Stores all the candidate profiles that matches with the selected filters the user selects	2.3.2, 2.3.3, 2.3.5	searchCandidates(), displayFilteredCandidates()
Candidate Search Manager	displayFilteredCandidates()	Function	Displays filtered candidate profiles and their resumes within 3 seconds of search request submission	2.3.5	candidateSearchResults
Candidate Search Manager	autoFillSearchBar()	Function	Main function that auto fills search bar by fetching suggestions from what the	2.3.6	fetchAutoFillSuggestions(), searchInput, autoFillSuggestions, AutoFillAPI

			user inputs		
Candidate Search Manager	fetchAutoFillSuggestions()	Function	Calls the Auto Fill API to retrieve suggestions from what the user inputted	2.3.6	autoFillSearchBar(), searchInput, autoFillSuggestions, AutoFillAPI
Candidate Search Manager	searchInput	Variable	Stores the input the user enters in the search bar	2.3.6	autoFillSearchBar(), fetchAutoFillSuggestions(), autoFillSuggestions
Candidate Search Manager	autoFillSuggestions	Variable	Stores the suggestions retrieved from the AutoFillAPI so it can be displayed to the user	2.3.6	autoFillSearchBar(), fetchAutoFillSuggestions(), searchInput, autoFillAPI
Candidate Search Manager	AutoFillAPI	Module	Fetches words from a normalized dictionary as a user types to suggest autocomplete words	2.3.6	autoFillSearchBar(), fetchAutoFillSuggestions(), autoFillSuggestions
Candidate Search Manager	searchQuery	Variable	Stores the search criteria the user entered	2.3.7	searchCandidates(), parseWildcardSearches()
Candidate Search Manager	parsedQuery	Variable	Stores the search criteria with wildcard symbols converted to RegEx	2.3.7	searchCandidates(), parseWildcardSearches()
Candidate Search Manager	parseWildcardSearches()	Function	Converts wildcard symbols such as * or ? into queries using Regex to accommodate flexible	2.3.7	/api/search/candidates, searchQuery, parsedQuery

			searching		
Candidate Search Manager	/api/search/candidates	API Endpoint	Handles incoming HTTP requests for candidate search based on matches, wildcards, and filters	2.3.1–2.3.8	searchCandidates(), activeSearchFilters, parseWildcardSearches()

EI Table 1.13.3 – Profile and Resume Engagement

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Profile and Resume Engagement	viewStudentProfile()	Function	Allows a company representative to view a student's profile information.	3.1.3.1, 3.1.3.4	StudentProfile, Resume
Profile and Resume Engagement	StudentProfile	Object	Stores student profile information including contact information, GPA, portfolio, resume, and skills.	3.1.3.1	Resume, ProfileView
Profile and Resume Engagement	Resume	Object	Stores resume information submitted by a student user.	3.5.1, 3.5.2	StudentProfile
Profile and Resume Engagement	followStudent()	Function	Allows a company representative to follow a	2.1.1	StudentProfile

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
			student user.		
Profile and Resume Engagement	ProfileView	Object	Records when a company representative views a student profile.	3.2.5	StudentProfile, ProfileViewNotification
Profile and Resume Engagement	ResumeMatch	Object	Represents a resume that satisfies company skill requirements	2.2.5, 2.5.2	validateAddress(), ValidatedAddress
Profile and Resume Engagement	ProfileViewNotification	Object	Stores notification data when a profile view triggers a student notification	3.2.5	ProfileView
Profile and Resume Engagement	ResumeRepository	Data Table	Stores submitted resumes and associated user tags.	3.1.5	Resume
Profile and Resume Engagement	StudentProfileRepository	Data Table	Stores student profile information available to company representatives.	3.1.3.1	StudentProfile
Profile and Resume Engagement	/api/profile/view	API Endpoint	Endpoint Handles requests to retrieve student profile information.	3.1.3.4	viewStudentProfile(), StudentProfile

EI Table 1.13.4 – Notification and News Feedback Analytics

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Analytic s Services Orchestrator	notificationAndNewsFeedbackAnalytics()	Function	Monitors , collects, and analyzes user interactions and reactions to app notifications and news feeds.	1.13.1–1.13.4	JobPostingAnalytics, SearchAndCandidateAnalytics, ProfileAndResumeEngagement
Analytic s Services Orchestrator	JobPostingAnalytics	Module	Collects and analyzes job posting engagement metrics.	1.13.1–1.13.4	notificationAndNewsFeedbackAnalytics()
Analytic s Services Orchestrator	SearchAndCandidateAnalytics	Module	Analyzes search activity and candidate interaction metrics.	1.13.1–1.13.4	notificationAndNewsFeedbackAnalytics()

Analytics Services Orchestrator	ProfileAndResumeEngagement	Module	Tracks profile views, resume interactions, and engagement trends.	1.13.1–1.13.4	notificationAndNewsFeedbackAnalytics()
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EI Table 1.13.5 – Performance Monitor

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Related Collaborators / Entities
Performance Monitor	PerformanceMonitor	Class	Consumes activity data and event streams to track system performance and produce alerts and reports for the API gateway.	4.1, 2.5.7	APIGateway, Database, JobPostingAnalytics, SearchAndCandidateAnalytics, ProfileAndResumeEngagement, NotificationAndNewsFeedAnalytics
Performance Monitor	ActivityData	Object	Inbound data objects passed from upstream analytics component containing system activity events to be monitored.	2.5.7	PerformanceMonitor, NotificationAndNewsFeedAnalytics
Performance Monitor	PerformanceAlert	Object	Outbound object produced when a monitored metric exceeds an acceptable threshold, sent	4.1	PerformanceMonitor, APIGateway

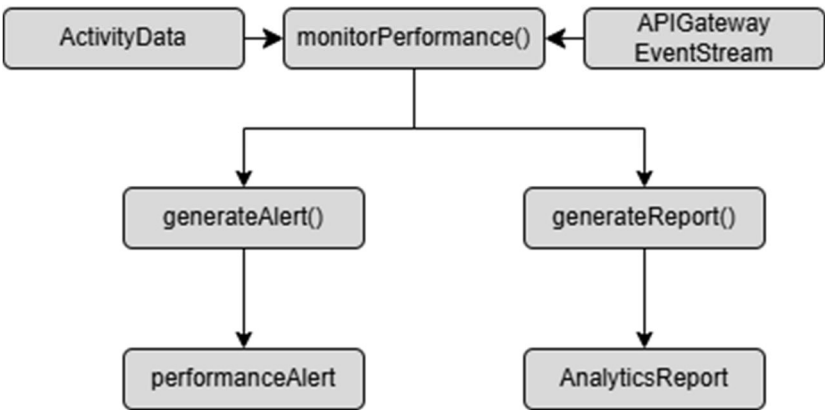
			to the API Gateway.		
Performance Monitor	AnalyticReport	Object	Outbound object containing aggregated performance data delivered to the API Gateway for display or storage.	2.5.7	PerformanceMonitor, APIGateway
Performance Monitor	monitorPerformance()	Method	Continuously evaluates incoming activity data and event stream metrics against performance thresholds.	4.1, 2.5.7	ActivityData, APIGatewayEventStream
Performance Monitor	generateAlert()	Method	Produces and dispatches a PerformanceAlert to the API Gateway when a metric breach is detected.	4.1	PerformanceAlert, APIGateway
Performance Monitor	generateReport()	Method	Aggregates monitored metrics into an AnalyticsReport and sends it to the API Gateway.	2.5.7	AnalyticsReport, APIGateway
Performance Monitor	APIGatewayEventStream	Data Store	Inbound stream of system events from the API Gateway consumed by the Performance Monitor as a data source.	4.1, 2.5.7	PerformanceMonitor, APIGateway
Performance Monitor	AccessPointSQLData	Data Store	SQL database access point supplying persistent system data	4.1, 2.5.7	PerformanceMonitor, Database

			consumed by the Performance Monitor.		
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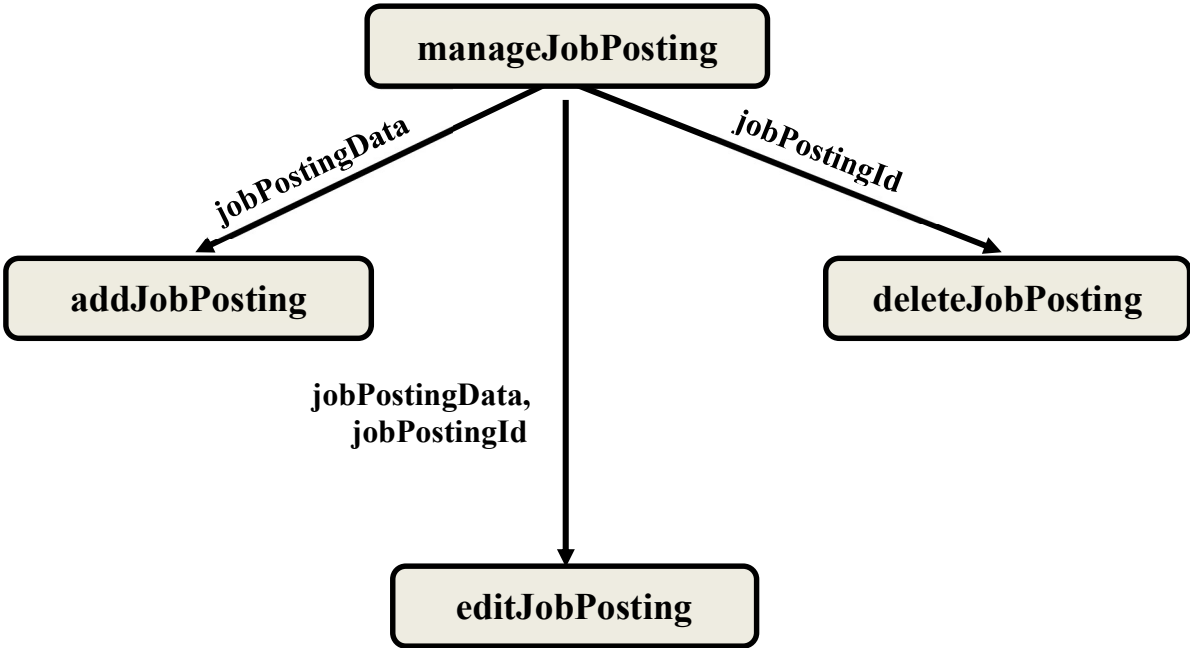
Structure Charts - Analytics Service

SC Figure 1.13.0.5 – Performance Monitor

SC 1.13.0.0.5 — Performance Monitor



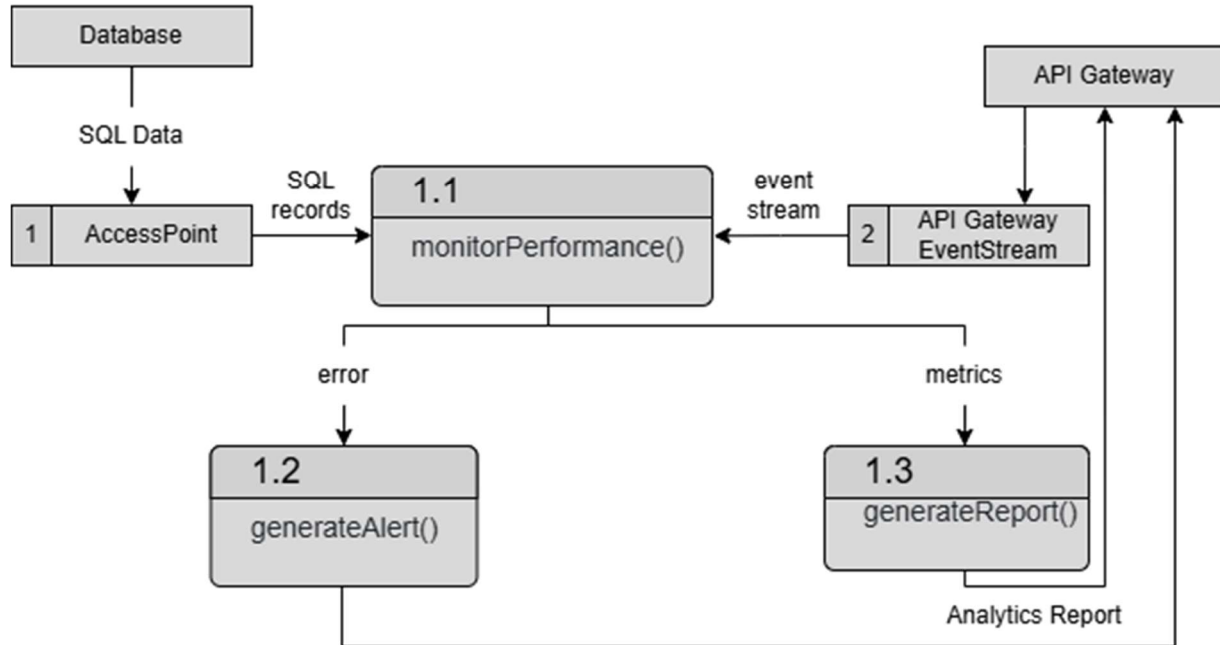
SC Figure 1.13.1.0.1 — manageJobPosting()



Data Flow Diagrams - Analytics Service

DFD Figure 1.13.0.0.5 – Performance Monitor

DFD 1.13.0.0.5 — Performance Monitor



DFD Figure 1.13.2.0.0.1 — Candidate Search Workflow

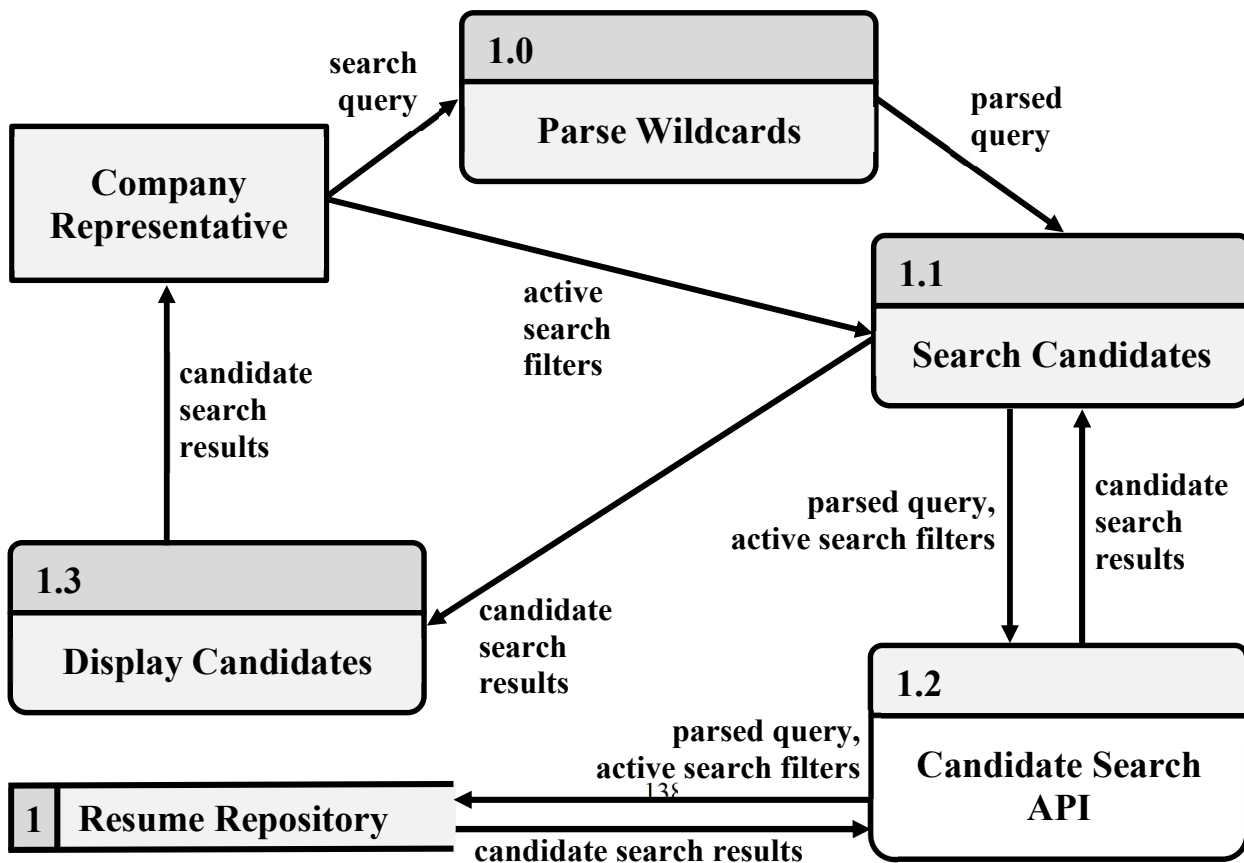
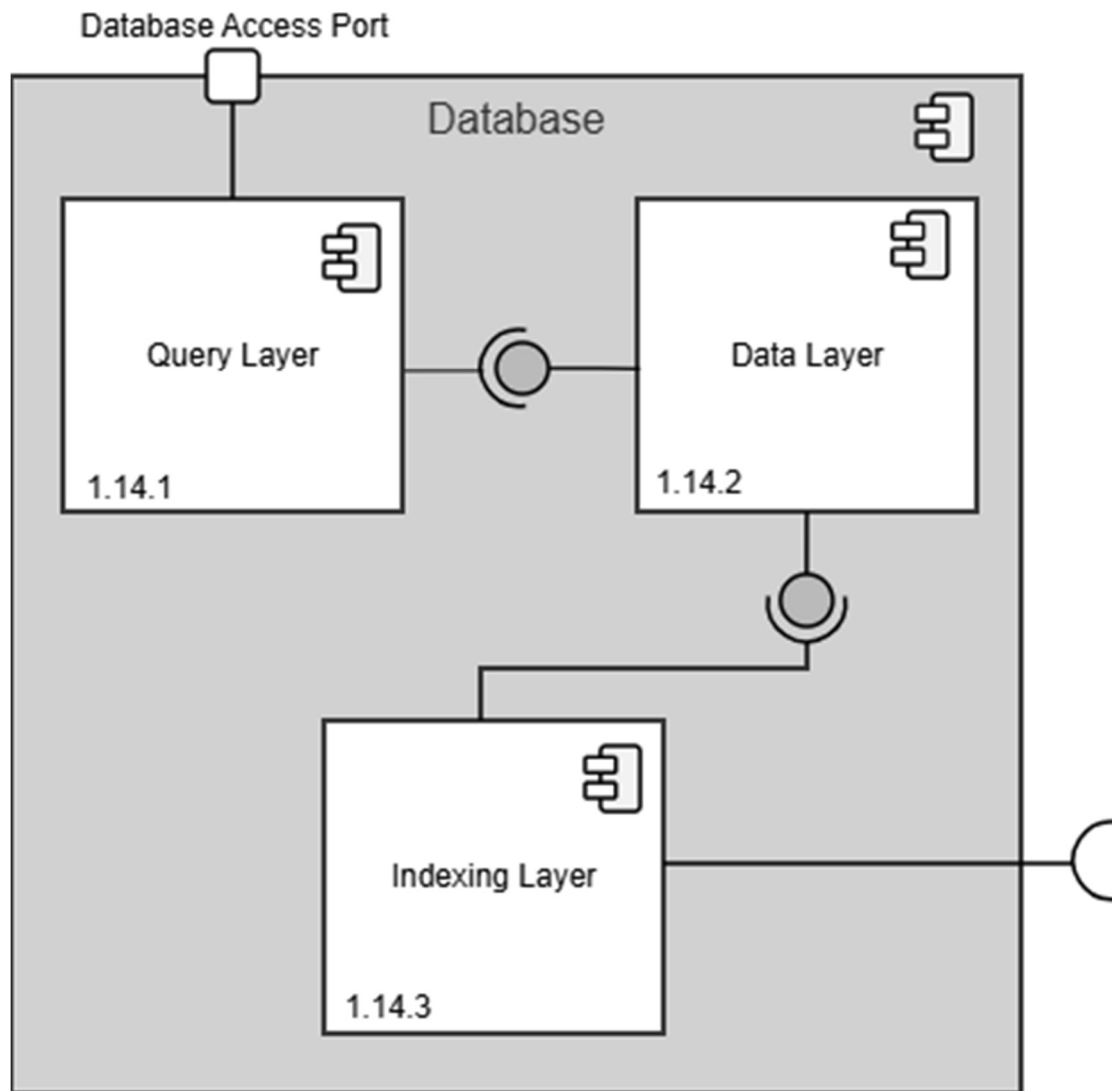


Figure 1.14 — Database Detail**Design Information – Design Information**

Field	Description / Purpose
View Name & Number	1.14 Database
Purpose	Describe the database.
Description	Basic service demonstrating the database.
Requirements	All SRS sections — provides persistent storage for all services. supports 2.1, 2.2, 2.3, 2.4, 2.5, 3.1, 3.2, 3.3, 3.4, 3.5
Elements	1.14.1 Query Layer
	1.14.2 Data Layer
	1.14.3 Indexing Layer
Navigation / References	1.5–1.13 (all service detail views)/ part of Data Storage Layer (Figure 1.1)
Viewpoint	Component Diagram

Entity Identification Tables – Database

EI Table 1.14.1 – Query Layer

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Query Layer	QueryProcessor	Class	Coordinates execution of database queries	3.1.3, 2.5.9	
Query Layer	SQLParser	Class	Converts SQLStatements into executable search queries	3.1.3, 2.5.9	QueryProcessor, ExecutionPlan
Query Layer	QueryOptimizer	Class	Determines the most efficient query execution strategy	3.1.3, 2.5.9	SQLParser, ExecutionPlan
Query Layer	ExecutionPlan	Object	Stores the generated plan used to execute a query	3.1.3, 2.5.9	QueryOptimizer, QueryExecutor
Query Layer	ResultSet	Object	Contains records returned from a completed query	3.1.3, 2.5.9	QueryExecutor
Query Layer	ExecuteQuery()	Method	Processes and executes an incoming query request	3.1.3, 2.5.9	QueryProcessor
Query Layer	FetchResults()	Method	Retrieves the query results for the service	3.1.3, 2.5.9	ResultSet
Query Layer	ParseSQL()	Method	Parse SQL syntax into internal structures	3.1.3, 2.5.9	SQLParser
Query Layer	OptimizeQuery()	Method	Generates an optimized execution plan	3.1.3, 2.5.9	QueryOptimizer

EI Table 1.14.2 – Data Layer

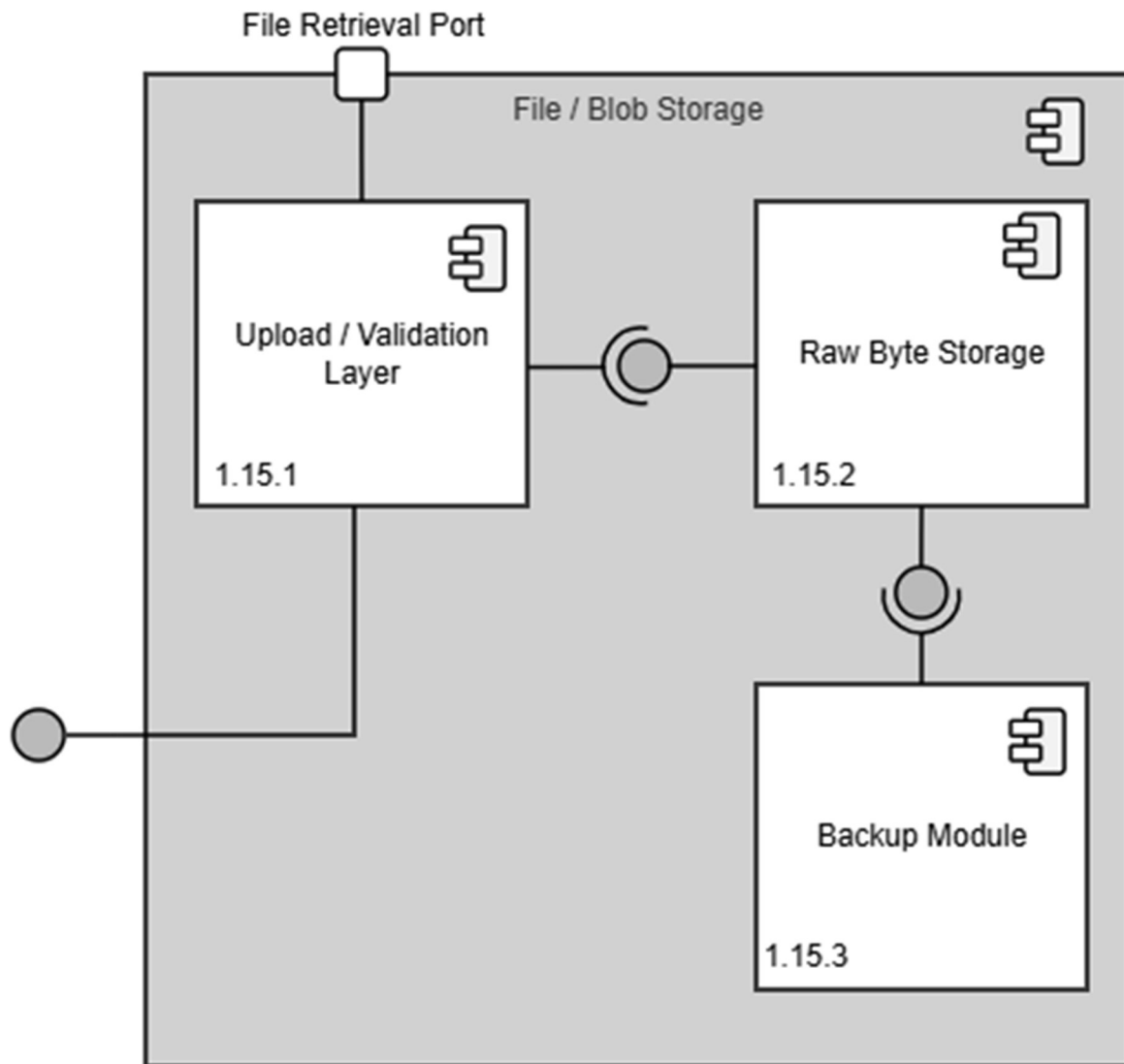
Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Data Layer	UserProfile	Data Table	Stores persistent student and company user profile data accessible by the system	3.1.3.1, 2.1.3	ResumeRecord, JobPosting, FollowRelation
Data Layer	ResumeRecord	Data Table	Stores structured resume form submissions linked to a student user tag	3.5.1, 3.5.2, 3.1.3.2	UserProfile, SkillTag, JobPosting
Data Layer	JobPosting	Data Table	Stores persistent job posting data created by company representatives	2.5.1, 2.5.9, 3.4.1	UserProfile, RequiredSkillsList, ApplicationRecord
Data Layer	RequiredSkillsList	Data Store	Stores the list of required skills associated with a specific job posting	2.5.2, 2.5.3, 2.5.4, 2.5.5	JobPosting, SkillTag
Data Layer	SkillTag	Object	Represents a single keyword skill tag attached to a resume or job posting	2.2.5, 2.3.3	ResumeRecord, RequiredSkillsList
Data Layer	FollowRelation	Data Table	Stores follower-followee relationships between student and company users	2.1.1, 2.2.4, 3.2.2	UserProfile, NotificationRecord
Data Layer	NotificationRecord	Data Table	Stores notification data pending delivery to a target user	2.2.1, 3.2.1, 3.2.4	FollowRelation, UserProfile
Data Layer	ApplicationRecord	Data Table	Stores student resume form submissions	2.1.2, 2.5.6, 3.1.3.	ResumeRecord, JobPosting, UserProfile

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
			sent to a specific job posting	2	
Data Layer	fetchRecord()	Function	Retrieves a single record from the appropriate data table by primary key	2.3.5, 3.3.2	UserProfile, ResumeRecord, JobPosting

EI Table 1.14.3 – Indexing Layer

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Indexing Layer	SearchIndex	Data Store	Stores pre-built indexed entries that map keywords to database record IDs	2.3.1, 3.3.1	IndexBuilder, QueryResolver
Indexing Layer	IndexBuilder	Module	Builds and updates the search index whenever relevant records are added or modified	2.3.5, 3.3.2	SearchIndex, JobPosting, ResumeRecord
Indexing Layer	QueryResolver	Function	Maps an incoming search query to matching index entries and returns record IDs	2.3.1, 2.3.8, 3.3.1	SearchIndex, FilterProcessor
Indexing Layer	FilterProcessor	Class	Applies user-selected filters to a set of index-resolved candidate records	2.3.2, 2.3.4, 3.3.4	QueryResolver, SearchFilter
Indexing Layer	SearchFilter	Object	Represents the collection of filter criteria selected for a given search request	2.3.2, 2.3.7, 3.3.3	FilterProcessor, QueryResolver
Indexing Layer	indexQuery	Variable	Stores the normalized search query string used to look up entries	2.3.1, 3.3.1	QueryResolver

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
			in the index		
Indexing Layer	WildcardMatcher	Module	Expands wildcard patterns in a query to match partial keyword entries in the index	2.3.7	QueryResolver, SearchIndex
Indexing Layer	AutocompleteIndex	Data Store	Stores prefix-keyed term suggestions used by the search bar autocomplete function	2.3.6	IndexBuilder, SearchIndex
Indexing Layer	indexResults	Variable	Stores the list of record IDs returned by the index lookup before data retrieval	2.3.5, 3.3.2	QueryResolver, FilterProcessor

Figure 1.15 — File / Blob Storage Detail**Design Information – File / Blob Storage**

Field	Description / Purpose
View Name & Number	1.15 File / Blob Storage
Purpose	Describe the file and blob storage.
Description	Basic service demonstrating file and blob storage.
Requirements	SRS 3.5 Resume (resume file storage), 2.1.3.1 (profile image uploads), 2.1.2 (company resume access)
Elements	1.15.1 Upload / Validation Layer 1.15.2 Raw Byte Storage 1.15.3 Backup Module
Navigation / References	1.14 Database; 1.6 Profile Management Service, 1.7 Resume Service; part of Data Storage Layer (Figure 1.1)
Viewpoint	Component Diagram

Entity Identification Tables – File / Blob Storage

El Table 1.15.1 – Upload / Validation Layer

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Upload/ Validation Layer	UploadManager	Class	Manages file uploads	3.1.3, 2.5.9	UploadManager
Upload/ Validation Layer	validateFileType()	Method	Checks that file is of an accepted type	3.1.3, 2.5.9	UploadManager, FileValidator
Upload/ Validation Layer	FileValidator	Class	Handles all file validation	3.1.3, 2.5.9	validateFileType()
Upload/ Validation Layer	FileMetadata	Data Store	Stores data related to a given file	3.1.3, 2.5.9	UploadManager

El Table 1.15.2 – Raw Byte Storage

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Raw Byte Storage	RawByteStorageModule	Module	Subsystem responsible for persistent file storage	3.1.3, 2.5.9	RawByteStorage Module
Raw Byte Storage	BlobStore	Class	Stores Binary Large Objects	3.1.3, 2.5.9	RawByteStorage Module
Raw Byte	StorageManager	Class	Main interface to	3.1.3,	RawByteStorage

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Storage			storage	2.5.9	Module
Raw Byte Storage	FileChunkManager	Class	Splits large files into chunks	3.1.3, 2.5.9	RawByteStorage Module
Raw Byte Storage	storeFile()	Method	Initializes file storage process	3.1.3, 2.5.9	StorageManager
Raw Byte Storage	retrieveFile()	Method	Initializes file retrieval process	3.1.3, 2.5.9	StorageManager
Raw Byte Storage	BlobStorage	Data Store	Main storage repository	3.1.3, 2.5.9	RawByteStorage Module
Raw Byte Storage	ChunkStorage	Data Store	Stores file chunks	3.1.3, 2.5.9	RawByteStorage Module

EI Table 1.15.3 – Backup Module

Subcomponent	Entity Name	Entity Type	Description / Purpose	Related Requirement(s)	Collaborators / Related Entities
Backup Module	BackupManager	Class	Container for all backup logic	3.1.3, 2.5.9	
Backup Module	BackupScheduler	Class	Responsible for automated system backups	3.1.3, 2.5.9	BackupManager
Backup Module	BackupStorage	Data Store	Stores backup files	3.1.3, 2.5.9	BackupManager
Backup Module	RestoreManager	Class	Manages file restoration requests	3.1.3, 2.5.9	BackupManager
Backup Module	SnapshotManager	Class	Creates system snapshots	3.1.3, 2.5.9	BackupManager

These sub-diagrams illustrate how each major component has been further decomposed into smaller, cohesive sub-components while preserving the explicit interfaces and ports defined at the system level. This decomposition supports modular design principles and enables independent development and maintenance of each subsystem.

3.3 Design Rationale

Discuss the rationale for selecting the architecture described in 3.1 including critical issues and trade/offs that were considered. You may discuss other architectures that were considered, provided that you explain why you didn't choose them.

4.0 DATA DESIGN

4.1 Data Description

Explain how the information domain of your system is transformed into data structures. Describe how the major data or system entities are stored, processed and organized. List any databases or data storage items.

4.2 Data Dictionary

Alphabetically list the system entities or major data along with their types and descriptions. If you provided a functional description in Section 3.2, list all the functions and function parameters. If you provided an OO description, list the objects and its attributes, methods and method parameters.

5.0 COMPONENT DESIGN

In this section, we take a closer look at what each component does in a more systematic way. If you gave a functional description in section 3.2, provide a summary of your algorithm for each function listed in 3.2 in procedural description language (PDL) or pseudocode. If you gave an OO description, summarize each object member function for all the objects listed in 3.2 in PDL or pseudocode. Describe any local data when necessary.

6.0 HUMAN INTERFACE DESIGN

6.1 Overview of User Interface

The following diagrams provide an approximation of the central UI elements that will allow the users to utilize the various services provided, including descriptions of their functionality and purposes as well as site navigation details.

The Career Bound interface uses role-aware navigation to simplify the user experience while maintaining role-specific functionality. Shared navigation labels such as Dashboard dynamically direct users to the appropriate interface based on their authenticated role. For example, student users are directed to the Student Dashboard, while company

representatives are directed to the Company Dashboard. This approach reduces interface clutter and maintains consistent navigation across the system.

6.2 Screen Images

UI View 1.1 — Login / Logout

Desktop View – Login / Logout

Logo	Search Bar	Profile/Menu
Dashboard Profile Jobs Resume Notifications News Feed	Login/Logout Email* NotMyLover@gmail.com... Password* Enter Password... Login New User? Register here	

Mobile View – Login / Logout

 CareerBound

Search Bar

Login/Logout

Email*

Password*

Login

New User? Register here

Design Information – Login / Logout

Field	Description / Purpose
View Name & Number	1.1 Login / Logout
Purpose / Design Concern	Provides a way for students and companies to log into their profiles.
Description	A website page which displays a log in field for the user to input their email and password to authenticate themselves.
Requirements	Authentication requirements supporting all protected system functionality under Sections 2.x and 3.x
Elements	Email Field
	Password Field
	Login Button
	Invalid Login Information Field
	Registration Redirect Button
Inputs	Email and Password fields, Login button click, Register link click, Forgot Password link click.
Outputs	Redirect to Dashboard (2.1 for students, 3.1 for companies) if user log in was successful. Displays an error message if invalid credentials were provided.
Navigation / References	Student Dashboard 2.1, Company Dashboard 3.1, Registration 1.2
Viewpoint	UI Wireframe / Mockup (Desktop & Mobile).

UI View 1.2 — Registration

Desktop View – Registration

Logo	Search Bar	Profile/Menu
Dashboard Profile Jobs Resume Notifications News Feed	Registration Back to Login Student / Employer* Employer ▼ First Name* Billy... Last Name* Jean... Email* NotMyLover@gmail.com... Password* Enter Password... ✓ 8 characters long ✓ Includes capital letter ✗ Includes special character Confirm Password* Enter Password... Submit	

Mobile View – Registration

 CareerBound  

Search Bar

Registration

Student / Employer*

Employer ▼

First Name*

Billy...

Last Name*

Jean...

Email*

NotMyLover@gmail.com...

Password*

Enter Password...

☒

 8 characters long

☒

 Includes capital letter

☐

 Includes special character

Confirm Password...

Submit

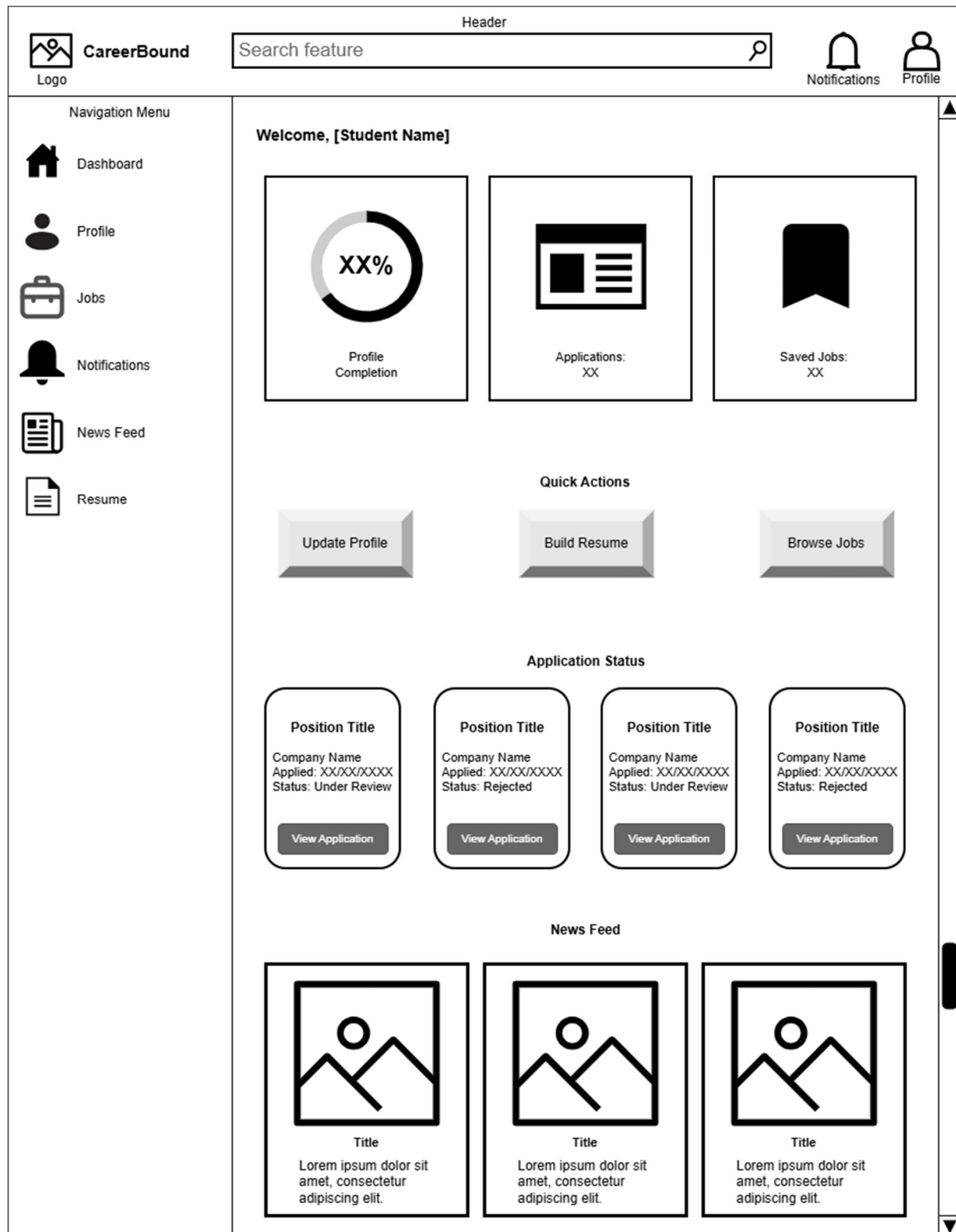
Back to Login

Design Information – Registration

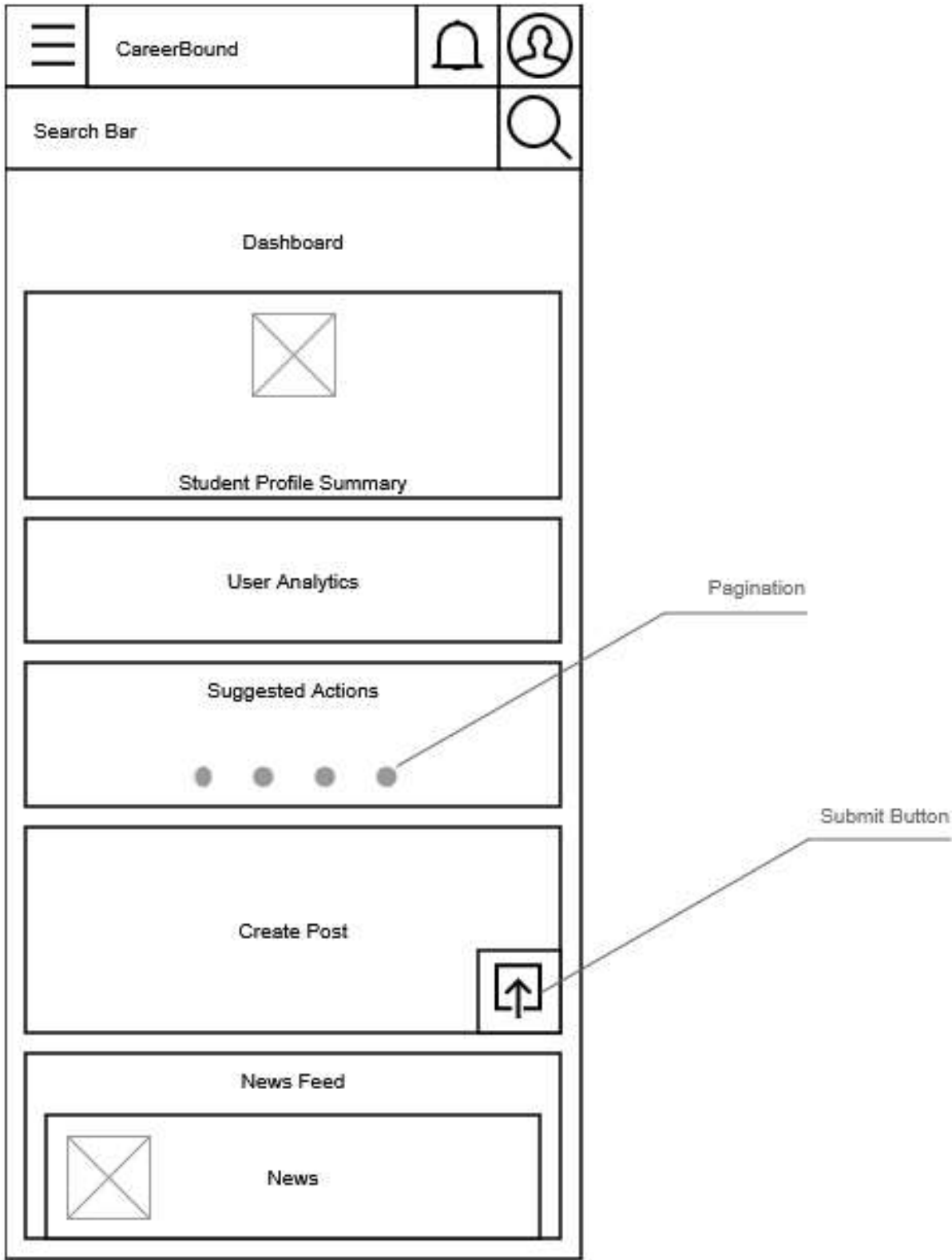
Field	Description / Purpose
View Name & Number	1.2 Registration
Purpose / Design Concern	Facilitates user account creation and role selection for new users.
Description	Displays a registration form allowing a user to select their role and establish account credentials.
Requirements	SRS 2.1 Company Profile Page, 3.1 Student User Profile
Elements	Back to Login button
	Student / Employer role selection dropdown
	Text fields
	Password validation checklist
	Submit button
Inputs	Role selection, First Name string, Last Name string, Email string, Password string, Confirm Password string, Submit action.
Outputs	Visual feedback on password requirement validation (status icons for "8 characters long", "Includes capital letter", and "Includes special character").
Navigation / References	1.1 Login / Logout (accessed via "Back to Login" button). 2.1 Student Dashboard or 3.1 Company Dashboard upon successful submission.
Viewpoint	UI Wireframe / Mockup (Desktop & Mobile).

UI View 2.1 — Student Dashboard

Desktop View – Student Dashboard



Mobile View – Student Dashboard



Design Information – Student Dashboard

Field	Description / Purpose
View Name & Number	2.1 – Student Dashboard
Purpose / Design Concern	Provides students with a centralized dashboard for managing career development activities, tracking applications, accessing important tools, and viewing personalized updates within the Career Bound system.
Description	Displays a summarized student profile; profile progress (desktop) provides analytics such as profile views, profile completion, matching jobs, post impressions, application summaries; application tracking cards (desktop); recommends actions (mobile) to take such as improve profile, create resume, apply for a job; provides a shortcut to post creation; and displays social media-style news feed containing career-related updates and announcements.
Requirements	3.1 Student User Profile, 3.2 Notifications, 3.3 Search Features, 3.4 News Feed, 3.5 Resume
Elements	Student Profile Summary Card (mobile)
	Student User Analytics Card (mobile)
	Suggested Actions Card (mobile)
	Quick Create Post Card (mobile)
	News feed List
	News feed Cards
	Submit Button
	Navigation menu (Sidebar for desktop, toggle for mobile)
	Header
	Search Feature
	Applications Summary Card (desktop)
	Saved Jobs Summary Card (desktop)
	Applications Status Cards (desktop)
	Notifications toggle button
	Profile Page hyperlink button
	Pagination
Inputs	Student Post, Click, Scroll, Search queries, navigation selections, notification interactions, profile selections, quick action button selections, application card selections, news feed interactions
Outputs	Profile completion percentage, application status information, saved job totals, recommended actions, job application details, career updates, company announcements, News feed content, New Student Post, Navigation redirects, Navigate to next action card
Navigation / References	Authentication, Student Profile, Resume Builder, Job Search, Job Details, Notifications, News Feed, Saved Jobs, Job Applications
Viewpoint	UI Wireframe / Mockup (Desktop & Mobile)

UI View 2.2 — Student Profile

Desktop View – Student Profile

CareerBound	Search Bar	 
Dashboard Profile Jobs Resume Notifications News Feed	Section Title	
	Profile Picture	Name: Email: Phone: Bachelors Degree: Skills: Linkedin Profile
	About Me	
	Edit Info Save	
	Delete Profile	

Mobile View – Student Profile

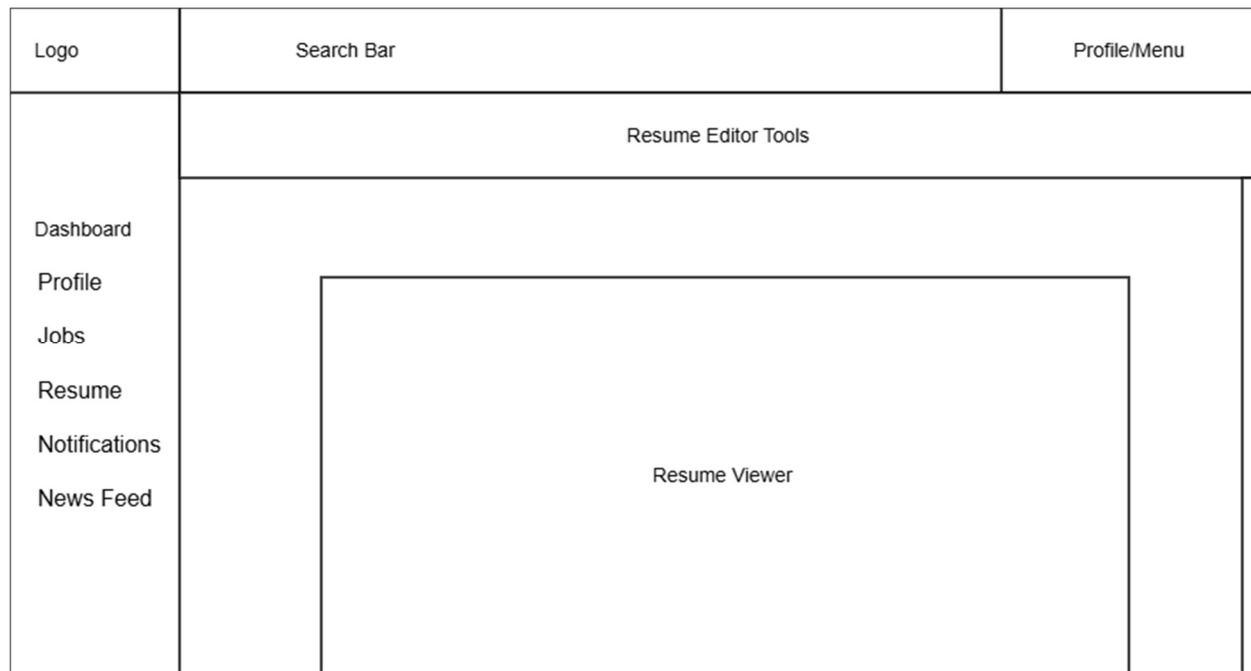
 CareerBound	 
Search Bar	[Filter]
Section Title	
Profile Picture	
Name: Email: Phone: Bachelors Degree: Skills: Linkedin Profile	
About Me	
Edit Info Save	Delete Profile

Design Information – Student Profile

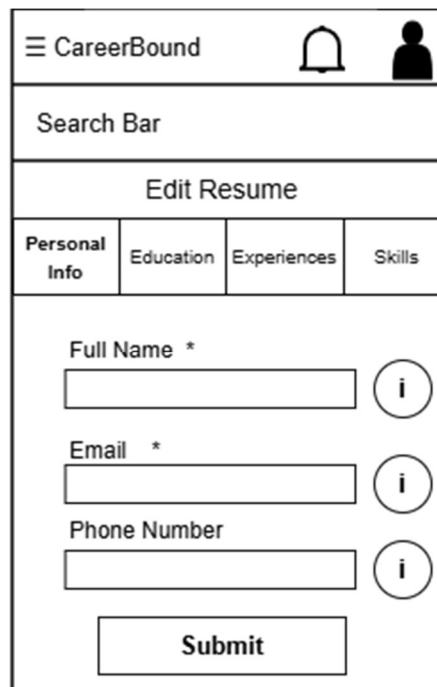
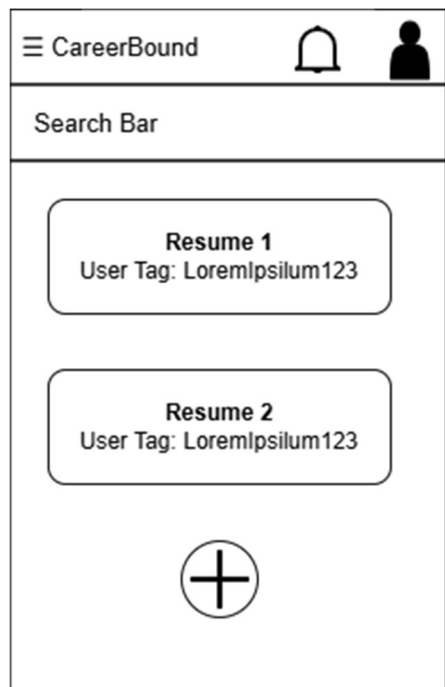
Field	Description / Purpose
View Name & Number	2.2 – Student Profile
Purpose / Design Concern	Allows students to view and manage their profile information.
Description	Displays student information such as name, contact information, program, skills, and resume-related details.
Requirements	3.1 Student User Profile (3.1.1–3.1.3.3)
Elements	Logo
	Search bar
	Menu/Profile
	Left Navigation
	Section Title
	Profile Picture
	User Information
	About Me
	Save/Delete buttons
Inputs	Updated student profile information, Upload CV
Outputs	Displayed student profile information
Navigation / References	Dashboard, Profile, Jobs, Resume, Notifications, News Feed
Viewpoint	UI Wireframe / Mockup (Desktop & Mobile)

UI View 2.3 — Resume Builder

Desktop View – Resume Builder



Mobile view – Resume Builder



Design Information – Resume Builder

Field	Description / Purpose
View Name & Number	2.3 Resume Builder
Purpose / Design Concern	Provides a space that allows students to view and edit their resume that is to be used.
Description	Displays student's resume and gives tools to make and save changes to the resume.
Requirements	SRS 3.5 Resume (3.5.1–3.5.5.2), 3.5.4 User Input, 3.5.5 Input Submission
Elements	Logo
	Search Bar
	Profile/Menu
	Dashboard
	Profile
	Jobs
	Resume
	Notifications
	News Feed
	Resume Editor Tools: Edit Resume Section Tabs/ Add New Resume Button/ Helpful info buttons/ Submit Resume Button/ Asterisks denoting Required Data Entry
	Resume Viewer: Resume List
Inputs	Resume Edits, Search terms, Resume data entries, Wildcard search
Outputs	Resume reflecting saved changes and recent changes made, Matching resume data terms, resume list search results, new resume data entries
Navigation / References	Profile/Menu, Dashboard, Profile, Jobs, Resume, Notifications, News Feed, Resume List, Edit Resume Screen
Viewpoint	UI Wireframe / Mockup (Desktop & Mobile)

UI View 2.4 — Job Search

Desktop View – Job Search

Logo	Search Bar	Profile/Menu
Dashboard Profile Jobs Resume Notifications News Feed	[Job Title] - [Company] [Employment Type] [Location] [Date Posted] Description: Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris [Job Title] - [Company] [Employment Type] [Location] [Date Posted] Description: Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris [Job Title] - [Company] [Employment Type] [Location] [Date Posted] Description: Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris	[Active Search Filters] X [Filter 1] X [Filter 2]

Mobile View – Job Search

 CareerBound



Search Bar

[Filter]

[Job Title] - [Company]
[Employment Type]
[Location]
[Date Posted]
[Save]

[Job Title] - [Company]
[Employment Type]
[Location]
[Date Posted]
[Save]

[Job Title] - [Company]
[Employment Type]
[Location]
[Date Posted]
[Save]

[Job Title] - [Company]
[Employment Type]
[Location]
[Date Posted]
[Save]

[Job Title] - [Company]
[Employment Type]
[Location]
[Date Posted]
[Save]

[Job Title] - [Company]
[Employment Type]
[Location]
[Date Posted]
[Save]

[Job Title] - [Company]
[Employment Type]
[Location]
[Date Posted]
[Save]

Design Information – Job Search

Field	Description / Purpose
View Name & Number	2.4 – Job Search
Purpose / Design Concern	Allows students to search for jobs and companies.
Description	Displays searchable job listings with filters and detailed results.
Requirements	SRS 2.3 Search Features, 3.3 Student Search Features
Elements	Search bar
	Filters
	Results list
	Save/apply buttons
Inputs	Search terms, filter selections, wildcard search
Outputs	Matching job postings and company search results
Navigation / References	Job Details Screen, Company Profile, Student Dashboard
Viewpoint	UI Wireframe / Mockup (Desktop & Mobile)

UI View 2.5 — Job Details

Desktop View – Job Details

Logo	Search Bar	Profile/Menu
Dashboard	[Job Title] [Company Name] [City, State] [Full-time] [Pay Rate] Apply Button	
Profile		
Jobs		
Resume		
Notifications		
News Feed		
	About the Role Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Responsibilities • Lorem ipsum dolor • Sit amet, consectetur adipiscing Nice to Have • Lorem ipsum dolor • Sit amet, consectetur adipiscing About [Company Name] [Company Website] [Company Size] [Industry] [Headquarters] Job Details [Job ID] [Department Name] [Experience Level] [In-Person/Remote/Hybrid] Required Skills [Skill 1] [Skill 2] [Skill 3] [Skill 4]	

Mobile View – Job Details

 CareerBound

Search Bar[Filter]

 Back to Search Results

[Job Title]
[Company Name]
[City, State]
[Employment Type]
[Pay Rate]

Apply Button

About the Role
Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum

Responsibilities

- Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod
- tempor incididunt ut labore et dolore magna aliqua. Ut enim ad

Job Details
[Job ID]
[Department Name]
[Experience Level]
[In-Person/Remote/Hybrid]

Nice to Have

- Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod
- tempor incididunt ut labore et dolore magna aliqua. Ut enim ad

About [Company Name]
[Company Website]
[Company Size]
[Industry]
[Headquarters]

Required Skills
[Skill 1]
[Skill 2]
[Skill 3]
...

166

Design Information – Job Details

Field	Description / Purpose
View Name & Number	2.5 – Job Details
Purpose / Design Concern	Allows students to get more information for Job Postings.
Description	Displays all information relevant to a given Job Postings.
Requirements	SRS 2.5.2, SRS 2.5.7, SRS 2.5.9
Elements	Back Button
	Job Data
Inputs	Tap/Click on Apply Button
Outputs	Application Submission Confirmation, Updated Application Status
Navigation / References	Job Search Screen, Company Profile, Student Dashboard
Viewpoint	UI Wireframe / Mockup (Desktop & Mobile)

UI View 3.1 — Company Dashboard

Desktop View – Company Dashboard

CareerBound [Company Logo] 🔍 Search Bar 🔔 Notifications 👤 Company Profile

Navigation

- Dashboard
- Profile
- Jobs
- Resume
- Notifications
- News Feed

3.1 Company Dashboard

Active Job Posting
08

Applications
42

Candidate Search
16

Notifications
05

Quick Actions

- Create Job Posting
- Candidate Search
- View Resumes

Resumes

Candidate	Job Posting	Status	Date
A. Rivera	Platform Engineer	Under Review	05/15/26
T. Nguyen	Data Analyst	New	05/14/26
A. Patel	UX Researcher	Interview	05/13/26
J. Kim	Product Manager	Complete	05/15/26

Job Posting Status

Active 08

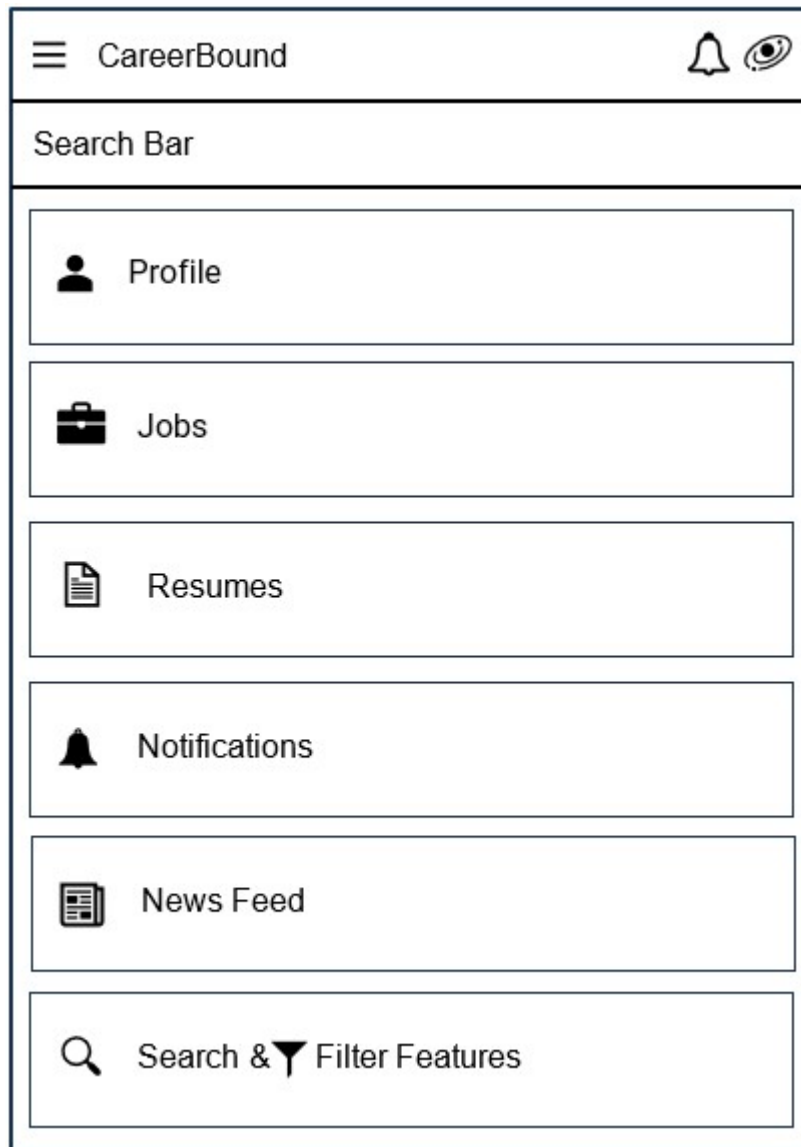
Under Review 03

Expiring Soon 02

News Feed

CareerBound Update
New company insights available for active Job Posting and Applications.

Hiring Trends
Candidate Search activity increased for software and analytics roles this week.

Mobile View – Company Dashboard**Design Information – Company Dashboard**

Field	Description / Purpose
View Name & Number	3.1 – Company Dashboard
Purpose / Design Concern	Company dashboard to help a company manage recruiting activity from desktop and mobile views.
Description	Provides a desktop and mobile view where company users can review job posting activity, applications, candidate search access, notifications, and company-related updates.
Requirements	Software Requirements Specification (SRS) for Career Bound: 2.1 Company Profile Page, 2.2 Notifications, 2.3 Search Features, 2.4 News Feed, 2.5 Job Postings

Field	Description / Purpose
	Reference: https://content.byui.edu/file/968cdd33-09eb-4430-99a4-131d25e138d4/1/Teach/SRS/430.SRS.CareerBound.pdf
Elements	Navigation menu: provides a company with a list of options to select from.
	Search bar: provides a company the ability to search.
	Company Profile area: provides a company with access to all company profile data.
	Notifications area: provides a company with access to company notifications.
	Candidate Search feature: provides a company the ability to search or filter job candidates.
	Job Posting area: provides a company access to job postings.
	Resume section: provides a company access to a list of candidates who have submitted job applications.
	News Feed section: provides a company access to news feed options
	Login / Logout: clicking the company logo provides a company with the ability to login or logout.
Inputs	Search query text, clicks on navigation items, clicks on quick action buttons, and selections related to applications or job postings.
Outputs	Company dashboard metrics, application status information, job posting status information, notifications, and news feed updates.
Navigation / References	1.1 Login / Logout, 3.2 Candidate Search, 3.3 Job Posting Creation, 4.1 Notifications, 4.2 News Feed
Viewpoint	UI Wireframe / Mockup (Desktop & Mobile)

UI View 3.2 — Candidate Search

Desktop View – Candidate Search

Logo	Search Bar	Profile/Menu	
Dashboard Profile Jobs Resume Notifications News Feed	 Search here Advanced Search Results for "Lorem Ipsum" 5 Results John Doe Senior  Jane Doe Junior  Joe Smith Senior  Sue Smith Junior  Mike Jackson Senior 		
	Filters		
	Education		
	Major		
	Technical Skills		

Mobile View – Candidate Search

The mockup shows a mobile application interface for candidate search. At the top is a header bar with a hamburger menu icon, the text 'CareerBound', a bell icon, and a user profile icon. Below the header is a 'Search Bar' section containing a search input field with a magnifying glass icon and the placeholder text 'Search here', followed by a funnel icon representing filters. The main content area is titled 'Results for "Lorem Ipsum"' and shows '5 Results'. On the left, there is a 'Filters' sidebar with three categories: 'Education', 'Major', and 'Technical Skills'. Each category has a list of checkboxes, with the first checkbox in each category being checked. The main area displays candidate cards. Three cards are visible: 'John Doe' (Senior), 'Jane Doe' (Senior), and 'Joe Smith' (Senior). Each card features a placeholder profile picture and the candidate's name and title. The 'Joe Smith' card is fully visible, while the others are partially obscured by the filters sidebar.

Design Information – Candidate Search

Field	Description / Purpose
View Name & Number	3.2 Candidate Search
Purpose / Design Concern	Allows company recruiters a way to find potential job seekers.
Description	A specialized search engine for the user to find candidates based on desired filters. Designed for both desktop users and mobile users.
Requirements	2.3.1, 2.3.2, 2.3.4, 2.3.8
Elements	Search Bar Site Navigation Search Results Grid Filter Categories Filter Checkboxes Search Result Cards Advanced Search Toggle

Field	Description / Purpose
	Candidate Name
	Candidate Profile Picture
	Candidate Year
	Filter Scroll Bar
Inputs	User will click on the search bar and enter a search item; user can choose a filter to toggle; the user can choose a candidate to open a new page.
Outputs	Candidate profile cards display candidate information; candidate profiles can be filtered.
Navigation / References	3.1 Company Dashboard, 3.3 Job Posting Creation
Viewpoint	UI Wireframe / Mockup (Desktop & Mobile)

UI View 3.3 — Job Posting Creation

Desktop View – Job Posting Creation

Desktop - 1

Logo	Search Bar	Profile/Menu
	Add job button Edit job button Remove job button	
Dashboard Profile Jobs Resume Notifications News Feed	Jobs	Job posting data Job posting data Free-form text area ☐ Toggle required skills posting Edit required skills Optional required skills list Filters Resume form submissions

Mobile View – Job Posting Creation

iPhone 17 - 1

 CareerBound



Search Bar

Add job button

Jobs

Edit job buttonRemove job button

Job posting data

Free-form text area

☒ Toggle required skillsEdit required skills

Optional required skills list

Filters

Resume form submissions

Design Information – Job Posting Creation

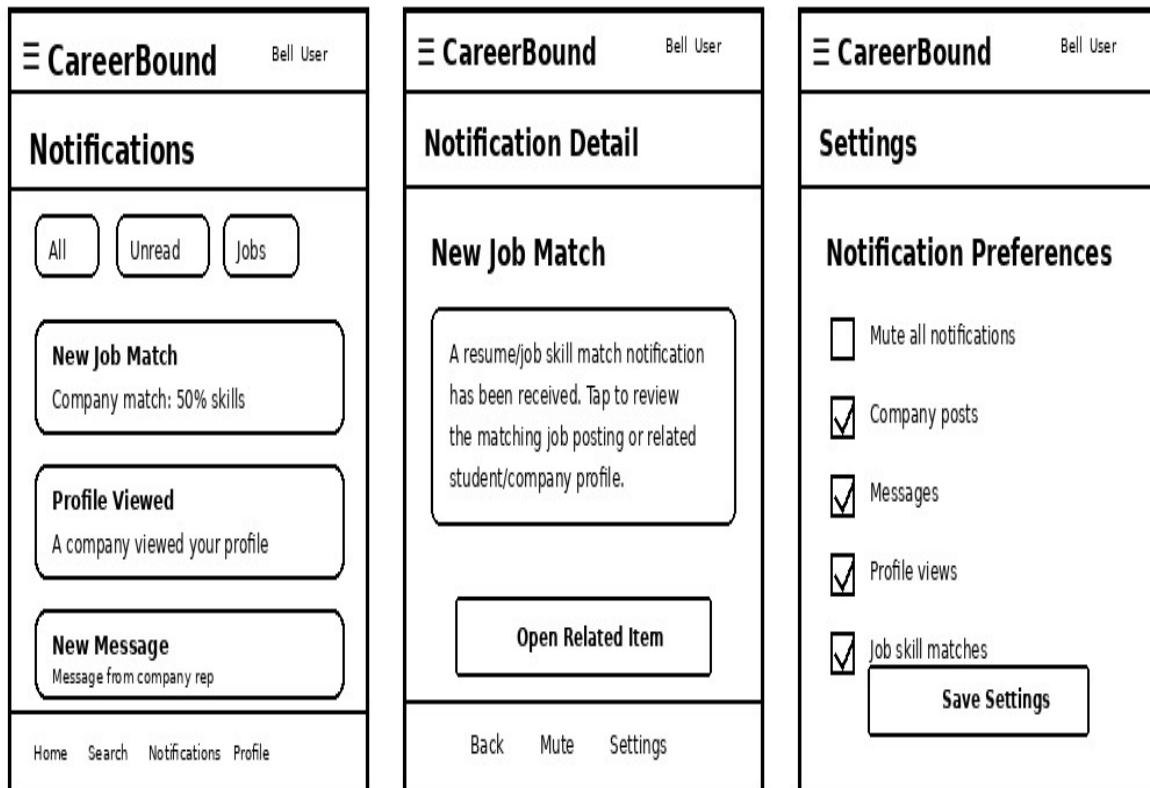
Field	Description / Purpose
View Name & Number	3.3 Job Posting Creation
Purpose / Design Concern	Allows company users to create, edit, and remove job postings, manage required skills, and review resume submissions in a centralized Jobs view.
Description	Displays a company-facing Jobs screen with job posting controls, a job posting data panel, a free-form text area, a required skills toggle, an editable required-skills list, a filters section, and a resume form submissions area in both desktop and mobile wireframe layouts.
Requirements	SRS 2.5 Job Postings (2.5.1--2.5.7)
Elements	Header with logo, search bar, and profile/menu
	Navigation menu with Dashboard, Profile, Jobs, Resume, Notifications, and News Feed
	Add Job Button
	Edit Job Button
	Remove Job Button
	Jobs Panel
	Job Posting data section
	Free-form text area
	Required skills toggle
	Edit Required Skills button
	Optional required skills list
	Filters section
	Resume form submission panel
Inputs	Company user selects Add Job, Edit Job, or Remove Job; enters or edits job posting text; toggles required skills on or off; edits the required skills list; applies filters; reviews submitted resumes.
Outputs	Displayed job posting information, updated required skills settings, filtered submission results, and visible resume form submissions for review.
Navigation / References	Related views include Company Dashboard (3.1), Candidate Search (3.2)
Viewpoint	UI Wireframe / Mockup (Desktop & Mobile)

UI View 4.1 — Notifications

Desktop View – Notifications

Logo	Search Bar	Profile/Menu
Dashboard	Notification ☒ Push Notifications News Feed New Post by companies you are following Receive notification from companies that you are following Personal message from company Company Representative Views profile	
Profile		
Jobs		
Resume		
Notifications	Notification Filters ☒ Create Filter Example Filter ✕	
News Feed		

Mobile View – Notifications



Design Information – Notifications

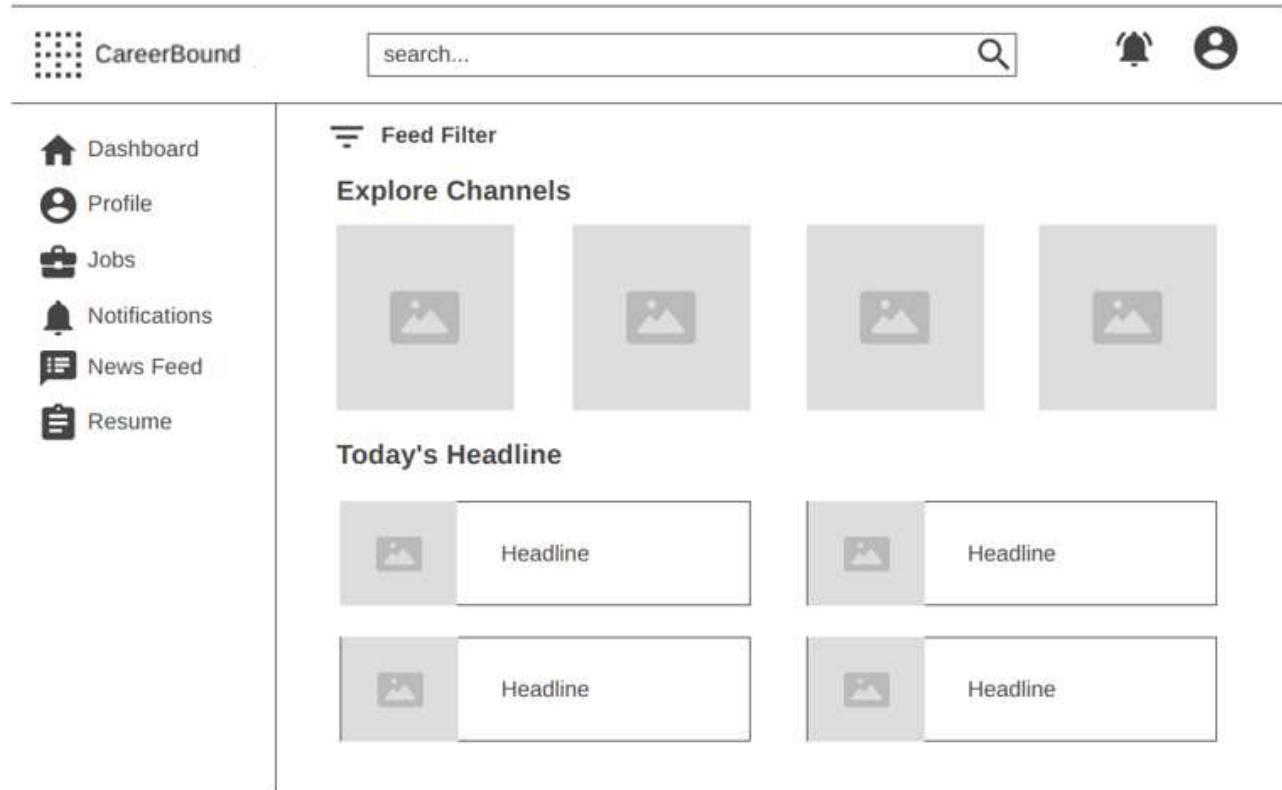
Field	Description / Purpose
View Name & Number	4.1 Notifications Screen
Purpose / Design Concern	The Notification Screen exist to allow the user to customize what notifications they receive from Career Bound. This is for the End User and satisfies section 2.2 / 3.2 in the SRS.CareerBound.docx
Description	The Notification Screen will allow the user to customize notifications about their News Feed, Company Viewing their profile, Push Notifications, New posts by company they follow, notifications from company they follow, personal messages, and allow the user to mute all notifications. Uses the filter component to create a list of companies to receive notifications from. Shows mobile notification screens for student users and company representatives, including the notification list, notification detail screen, push notification handling, and notification preferences.

Field	Description / Purpose
Requirements	1.2.1 Application Scope; 2.2 Notifications; 3.2.1. There shall be a notification push sent to the student user instantly after a notification trigger. 3.2.2. The student user shall receive a notification when a company they are following makes a post or sends out a notification to its followers. 3.2.3. The student user shall receive a notification when a company sends them a personal message. 3.2.4. Notifications may be pushed to a student user's phone outside of the system 3.2.5. Student users may receive a notification when a company representative views his/her profile. 3.2.6. All users may choose to mute notifications 3.2.7. There shall be settings allowing the student user to add, alter and remove filters for notifications.
Elements	Search Bar App navigation Toggle Switch Create Filter Button Edit Pen Icon Delete X Icon Notification Card: Notification Screen; Notification List; Notification Detail View; Push Notification Handler; Notification Preferences; Mute Notification Control; Notification Filter Card: Filter Settings; Profile/Menu API Client Adapter / REST/HTTP API Port
Inputs	Toggle Switch, Click Button, Click Text, Filter Companies, Search, Selected notification, filter settings, mute setting changes, push notification trigger, and mobile navigation actions
Outputs	Notifications Settings, Filtered Company List, Notification list, notification detail information, filtered notification results, push notification display, and updated notification settings
Navigation / References	Filter Create Screen, Dashboard Screen, Profile Screen, Jobs Screen, Resume Screen, News Feed Screen, Mobile Client, Notification Screen, Notification Detail View,

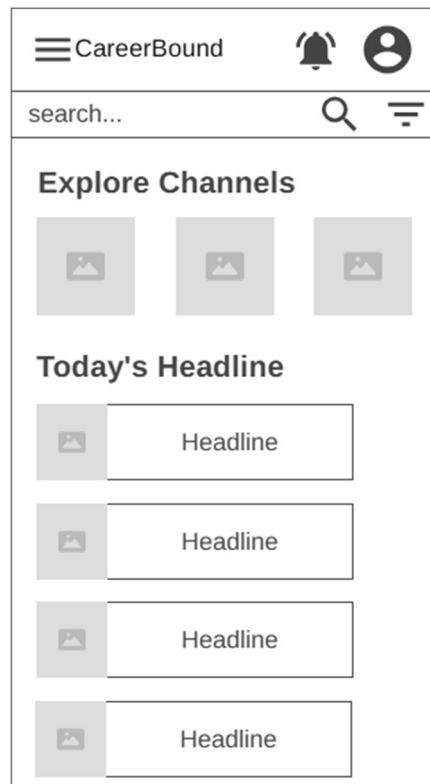
Field	Description / Purpose
	Notification Preferences, API Gateway / Orchestrator
Viewpoint	UI Wireframe / Mockup (Desktop & Mobile)

UI View 4.2 — News Feed

Desktop View – News Feed



Mobile View – News Feed



Design Information – News Feed

Field	Description/Purpose
View Name & Number	4.2 – News Feed
Purpose / Design Concern	Allows students to filter and view their news feed.
Description	Displays relevant news for students, which can be filtered by different tags to view specific news reports.
Requirements	2.4 News Feed (2.4.1–2.4.6), 3.4 News Feed (3.4.1–3.4.2)
Elements	Logo
	Notifications
	Search Bar
	Menu/Profile
	Feed Filter
	Explore Channels
	Headlines
Inputs	Current news, current location
Outputs	Updated news feed, current location displayed
Navigation / References	Dashboard, Profile, Jobs, Resume, Notifications, News Feed
Viewpoint	UI Wireframe / Mockup (Desktop & Mobile)

6.3 Screen Objects and Actions

A discussion of screen objects and actions associated with those objects.

7.0 REQUIREMENTS MATRIX

The following table provides a cross-reference that traces each major component in the system architecture to the specific SRS requirements it fulfills. Below this SRS Requirements Mapping Table you will find Requirements Traceability Matrices for quick visual SRS Requirements tracing.

SRS Requirements Mapping Table

Component	SRS Sections / Requirements It Directly Supports
Client Applications (Parent)	• 1.2 Application Scope (web and mobile interfaces that share the same features) • Overall client-side architecture, shared UI patterns, role-based views, and coordination between web and mobile clients
Web Client	• Primary web-based user interface (browser UI) • Implements all web UI elements for Company and Student features (sections 2.x and 3.x) • Profile pages, search bar, forms, news feed display, notifications display, resume form, job browsing, etc. (1.2.1)
Mobile Client	• Mobile application user interface • Implements the same features and characteristics as the Web Client (per SRS 1.2.1) • Mobile-optimized UI for all features in sections 2.x and 3.x (profiles, search, news feed, notifications, resume form, job browsing, etc.)
API Gateway / Orchestrator	• Cross-cutting orchestration for all client–service communication (implicit in 1.2 and all functional requirements in 2.x / 3.x)
Authentication Service	• Foundational requirement for all logged-in features (profiles, following, notifications, job postings, resume access, etc.) • Required for every section under 2.x and 3.x
Profile Management Service	• 2.1 Company Profile Page (2.1.1–2.1.3.10) • 3.1 Student User Profile (3.1.1–3.1.3.3)
Resume Service	• 3.5 Resume (3.5.1–3.5.5.2) • Referenced in 2.1.2, 2.2.5, 2.3.3, 2.3.5, 2.5.6, 3.1.3.1–3.1.3.3
Job Posting Service	• 2.5 Job Postings (2.5.1–2.5.7) • Referenced in 3.3.1, 3.4.1, 2.2.5
Search Service	• 2.3 Search Features (2.3.1–2.3.8) • 3.3 Search Features (3.3.1–3.3.2)

Component	SRS Sections / Requirements It Directly Supports
Communication Services (Notifications + News Feed)	• 2.2 Notifications (2.2.1–2.2.5) • 2.4 News Feed (2.4.1–2.4.6) • 3.2 Notifications (3.2.1–3.2.7) • 3.4 News Feed (3.4.1–3.4.2)
Recommendation Service	• Job and candidate recommendations based on profiles, resumes, and job postings • Supports 2.3 Search Features, 2.5 Job Postings, 3.3 Search Features, and 3.1/3.5 profile/resume data
External Services	• External integrations (USPS address validation in 2.1.3.9, autocomplete API in 2.3.6, push notification services in 2.2, etc.) • Any third-party APIs referenced throughout the SRS
Analytics Service	• Provides reporting and analytics on job posting performance (views, applications, 2.5), candidate search effectiveness (filters, results, 2.3), profile and resume engagement (2.1, 3.1, 3.5), and notification / news feed activity (2.2, 2.4, 3.2, 3.4) • Monitors key performance metrics (e.g., search results returned within 3 seconds per 2.3.5) • Supports data-driven platform improvements, usage insights, and overall system monitoring
Data Storage Layer (Parent)	• Provides the single unified SQL / Data Access Port used by every service in the system • Acts as the central enclosing component that coordinates Database and File/Blob Storage • Manages system-wide data concerns (transactions, consistency, backups, scalability, reliability) • 4. Hardware requirements (implicit backend storage, servers, infrastructure)
Database	• All persistent data storage for profiles, resumes, job postings, notifications, news feed, authentication, etc. • Referenced in every “Data X.X.X” entry throughout the SRS
File / Blob Storage	• Storage of uploaded resumes, profile images, attachments, and other binary files • Supports 3.5 Resume submission and any file-related requirements

The following Requirements Traceability Matrices map the requirements defined within the Software Requirements Specification (SRS) to the CareerBound system architecture components responsible for supporting, implementing, or interacting with those requirements.

These matrices are intended to:

- verify that all documented requirements are addressed within the system design,
- demonstrate traceability between requirements and architectural components,

- support validation and quality assurance activities, and
- ensure consistency between the SRS and the Software Design Description (SDD).

An “X” within a matrix cell indicates that the corresponding component directly supports, implements, manages, or participates in fulfilling the specified requirement.

The matrices are organized by major SRS requirement sections for improved readability and maintainability.

7.1 Company Profile Requirements Matrix (SRS 2.1)

Component	2.1.1 Profile	2.1.2 Resume	2.1.3.1 Contact	2.1.3.2 Location	2.1.3.3 Skills	2.1.3.4 Following	2.1.3.5 Posting	2.1.3.6 Analytics	2.1.3.7 Messaging	2.1.3.8 Search	2.1.3.9 Validation	2.1.3.10 Security
Web Client	X	X	X	X	X	X	X	X	X	X	X	X
Mobile Client	X	X	X	X	X	X	X	X	X	X	X	X
API Gateway / Orchestrator	X	X	X	X	X	X	X	X	X	X	X	X
Authentication Service	X	X	X	X	X	X	X	X	X	X	X	X
Profile Management Service	X	X	X	X	X	X	X	X	X	X	X	X
Resume Service		X			X			X				
Job Posting Service				X		X						
Search Service		X				X				X	X	
Communication Services			X			X			X			
Recommendation Service		X				X				X		
External Services											X	

Component	2.1.1 Profile	2.1.2 Resume	2.1.3.1 Contact	2.1.3.2 Location	2.1.3.3 Skills	2.1.3.4 Following	2.1.3.5 Posting	2.1.3.6 Analytics	2.1.3.7 Messaging	2.1.3.8 Search	2.1.3.9 Validation	2.1.3.10 Security
Analytics Service		X		X	X	X			X			
Database	X	X	X	X	X	X	X	X	X	X	X	X
File / Blob Storage		X				X						

7.2 Notification Requirements Matrix (SRS 2.2)

Component	2.2.1 Alerts	2.2.2 Updates	2.2.3 Push	2.2.4 Delivery	2.2.5 Preferences
Web Client	X	X	X	X	X
Mobile Client	X	X	X	X	X
API Gateway / Orchestrator	X	X	X	X	X
Authentication Service	X	X	X	X	X
Profile Management Service					
Resume Service					
Job Posting Service					

Component	2.2.1 Alerts	2.2.2 Updates	2.2.3 Push	2.2.4 Delivery	2.2.5 Preferences
Search Service					
Communication Services	X	X	X	X	X
Recommendation Service					
External Services			X	X	
Analytics Service	X	X	X	X	X
Database	X	X	X	X	X
File / Blob Storage					

7.3 Search Requirements Matrix (SRS 2.3)

Component	2.3.1 Candidate	2.3.2 Company	2.3.3 Filters	2.3.4 Wildcards	2.3.5 Performance	2.3.6 Autocomplete	2.3.7 Ranking	2.3.8 Resume
Web Client	X	X	X	X	X	X	X	X
Mobile Client	X	X	X	X	X	X	X	X
API Gateway / Orchestrator	X	X	X	X	X	X	X	X

Component	2.3.1 Candidate	2.3.2 Company	2.3.3 Filters	2.3.4 Wildcards	2.3.5 Performance	2.3.6 Autocomplete	2.3.7 Ranking	2.3.8 Resume
Authentication Service	X	X	X	X	X	X	X	X
Profile Management Service		X	X					
Resume Service			X					
Job Posting Service	X	X	X	X	X			
Search Service	X	X	X	X	X	X	X	X
Communication Services								
Recommendation Service		X	X	X			X	X
External Services						X	X	
Analytics Service	X	X	X	X	X	X	X	X
Database	X	X	X	X	X	X	X	X
File / Blob Storage								

7.4 News Feed Requirements Matrix (SRS 2.4)

Component	2.4.1 Posts	2.4.2 Updates	2.4.3 Media	2.4.4 Comments	2.4.5 Reactions	2.4.6 Feed
Web Client	X	X	X	X	X	X
Mobile Client	X	X	X	X	X	X
API Gateway / Orchestrator	X	X	X	X	X	X
Authentication Service	X	X	X	X	X	X
Profile Management Service						
Resume Service						
Job Posting Service						
Search Service						
Communication Services	X	X	X	X	X	X
Recommendation Service						
External Services						
Analytics Service	X	X	X	X	X	X
Database	X	X	X	X	X	X
File / Blob Storage			X			

7.5 Job Posting Requirements Matrix (SRS 2.5)

Component	2.5.1 Create	2.5.2 Edit	2.5.3 Delete	2.5.4 Skills	2.5.5 Details	2.5.6 Apply	2.5.7 Status	2.5.8 Filters	2.5.9 Data
Web Client	X	X	X	X	X	X	X	X	X
Mobile Client	X	X	X	X	X	X	X	X	X
API Gateway / Orchestrator	X	X	X	X	X	X	X	X	X
Authentication Service	X	X	X	X	X	X	X	X	X
Profile Management Service									
Resume Service						X	X	X	
Job Posting Service	X	X	X	X	X	X	X	X	X
Search Service	X	X	X	X			X	X	X
Communication Services									
Recommendation Service	X	X	X	X	X	X	X	X	X
External Services									
Analytics Service	X	X	X	X	X	X	X	X	X
Database	X	X	X	X	X	X	X	X	X
File / Blob Storage			X						

7.6 Student Profile Requirements Matrix (SRS 3.1)

Component	3.1.1 Student	3.1.2 Accolades	3.1.3.1 GPA	3.1.3.2 Skills	3.1.3.3 Experience	3.1.3.4 Resume
Web Client	X	X	X	X	X	X
Mobile Client	X	X	X	X	X	X
API Gateway / Orchestrator	X	X	X	X	X	X
Authentication Service	X	X	X	X	X	X
Profile Management Service	X	X	X	X	X	X
Resume Service			X	X	X	X
Job Posting Service						
Search Service						
Communication Services						
Recommendation Service		X	X	X	X	X
External Services						
Analytics Service	X	X	X	X	X	X
Database	X	X	X	X	X	X
File / Blob Storage			X	X		X

7.7 Shared Notification Requirements Matrix (SRS 3.2)

Component	3.2.1 Alerts	3.2.2 Messages	3.2.3 Push	3.2.4 Delivery	3.2.5 Tracking	3.2.6 Preferences	3.2.7 History
Web Client	X	X	X	X	X	X	X
Mobile Client	X	X	X	X	X	X	X
API Gateway / Orchestrator	X	X	X	X	X	X	X
Authentication Service							
Profile Management Service							
Resume Service							
Job Posting Service							
Search Service							
Communication Services	X	X	X	X	X	X	X
Recommendation Service							
External Services			X	X		X	
Analytics Service	X	X	X	X	X	X	X
Database	X	X	X	X	X	X	X
File / Blob Storage							

7.8 Student Search Requirements Matrix (SRS 3.3)

Component	3.3.1 Jobs	3.3.2 Companies	3.3.3 Matches	3.3.4 Filters
Web Client	X	X	X	X
Mobile Client	X	X	X	X
API Gateway / Orchestrator	X	X	X	X
Authentication Service				
Profile Management Service		X		X
Resume Service				
Job Posting Service	X		X	
Search Service	X	X	X	X
Communication Services				
Recommendation Service	X	X	X	X
External Services				
Analytics Service	X	X	X	X
Database	X	X	X	X
File / Blob Storage				

7.9 Shared News Feed Requirements Matrix (SRS 3.4)

Component	3.4.1 Feed	3.4.2 Updates
Web Client	X	X
Mobile Client	X	X
API Gateway / Orchestrator	X	X
Authentication Service		
Profile Management Service		
Resume Service		
Job Posting Service		
Search Service		
Communication Services	X	X
Recommendation Service		
External Services		
Analytics Service	X	X
Database	X	X
File / Blob Storage		

7.10 Resume Requirements Matrix (SRS 3.5)

Component	3.5.1 Builder	3.5.2 Upload	3.5.3 Storage	3.5.4.1 Fields	3.5.4.2 Help	3.5.5.1 Submit	3.5.5.2 Review	3.5.5.3 Required
Web Client	X	X	X	X	X	X	X	X
Mobile Client	X	X	X	X	X	X	X	X
API Gateway / Orchestrator	X	X	X	X	X	X	X	X
Authentication Service	X	X	X	X	X	X	X	X
Profile Management Service			X				X	
Resume Service	X	X	X	X	X	X	X	X
Job Posting Service								
Search Service								
Communication Services								
Recommendation Service		X	X				X	
External Services								
Analytics Service	X	X	X	X	X	X	X	X
Database	X	X	X	X	X	X	X	X
File / Blob Storage	X	X	X			X	X	

7.11 Hardware Requirements Matrix (SRS 4.0)

Component	4.1 Performance	4.2 Browsers	4.3 Devices	4.4 Supported
Web Client	X	X	X	X
Mobile Client	X	X	X	X
API Gateway / Orchestrator	X			
Authentication Service	X			
Profile Management Service	X			
Resume Service	X			
Job Posting Service	X			
Search Service	X			
Communication Services	X			
Recommendation Service	X			
External Services			X	X
Analytics Service	X			
Database	X			
File / Blob Storage	X			

8.0 APPENDICES

This section is optional.

Appendices may be included, either directly or by reference, to provide supporting details that could aid in the understanding of the Software Design Document.